2026 PRELIMS 99% MASTERY · CONSOLIDATED CHAPTER-WISE FACTS

ENVIRONMENT & ECOLOGY

पर्यावरण एवं पारिस्थितिकी — संपूर्ण उच्च-प्राथमिकता संकलन

Ecosystem Dynamics, Biodiversity Hotspots, National Parks, Biosphere Reserves, Ramsar Wetlands, Climate Conventions (UNFCCC, CBD), Pollution Acts.

UPPCS & UKPCS Prelims 2026 • High-Yield Master Revision Series • 99% Coverage Edition

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Consolidated High-Yield Revision Matrix • All 35 Chapters

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Executive Overview: Master Current Affairs & Memory Anchors

📰 2026 Prelims Apex Environment Current Affairs Matrix

YEAR / REF	KEY HIGH-YIELD FACT	EXAM RELEVANCE
2024	**India Expands Ramsar Network to 85 Wetlands:** Added 5 new sites on World Wetlands Day 2024 (Magadi Kere, Ankasamudra, Aghanashini in Karnataka; Karaivetti, Longwood Shola in Tamil Nadu). Tamil Nadu leads India with 16 sites; UP is 2nd with 10 sites.	Ramsar Wetlands, State-wise Distribution
2023	**Status of Tigers in India 2022 Release:** Tiger population reached **3,682** (75% of global wild tigers). Madhya Pradesh leads with 785 tigers, Karnataka 563. Ranipur notified as UP's 4th Tiger Reserve (53rd in India).	Project Tiger (1973), Demographics
2023	**COP 28 UAE Consensus (Dubai):** Historic agreement on operationalizing the **Loss & Damage Fund** and calling for "transitioning away from fossil fuels" in energy systems.	UNFCCC Milestones, Climate Finance
2022	**Kunming-Montreal Global Biodiversity Framework (CBD COP 15):** Adopted the historic **30×30 target** (protect 30% of degraded land and oceans globally by 2030).	Global Biodiversity Targets (CBD)
2022	**Wildlife (Protection) Amendment Act, 2022:** Rationalized wildlife protection schedules from 6 to 4; formally aligned domestic law with CITES Appendix listings.	WPA 1972 Statutory Architecture
Rolling	**Mission LiFE (Lifestyle for Environment):** India-led global mass movement promoting mindful consumption; introduced at COP 26 Glasgow, launched June 2022.	Sustainable Development Policy

🧠 Master Memory Anchors & Formula Mnemonics

- **4 Biodiversity Hotspots of India: H-I-W-S** (Himalaya, Indo-Burma, Western Ghats–Sri Lanka, Sundaland [Nicobar]).
- **6 Kyoto Protocol Greenhouse Gases: C-M-N-H-P-S** (Carbon dioxide, Methane, Nitrous oxide, Hydrofluorocarbons, Perfluorocarbons, Sulphur hexafluoride; NF_3 added in Doha Amendment).
- **Trophic Energy Transfer Rule: Lindeman's 10% Law (1942)** — Only ~10% of energy is transferred to the next higher trophic level; 90% lost as metabolic heat. Energy pyramids are *always upright*.
- **WPA Schedules (Post-2022 Amendment):**
  - **Schedule I:** Animal species enjoying highest level of protection (Blackbuck, Lion-tailed Macaque, Great Indian Bustard).
  - **Schedule II:** Animal species with lesser protection.
  - **Schedule III:** Protected Plant species.
  - **Schedule IV:** Specimens listed under CITES Appendices.
- **Ozone Column Units:** Measured in **Dobson Units (DU)** (1 DU = 0.01 mm thickness at STP; normal ozone column ~300 DU; <220 DU is classified as an ozone hole).
- **Montreux Record:** Ramsar sites undergoing adverse ecological changes. In India: **Keoladeo National Park** (Rajasthan) and **Loktak Lake** (Manipur); Chilika Lake was removed post-restoration.

Chapter 01: Environment (पर्यावरण) Basics

📰 Current Affairs & High-Yield Updates

YEAR	FACT	WHY ASKED
2023	**Amrit Dharohar Scheme** — wetlands conservation; launched on WED 2023	Wetlands + governance
2023	**National Green Hydrogen Mission**	SD / climate policy overlap
2019	**NCAP** (राष्ट्रीय स्वच्छ वायु कार्यक्रम) — 132 cities; 20–30% PM2.5/PM10 reduction target by 2024	Urban (नगरीय) environment governance

🎯 Consolidated Must-Score Facts

1. Environment means the sum of **biotic and abiotic** factors that affect living organisms. Notes do not treat "only physical" or "only living" as the full definition.
2. The four (चातुर्याम) spheres are **lithosphere, hydrosphere, atmosphere, and biosphere**. Biosphere roughly spans from about **200 m below soil** to about **6 km** above sea level.
3. **Ecology** (पारिस्थितिकी) is the **study** of organism–environment links. An **ecosystem** (पारिस्थितिकी तंत्र) is the **functional unit**. A **habitat** (वास स्थान) is the **place**. A **niche** is the **role**.
4. Abiotic factors include light, temperature, water, soil, gases, pH, and salinity. **Edaphic** means soil; **climatic** means weather; **topographic** means slope and altitude.
5. Biotic structure runs **producers → consumers → decomposers**. **Rhizobium** fixes nitrogen in legumes.
6. **Liebig's law** says growth is limited by the **scarcest** essential resource. **Shelford's law** says a species survives only inside a **min–max tolerance** range.
7. Land is about **29%** of Earth's surface and water about **71%**. **Lentic** means standing water; **lotic** means flowing water. Mangroves and wetlands are **transitional**.
8. Human–environment thought moves **environmental determinism (Ratzel) → possibilism (Vidal de la Blache) → neo-determinism (Griffith Taylor)**.
9. **IPAT** is Impact = Population (जनसंख्या) × Affluence × Technology. Hardin's **Tragedy of the Commons** dates to **1968**.
10. Sustainable development is defined in the **Brundtland Report (1987)** (ब्रंटलैंड रिपोर्ट). **Rio 1992** produced **Agenda 21**, **UNFCCC** (यूएनएफसीसीसी), and **CBD**.
11. India's **LiFE** Mission launched in **2022**; the idea was floated at **COP26 in 2021**, not COP25.
12. **Chipko** (चिपको) began in **1973** at **Reni, Uttarakhand (उत्तराखंड)**. **JFM** guidance is **1990**. **FRA** is **2006**.
13. **Article 48A** is the State DPSP (राज्य के नीति निदेशक तत्व) on environment. **Article 51A(g)** is the citizen's Fundamental Duty. **Article 48** is about agriculture and cows — not environment.
14. **Article 21** (अनुच्छेद 21) has been read to include a **healthy environment**. The **42nd (42वां) Amendment (1976)** added 48A and 51A(g).
15. The **Environment (Protection) Act, 1986** is India's post-**Bhopal (1984)** umbrella law.
16. **MoEF** began in **1985** and became **MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) in 2014**. Headquarters is New Delhi.
17. **Stockholm 1972** (स्टॉकहोम सम्मेलन) was the first UN human-environment conference and led to **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) at **Nairobi**.
18. World Environment Day (विश्व पर्यावरण दिवस) is **5 June**. Earth Day (पृथ्वी दिवस) is **22 April**. World Wildlife Day (विश्व वन्यजीव दिवस) is **3 March**, not 22 March.
19. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February**. International Ozone Day is **16 September**. Desertification (मरुस्थलीकरण) and Drought (सूखा) Day is **17 June**. International Day for Biological Diversity is **22 May**.
20. UNEP headquarters is **Nairobi, Kenya**. CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) and MoEFCC sit in **New Delhi**.
21. The **Brundtland Commission (WCED)** worked **1983–1987** and defined sustainable development in *Our Common Future*.
22. Uttar Pradesh (उत्तर प्रदेश) mainly falls in the **Gangetic Plain** biogeographic zone (**Zone 7**).
23. Classic UP river–city facts include Ganga (गंगा) at Varanasi (वाराणसी) and Kanpur (कानपुर), Yamuna (यमुना) at Agra, and Gomti at Lucknow (लखनऊ).
24. Pollution hotspots often tested for UP include **Kanpur tanneries**, NCR air (Ghaziabad/Noida), and Ganga ghats at **Varanasi** (वाराणसी).
25. **Ganga Action Plan** (गंगा एक्शन प्लान) began in **1985**. **Namami Gange** (नमामि गंगे) launched in **2014**.
26. A standard triad classifies environment as **physical**, **biological**, and **cultural**. **Operational environment** is **not** part of that triad.
27. Natural versus anthropogenic, terrestrial versus aquatic, and internal versus external are further common classification axes.
28. Micro-, meso-, and macro-scales describe environment size from a pond edge to a region or the globe. Competitive papers usually default to the **natural/physical** sense of environment.
29. Agenda 21 belongs to **Rio 1992**, not 1995. Mixing Brundtland 1987 with Agenda 21's year is a common trap.
30. **Carrying capacity** is the maximum load an environment can sustain indefinitely. Resource extraction must stay **below** regeneration, and waste must stay **within** absorption capacity — not the reverse.
31. **Ecological footprint** (पारिस्थितिक पदचिह्न) measures the land/water area needed to support consumption, in **global hectares (gha)**.
32. **Pedosphere** (soil) is usually grouped with the lithosphere or with edaphic abiotic factors. It is not counted as a fifth sphere.
33. **Micro-scale** environment covers immediate surroundings such as a room or a pond edge.
34. **Meso-scale** covers a village or city.
35. **Macro-scale** covers regional or global systems.
36. **Total environment** can be natural, social, economic, or cultural. Competitive papers usually default to **natural/physical** environment.
37. **UNEP** (United Nations Environment Programme) was created at Stockholm. Its headquarters is in **Nairobi, Kenya**.
38. **World Environment Day (WED)** is observed every year on **5 June**. The date marks the opening day of the Stockholm Conference in 1972.

39. **MoEF** was established in **1985**. It was renamed **MoEFCC in 2014** when climate change (जलवायु परिवर्तन) was added to its mandate.
40. **Article 48A** is a Directive Principle. The State shall protect the environment and safeguard forests and wildlife. The **42nd Amendment (1976)** added it.
41. **Article 51A(g)** is a Fundamental Duty. Every citizen shall protect and improve the natural environment. The same amendment added it.
42. **Article 21** guarantees the Right to Life. The Supreme Court expanded it to include the **right to a wholesome and healthy environment** in cases such as *M.C. Mehta* and *Subhash Kumar*.
43. **Liebig's Law of Minimum** states that growth is controlled by the **scarcest essential resource**. That resource is the limiting factor.
44. **Shelford's Law of Tolerance** (शेलफोर्ड का सहनशीलता नियम) states that each species survives only within a **minimum–maximum range** for each factor. Beyond that range, stress or death follows even if other factors are abundant.
45. **N_2** makes up about **78%** of air, but most life cannot use it directly. Organisms need **nitrogen fixation** first.
46. **Rhizobium** bacteria live in legume root nodules and fix atmospheric N_2. Pulses, gram, and soybean are common legume examples.
47. **Azotobacter** is a free-living soil bacterium that fixes nitrogen.
48. **Cyanobacteria** (blue-green algae) also fix nitrogen in soil and water.
49. **Humus** is organic matter in soil. It comes from decomposed plant and animal material, so it is **not** an abiotic component.
50. **Dissolved Oxygen (DO)** is a key aquatic abiotic indicator. Fish die in polluted rivers when DO falls too low.

📊 Quick Reference Grid

Core tags

ITEM	LOCK
Ecology / Ecosystem / Habitat / Niche	Study / functional unit / place / role
Four spheres	Lithosphere · hydrosphere · atmosphere · biosphere
Biosphere span	~200 m below soil → ~6 km above sea level
Edaphic / Climatic / Topographic	Soil / weather / slope–altitude
Liebig / Shelford	Scarcest resource / min–max tolerance
Lentic / Lotic	Standing / flowing water
Thought chain	Ratzel determinism → Vidal possibilism → Griffith Taylor neo-determinism
IPAT / Tragedy of Commons	P×A×T / Hardin **1968**

Milestones & law

ITEM	LOCK
Brundtland / Rio	**1987** SD definition / **1992** Agenda 21 + UNFCCC + CBD
LiFE	Idea COP26 **2021**; launch **2022**
Chipko / JFM / FRA	**1973** Reni / **1990** / **2006**
Art. 48A / 51A(g) / 48	State DPSP env / citizen duty / agri–cows (**not** env)
42nd Amd / EPA	**1976** adds 48A+51A(g) / EPA **1986** post-Bhopal
MoEF → MoEFCC	**1985** → **2014**; HQ New Delhi
Stockholm / UNEP	**1972** → UNEP **Nairobi**
GAP / Namami Gange	**1985** / **2014**

Days (quick)

ITEM	LOCK
WED / Earth / Wildlife	5 Jun / 22 Apr / **3 Mar** (≠ 22 Mar)
Wetlands / Ozone / Desertif. / Biodiversity	2 Feb / 16 Sep / 17 Jun / 22 May


Earth's Four Interconnected Spheres & Biosphere Envelope

Chapter 02: Ecology (पारिस्थितिकी) & Ecosystem (पारिस्थितिकी तंत्र)

🎯 Consolidated Must-Score Facts

1. **Ecology** is the **study** of organism–environment (पर्यावरण) interactions. **Haeckel (1866)** coined the term. **Ramdeo Misra** is called the father of Indian ecology. **Bionomics** (बायोनॉमिक्स) is synonymous with ecology; it does **not** mean "management of life."
2. An **ecosystem** (पारिस्थितिकी तंत्र) is a **functional unit** of biotic and abiotic interaction with energy flow and nutrient cycling. **Tansley (1935)** coined the word.
3. Ecosystems are normally **open systems**. Calling an ecosystem a closed system is wrong.
4. Four (चातुर्याम) basic components often listed together are **abiotic materials, producers, consumers, and decomposers**.
5. The seven ecological levels run **organism → population (जनसंख्या) → community → ecosystem → landscape → biome (जीवोम) → biosphere**. A frequent shorter order asked is **Population → Community → Ecosystem → Landscape**.
6. The **biosphere** is the largest ecosystem of Earth. Among ecosystem *types*, the **marine/ocean** ecosystem covers the largest area and is cited as the **most stable**.
7. Ecosystems may be natural or anthropogenic, terrestrial or aquatic, and **lentic** (standing) or **lotic** (flowing).
8. Artificial / anthropogenic examples include **farmland / rice field**, orchard, garden, aquarium, and human-made reservoir.
9. A pond's vertical/horizontal zones include **littoral → limnetic → profundal**. Marine zones include littoral, neritic, pelagic, benthic, and abyssal.
10. Structure runs **producers → primary/secondary/tertiary consumers → decomposers and detritivores**. Producers generally outnumber primary consumers.
11. Core ecosystem functions are **productivity, decomposition, nutrient cycling, and unidirectional energy flow**.
12. A **niche** is a species' **role**, not its place. A **habitat** (वास स्थान) is the place. **Joseph Grinnell (1917)** coined the ecological-niche concept.
13. A **fundamental niche** (मूल (मौलिक) निकेत) is the full potential role without competitors. A **realized niche** (वास्तविक निकेत) is the narrower role after competition.
14. **Gause's competitive exclusion** says two species with identical niches cannot stably coexist when resources are limited.
15. **Liebig** = limiting factor; **Shelford** = tolerance range; **Lindeman (1942)** ≈ **10%** energy transfer between trophic levels.
16. **NPP = GPP − plant respiration**. Plants capture only about **1%** of incident sunlight on leaves.
17. Decreasing productivity order often tested: **Mangroves > grasslands > lakes > oceans**.
18. **Primary succession** (प्राथमिक अनुक्रमण) starts on a lifeless substrate. **Secondary succession** (द्वितीयक अनुक्रमण) starts where **soil already exists** after disturbance.
19. Full succession phases: **Nudation → Migration → Ecesis → Reaction → Stabilisation**. The full community sequence is a **sere** (क्रमक). UPPCS 2024 often starts from Migration when Nudation is omitted.
20. Pioneers such as lichen and moss lead through seral stages to a **climax community** (चरम समुदाय). **Monoclimax** theory is linked to **F.E. Clements**. **Hydrosere** is wet; **xerosere** is dry.
21. In grasslands, trees often fail to replace grasses because of **water limits and fire**.
22. An **r-strategist** produces many offspring with fast growth and short life. A **K-strategist** produces few offspring with slow growth and long life.
23. Population growth models contrast the **J-curve** (exponential) with the **S-curve** (logistic) that levels near **carrying capacity**.
24. MEA (**2005**) ecosystem services are usually **provisioning, regulating, supporting, and cultural**. Some WDR/UPSC (संघ लोक सेवा आयोग) framings add **preserving**. Supporting examples include nutrient cycling and (in that framing) crop pollination.
25. Ecosystem balance is a **dynamic equilibrium**. **Resilience** is the capacity to recover after disturbance. **Deforestation** is a main cause of ecological imbalance in India.
26. **Eutrophication** (सुपोषण) is nutrient overload that drives algal bloom and a dissolved-oxygen crash.
27. Energy flow is **one-way**. Nutrients **cycle**. Saying energy cycles is a classic wrong option.
28. A **community** is all populations of different species in an area. A **population** is one interbreeding species in one area.
29. An **ecotone** (इकोटोन) is the transition zone between two communities or ecosystems.
30. Aquatic clines: **pycnocline** = density gradient; **halocline** = salinity gradient; **thermocline** = temperature-with-depth gradient.
31. In the **phosphorus cycle**, weathering of rocks is the main source releasing the nutrient into the cycle.
32. **Upwelling** brings cold, nutrient-rich water to the surface and raises marine productivity.
33. **Eucalyptus** is often treated as an environmental hazard because it draws heavy groundwater and can lower the water table.
34. UP lotic examples include the **Ganga (गंगा), Yamuna (यमुना), and Gomti**. Terai (तराई) forest and Vindhyan scrub are terrestrial examples.
35. UP wetland examples include **Sur Sarovar (Agra)** and **Nawabganj (Unnao)**. Western UP sugarcane belts are classic agro-ecosystems.
36. Odum is linked to modern ecosystem ecology; keep him separate from Haeckel (term ecology) and Tansley (term ecosystem). **Deep ecology** was coined by **Arne Næss (1973)**. **Ecological Transition** was used by **John W. Bennett**.
37. **Allen's Rule:** colder climates → smaller appendages (less heat loss). **Bergmann's Rule:** colder climates → larger body size (heat conservation). Do not swap the pair.
38. **Gloger's Rule:** darker pigmentation is more common in warm, humid regions.
39. **Limiting similarity** means coexisting species differ enough in niche use that they do not compete for exactly the same resource at the same time.
40. **Ecology** (from Greek *Oikos* = house + *Logos* = study) is the study of relationships between living organisms and their environment.
41. **Ernst Haeckel** coined the term **Ecology** in **1866**.
42. **Bionomics** (Greek *bio* = life; *nomos* = law) is a comprehensive study of an organism in relation to its environment. It is **synonymous with ecology**.

43. **Ramdeo Misra** is regarded as the father of **ecology in India** (ecology teaching at BHU).
44. **Eugene P. Odum** is called the father of modern **ecosystem ecology** (*Fundamentals of Ecology*, 1953).
45. **Arne Næss** coined **deep ecology** in **1973**.
46. **John W. Bennett** used the concept of **Ecological Transition** in *The Ecological Transition: Cultural Anthropology and Human Adaptation*.
47. **Organ** is anatomy, **not** an ecological organisation level.
48. **Autoecology** studies one species in its environment.
49. **Synecology** studies communities and ecosystems together.
50. **A.G. Tansley** introduced the term **ecosystem** in **1935**.
51. **Ecology Terminological Origins:** Term 'Ecology' (*Oekologie*) coined in 1866 by German biologist **Ernst Haeckel** (from Greek *oikos* meaning home/household and *logos* meaning study); term 'Ecosystem' coined in 1935 by British ecologist **A.G. Tansley**; term 'Biosphere' introduced by geologist **Eduard Suess (1875)**; **Eugene Odum** is regarded as the 'Father of Modern Ecology'.
52. **Ecological Niche Concept:** First formulated by **Joseph Grinnell (1917)** and refined by **G. Evelyn Hutchinson (1957)**; represents the specific functional role and physical position of an organism within its ecosystem ('Address is habitat, profession is niche'); two species cannot coexist indefinitely in the same niche if they compete for identical resources (**Gause's Principle of Competitive Exclusion**).
53. **Ecotone & Edge Effect: Ecotone** is a transitional zone of junction between two diverse ecological communities/biomes (e.g. mangrove between terrestrial and marine; marshland between aquatic and dry land); exhibits high species richness and population density known as the **Edge Effect**; organisms occurring primarily or most abundantly in this transition zone are called **Edge Species** (e.g. birds in forest borders).
54. **Ecological Succession Stages:** Directional and predictable process of community change over time; (1) **Primary Succession:** Initiated on previously uncolonized, barren, soil-less surfaces (bare rock, cooled lava, sand dune); pioneer species on bare rock are **Lichens and Mosses** (secrete acids to weather rock into soil); (2) **Secondary Succession:** Occurs in areas where an existing community has been destroyed (forest fire, abandoned farmland, flood) but where soil and seeds already exist (much faster); final stable, self-perpetuating terminal community is the **Climax Community**.
55. **Ecological Pyramids (Charles Elton, 1927):** (1) **Pyramid of Numbers:** Upright in grassland and pond ecosystems; **Inverted in parasitic food chains and single tree ecosystem** (one large tree supports hundreds of herbivorous birds, which host thousands of lice/parasites); (2) **Pyramid of Biomass:** Upright in terrestrial ecosystems; **Inverted in aquatic / ocean ecosystems** (biomass of primary producers [phytoplankton] at any given time is much smaller than the biomass of consumers [zooplankton, fish] due to rapid turnover rate of microscopic algae); (3) **Pyramid of Energy: ALWAYS UPRIGHT** without any exception in all ecosystems, governed by the Second Law of Thermodynamics.
56. **Lindeman's 10% Energy Transfer Law (1942):** Formulated by Raymond Lindeman; states that during the transfer of organic energy from one trophic level to the next higher level, only about **10% of the energy is conserved and stored as biomass**, while the remaining **90% is dissipated as heat** during metabolic respiration and waste.
57. **Bioaccumulation vs Biomagnification:** (1) **Bioaccumulation:** Gradual buildup of a persistent, fat-soluble, non-biodegradable chemical substance (like DDT, heavy metals) within the tissues of a single individual organism over its lifetime directly from its environment; (2) **Biomagnification / Bioamplification:** Progressive increase in concentration of the toxic substance at **successively higher trophic levels** across a food chain (e.g. DDT concentration increases from water [0.000003 ppm] -> zooplankton [0.04 ppm] -> small fish [0.5 ppm] -> large fish [2 ppm] -> fish-eating birds [25 ppm], causing eggshell thinning and population collapse of eagles/pelicans).

📊 Quick Reference Grid

Founders / levels

ITEM	LOCK
Ecology term / Indian ecology	Haeckel **1866** / Ramdeo Misra
Ecosystem term	Tansley **1935**; normally **open** system
Levels (short)	Population → Community → Ecosystem → Landscape
Niche coined	Grinnell **1917** (role ≠ place)
Lindeman	**10%** energy transfer (**1942**)
NPP	GPP − plant respiration; plants ~**1%** of sunlight

Succession / services

ITEM	LOCK
Primary vs secondary	Lifeless substrate / soil already present
Sere phases	Nudation → Migration → Ecesis → Reaction → Stabilisation
Monoclimax	F.E. Clements
r vs K	Many offspring–short life / few–long life
MEA 2005 services	Provisioning · regulating · supporting · cultural
Eutrophication	Nutrient overload → bloom → DO crash
Energy vs nutrients	Energy **one-way**; nutrients **cycle**
Ecotone / Deep ecology	Transition zone / Arne Næss **1973**

Chapter 03: Food Chain (खाद्य श्रृंखला) & Energy Flow

🎯 Consolidated Must-Score Facts

1. A **food chain** is a single linear who-eats-whom sequence. A **food web** is a network of interconnected chains and is more stable than one chain alone.
2. A **grazing food chain** starts from **living green plants**. A **detritus food chain** starts from **dead organic matter**.
3. Classic terrestrial chain: **Grass → Grasshopper → Frog → Snake**. Classic human-linked chain: **Grass → Goat → Human**.
4. Classic marine chain: **Diatoms → Crustaceans → Herrings**.
5. **Trophic levels** are feeding ranks counted from producers: **T1 producers, T2 herbivores, T3–T5 carnivores**.
6. The **Sun is the energy source**, not a trophic level. Never count sunlight as T1. **Solar energy** is the driving force of the ecosystem.
7. Energy flow is **unidirectional**: Sun → producers → consumers → heat. Nutrients cycle; **energy does not**.
8. About **10%** of energy passes to the next trophic level (**Lindeman, 1942**). About **90%** is lost as heat, respiration, and movement. Energy **decreases** upward.
9. Food chains stay short, usually about **3–5 levels**, because usable energy runs out. Food chains are **not** found within a single species.
10. **Humans** are **primary as well as secondary** consumers (plants and meat).
11. Main marine primary producers are **phytoplankton**.
12. Primary consumers are mainly herbivores such as **deer**; **ants** may also act as primary consumers (and scavengers/decomposers in broader roles). Fox and tiger are not primary consumers.
13. Decomposers are **fungi and bacteria**, not viruses.
14. The **energy pyramid** (kcal/m^2/yr) is **always upright**. An inverted energy pyramid is never correct.
15. A **biomass pyramid** can invert in **pond / marine** systems. A **numbers pyramid** can invert in a parasitic chain (one tree, many insects).
16. **Biomagnification** raises toxin concentration up the chain. For DDT, the **snake** (tertiary consumer) shows higher concentration than grasshopper or toad.
17. Butterfly decline harms **pollination** and can cut populations of predators such as wasps, spiders, and birds. It does not by itself force a drastic rise in fungal infections of crops.
18. Grazing does **not** start from dead matter. Detritus does **not** start from living green plants.
19. More links in a food web raise stability, but energy still flows **one way**.
20. Decomposers and detritivores close the detritus path and recycle nutrients back to producers.
21. Apex predators sit at the top trophic levels and are often scarce because little energy remains.
22. Dudhwa (दुधवा)'s grass → deer → tiger chain is a classic UP **grazing** example with tiger as apex predator.
23. A Ganga (गंगा) algae → fish → dolphin web is a standard UP **lotic aquatic** food-web example.
24. Terai (तराई) sal leaf litter → earthworm is a UP **detritus** start. Do not call it grazing.
25. Western UP sugarcane → pest → predator is a short **cropland** chain of about three levels.
26. Vulture decline after veterinary **diclofenac** (डाइक्लोफेनाक) disrupts the **detritus / scavenging** pathway.
27. Ecological pyramids may show energy, biomass, or numbers — always note which quantity is plotted.
28. Saying "energy is cyclic like nutrients" is a frequent false statement.
29. Match pyramid type to exception: energy never inverts; biomass may invert at sea/pond; numbers may invert with parasites.
30. Keep Lindeman's **10% rule** as an approximate key fact, not an exact laboratory constant for every ecosystem.
31. **Omnivores** such as bears, crows, and humans link grazing and detritus pathways. They can occupy **more than one trophic level** at once.
32. **Keystone predators** such as tigers regulate entire webs. Removing them causes trophic cascade across several chains.
33. **T1 (Producers)** are autotrophs such as green plants, algae, and phytoplankton. They fix solar energy into chemical energy.
34. **T2 (Primary consumers)** are herbivores that eat producers directly. Examples include deer, zooplankton, and caterpillars.
35. **T3 (Secondary consumers)** are carnivores that eat primary consumers. Examples include frogs and small fish.
36. **T4 (Tertiary consumers)** eat secondary consumers. Examples include snakes and large fish.
37. **T5 (Quaternary consumers)** are top predators with few or no natural enemies. Tigers and eagles appear here when the chain is long enough.
38. **Omnivores** eat both plants and animals. They span **multiple trophic levels** simultaneously.
39. **Decomposers** break down dead matter from **all levels**. They recycle nutrients but are usually drawn **beside** the main chain, not as the top of an energy pyramid.
40. **Detritivores** such as earthworms fragment dead material and bridge the detritus chain toward decomposers.
41. **Biomagnification** means a non-degradable toxin (classic example **DDT**, also mercury) rises in concentration at successive trophic levels and lodges in fatty tissues. The highest concentration appears in top consumers such as a **snake** in a grasshopper–toad–snake chain.
42. **Energy flow** is the movement of solar energy through living systems via food chains and food webs.
43. **Nutrients** such as carbon and nitrogen **cycle** through decomposers and return to producers. **Energy does not cycle.**
44. **Raymond Lindeman (1942)** showed that roughly **10%** of energy or organic matter passes to the next higher trophic level.
45. **Ecological efficiency** equals energy at the higher level divided by energy at the lower level, multiplied by 100. The NCERT standard is about **10%**.
46. **Charles Elton (1927)** developed the early pyramid concept in trophic studies.
47. **Decomposers** are usually omitted from standard pyramid diagrams. They act on all levels and return nutrients to producers.
48. **Biomagnification** of toxins such as DDT increases concentration up the food chain. Top predators suffer most even as energy decreases upward.

📊 Quick Reference Grid

Chain types

ITEM	LOCK
Food chain vs web	Linear sequence / interconnected network (more stable)
Grazing vs detritus	Living green plants / dead organic matter
Trophic ranks	T1 producers → T2 herbivores → T3–T5 carnivores
Sun	Energy **source** — **not** a trophic level
Lindeman 10%	~10% up; ~90% lost; energy ↓ upward
Humans	Primary **and** secondary consumers
Decomposers	Fungi + bacteria (≠ viruses)

Pyramids / toxins

ITEM	LOCK
Energy pyramid	**Always upright**
Biomass / numbers invert	Pond–marine / parasitic chain
Biomagnification	Toxin ↑ up chain (DDT highest in apex)
Diclofenac	Vulture crash → scavenging pathway break
UP grazing example	Dudhwa grass → deer → tiger

Chapter 04: Biodiversity

🎯 Consolidated Must-Score Facts

1. Biodiversity is variability among living organisms at **genetic, species, and ecosystem** levels — the range of plants and animals in an area.
2. **Walter G. Rosen (1985)** coined the word *biodiversity*. **E.O. Wilson** popularised it. **Whittaker (1960)** advanced **alpha, beta, and gamma** diversity.
3. **Genetic diversity** is variation **within** a species. **Species diversity** is variety **among** species. **Ecosystem diversity** is variety of habitats and ecosystems.
4. **Alpha** = mean species diversity at a local habitat (वास स्थान). **Beta** = change between ecosystems. **Gamma** = overall regional diversity.
5. The most significant aspect of biodiversity is **maintenance of the ecosystem**, not food or drugs alone.
6. Species richness is generally highest in the **tropics / tropical rainforest** (optimum biome (जीवोम)) and **increases toward the equator**; it decreases toward the poles and with altitude.
7. On land, tropical rainforest is richest; in the sea, **coral reefs** are among the richest. UPPCS has accepted both rainforest and coral reefs for "relatively higher diversity."
8. India holds about **2.4%** of world land but about **7–8%** of global species (rough shares often cited: mammals ~7.6%, birds ~12.6%, reptiles ~6.2%, amphibians ~4.4%).
9. India's four (चातुर्याम) biodiversity hotspots are **Himalaya (हिमालय), Indo-Burma, Western Ghats–Sri Lanka, and Sundaland**. **Andamans** fall under **Indo-Burma**; **Nicobar** (निकोबार) under **Sundaland**. Eastern Ghats (पूर्वी घाट) / Vindhyas / Naro Hills are **not** hotspots.
10. A hotspot needs about **≥1500 endemic vascular plants** and **≥70%** original habitat lost. Criteria also stress **species richness, endemism, and threat**. There are **36** global hotspots (~2.5% of land; >50% endemic plants).
11. **Silent Valley (साइलेंट वैली) (Kerala)** is among India's richest rainforest biodiversity sites (NP **1984**; Silent Valley Project = Kerala). **Valley of Flowers** (फूलों की घाटी) is in **Chamoli (चमोली), Uttarakhand (उत्तराखंड)**. Himalayan richness reflects confluence of biogeographic zones.
12. The top cause of biodiversity loss is **habitat destruction**. HIPPO = Habitat, Invasive species (आक्रामक विदेशी प्रजाति), Pollution, Population (जनसंख्या) pressure, Over-exploitation. Vegetarianism is **not** a threat. Fewer trophic levels reduce diversity.
13. India aims for **Land Degradation (भू-क्षरण) Neutrality by 2030**.
14. **In-situ** (स्थान पर) = natural habitat (NP, sanctuary, BR, sacred grove). **Ex-situ** (स्थानांतरित) = outside habitat (zoo, botanical garden (वनस्पति उद्यान), seed bank (बीज बैंक)). Cryo-banks use **liquid nitrogen**. Biosphere reserves (जैवमंडल आरक्षित क्षेत्र) best join biodiversity with traditional human life / cultural diversity.
15. The **CBD (1992, Rio)** rests on three pillars: conserve, sustainable use, fair benefit-sharing. **Cartagena Protocol** = biosafety / LMOs. **Nagoya (2010)** = ABS.
16. India's **Biological Diversity Act, 2002** creates **NBA (Chennai, 2003)**, SBBs, BMCs, and **PBRs**. BMCs are central to Nagoya ABS, including collection fees.
17. On the IUCN Red List, **Threatened** = **VU + EN + CR**. **Endangered** = **EN** only. Red Data Book (रेड डेटा बुक) lists threatened species. A species not seen for **50 years** may be treated as extinct in older definitional stems.
18. An **endemic** species occurs only in one area. An **invasive alien** is non-native and harmful (**Prosopis juliflora** reduces native biodiversity).
19. Keystone / flagship / umbrella / indicator roles stay distinct.
20. International Day for Biological Diversity is **22 May** (earlier linked to **29 December**). UN **Decade on Biodiversity = 2011–2020**.
21. India's first **National Centre for Marine Biodiversity (NCMB)** is at **Jamnagar (Gujarat)**.
22. The **Kunming–Montreal GBF (2022)** includes the **30×30** target. CBD COP-10 = **Nagoya 2010**; COP-11 = **Hyderabad (हैदराबाद) 2012**.
23. Hotspot concept = **Norman Myers** (not BirdLife). **BirdLife International** identifies Important Bird and Biodiversity Areas.
24. **TRAFFIC** (WWF (डब्ल्यूडब्ल्यूएफ)–IUCN alliance, 1976) monitors wildlife trade (पण्याध्यक्ष) — not a UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) bureau.
25. India coral reefs: **Lakshadweep (लक्षद्वीप) (atolls), Gulf of Mannar, Gulf of Kachchh, Andaman (अंडमान) & Nicobar** — **not** Sundarbans (सुंदरबन) (mangroves). Coral bleaching (प्रवाल विरंजन) mainly from **warming seawater / global warming (भूमंडलीय तापन)**.
26. **Diclofenac** (डाइक्लोफेनाक) veterinary painkiller caused India's vulture crash. **Dugong** = herbivorous marine mammal (Schedule I); Gulf of Mannar / Palk. **Indian wild ass** = Rann of Kachchh. **Lion-tailed macaque** = Western Ghats (Kerala, Karnataka, Tamil Nadu (तमिलनाडु)).
27. India has **80 Ramsar sites**. **Tamil Nadu 16** (highest), **Uttar Pradesh (उत्तर प्रदेश) 10**, **Punjab 6**. **Montreux Record** lists Ramsar sites with adverse ecological change from human interference.
28. UP Terai (Dudhwa (दुधवा), Pilibhit) sits on the **Himalaya hotspot** fringe. Key UP species: tiger, rhino, barasingha, Gangetic dolphin; state bird **Sarus crane**.
29. Common UP invasives: water hyacinth, Parthenium, Lantana.
30. NBA HQ = **Chennai**. CBD ≠ Nagoya year. Threatened ≠ Endangered.
31. **Sea buckthorn** (Ladakh (लद्दाख) / cold Himalaya) checks erosion and is nutritious — not primarily a biodiesel (बायो-डीजल) timber crop.
32. **TEEB** makes nature's economic values visible for decision-makers. **Wetlands International** is an independent NGO (HQ Netherlands), not a Ramsar intergovernmental organ.
33. **CEBPOL** (Centre for Biodiversity Policy and Law) sits with NBA, **Chennai** (India–Norway).
34. Maximum butterflies in tropics because they **cannot tolerate low temperatures**.
35. **Gulf of Mannar Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र)** is often cited as India's richest marine biodiversity region (21 islands mosaic).
36. Afforestation does **not** cause biodiversity loss; road expansion, urbanisation, and agriculture expansion do (UKPCS-style stems).
37. **Biodiversity** is the variety and variability of life at **genetic**, **species**, and **ecosystem** (पारिस्थितिकी तंत्र) levels in a given area — the diverse range of plants and animals in an ecosystem.
38. **International Day for Biological Diversity** is **22 May** (UN moved it from the older **29 December** CBD entry-into-force date). The UN declared **2011–2020** the **Decade on Biodiversity**.
39. **Genetic diversity** is variation in genes and alleles **within** a species.

40. **Species diversity** depends on **species richness** (how many species) and **evenness** (how evenly individuals are spread).
41. **R.H. Whittaker (1960)** advanced the **alpha–beta–gamma** scale for species diversity.
42. **Alpha diversity** (अल्फा विविधता) is the **mean species diversity in habitats at a local scale**.
43. **Beta diversity** (बीटा विविधता) is the comparison / change of species **between** ecosystems.
44. **Gamma diversity** is overall diversity across different ecosystems in a **region**.
45. **Ecosystem diversity** is the variety of habitats and ecosystems in a region such as forests, wetlands, grasslands, and deserts.
46. **17 megadiverse countries** together hold most of the world's species. **India** is one of them.
47. **Russia is not** on the standard megadiverse list of 17.
48. **Silent Valley (Kerala, Palakkad)** is remnant moist evergreen rainforest of the Western Ghats.
49. **Direct values** include food, fuel, fibre, medicine, and timber taken from nature.
50. **Indirect values** include soil formation, prevention of soil erosion, recycling of waste, pollination of crops, climate regulation, and flood control.
51. **Biodiversity Terminology & Levels:** Term 'Biological Diversity' coined by **Raymond F. Dasmann (1968)**; contracted to 'Biodiversity' by **W.G. Rosen (1985)**; popularized globally by entomologist **Edward O. Wilson** ('Father of Biodiversity'); 3 fundamental levels: (1) Genetic Diversity (variation within species), (2) Species Diversity (richness and evenness), (3) Ecosystem / Ecological Diversity.
52. **Whittaker's Alpha, Beta & Gamma Diversity (1960):** (1) **Alpha ($□lpha$) Diversity:** Species diversity within a particular local habitat or ecosystem; (2) **Beta ($□eta$) Diversity:** Rate of species turnover or change between two different ecosystems / along an environmental gradient; (3) **Gamma (γ) Diversity:** Overall species diversity of a large landscape, region, or biome.
53. **Latitudinal Diversity Gradient:** Species richness systematically **decreases as one moves from the Equator towards the Poles**; Tropics (between 23.5°N and 23.5°S) harbor apex biodiversity because of prolonged evolutionary stability (no glaciations), constant solar energy, and high annual rainfall.
54. **Biodiversity Hotspots (Norman Myers, 1988):** Critical regions characterized by exceptional concentrations of endemic species undergoing rapid habitat loss; to qualify as a Hotspot under Conservation International criteria, a region must strictly satisfy 2 criteria: (1) Must contain at least **1,500 species of vascular plants (>0.5% of world's total) as endemics**; (2) Must have **lost at least 70% of its original primary native habitat** (i.e. <=30% of original vegetation remains).
55. **Four Biodiversity Hotspots of India:** (1) **Himalaya Hotspot:** Entire Indian Himalayan range extending into Pakistan, Nepal, Bhutan, and China; (2) **Indo-Burma Hotspot:** North-Eastern India south of Brahmaputra (Assam, Meghalaya, Manipur, Mizoram, Nagaland, Tripura, and Andaman Islands); (3) **Western Ghats and Sri Lanka:** Sahyadris of Maharashtra, Goa, Karnataka, Kerala, and TN; (4) **Sundaland:** Includes the **Nicobar Islands** (extending into Indonesia, Malaysia, Singapore, and Brunei).
56. **In-situ vs Ex-situ Conservation Taxonomy:** (1) **In-situ ('On-site') Conservation:** Conserving endangered plants and animals in their natural wild habitats; includes: **National Parks, Wildlife Sanctuaries, Biosphere Reserves, Tiger Reserves, Ramsar Wetlands, Sacred Groves (*Devrais* in Maharashtra, *Orans* in Rajasthan, *Kavu* in Kerala)**; (2) **Ex-situ ('Off-site') Conservation:** Conserving species outside their natural habitats in human-controlled settings; includes: **Zoological Parks, Botanical Gardens, Seed Banks, Cryopreservation (-196°C in liquid nitrogen), Pollen Banks, Gene Banks, In-vitro fertilization clinics**.
57. **IUCN Red List Categories:** Established in 1964 by International Union for Conservation of Nature (headquarters in **Gland, Switzerland**); 9 categories: Extinct (EX), Extinct in the Wild (EW), **Critically Endangered (CR)** (reduction in population >=90% over 10 years or 3 generations, or <50 mature individuals), **Endangered (EN)** (reduction >=70%, or <250 individuals), **Vulnerable (VU)** (reduction >=50%, or <1,000 individuals), Near Threatened (NT), Least Concern (LC), Data Deficient (DD), Not Evaluated (NE).

📊 Quick Reference Grid

Concepts

ITEM	LOCK
Term / populariser	Rosen **1985** / E.O. Wilson
Alpha / Beta / Gamma	Local / between ecosystems / regional
Genetic / Species / Ecosystem	Within species / among species / habitats
Richest land / sea	Tropical rainforest / coral reefs
India land vs species	~**2.4%** land; ~**7–8%** species
4 Indian hotspots	Himalaya · Indo-Burma · WG–SL · Sundaland
Andaman / Nicobar	Indo-Burma / Sundaland
Hotspot criteria	≥1500 endemic plants + ≥70% habitat lost; **36** global

Law / lists

ITEM	LOCK
In-situ / Ex-situ	Natural habitat / outside (zoo, seed bank)
CBD / Cartagena / Nagoya	1992 Rio / biosafety LMOs / ABS **2010**
BDA / NBA	**2002** / NBA **Chennai 2003**
Threatened (IUCN)	**VU + EN + CR** (Endangered ≠ all Threatened)
Biodiversity Day	**22 May**
Kunming–Montreal	**2022**; 30×30
India Ramsar	**80** sites; TN highest; UP **10** class fact
Hotspot concept	Norman Myers (≠ BirdLife)

Chapter 05: Habitat (वास स्थान), Flora (वनस्पति) & Fauna (प्राणीजात)

🎯 Consolidated Must-Score Facts

1. A **habitat** (वास स्थान) is the physical **place** where an organism lives. A **niche** is the **role**. An **ecosystem** (पारिस्थितिकी तंत्र) is the functional unit. **Environment** (पर्यावरण) is the total surroundings.
2. Habitats are commonly classed as **terrestrial, aquatic, or transitional** (mangrove/wetland). Transitional habitats are not purely terrestrial.
3. Adaptation labels include **arboreal** (trees), **fossorial** (burrows), **cursorial** (open plains), and **xericole** (desert).
4. **Habitat loss** is the number-one biodiversity threat. **Fragmentation** splits patches; **corridors** reconnect them.
5. **MISHTI** is the Mangrove Initiative for Shoreline Habitats & Tangible Income from Budget **2023–24** under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय). It is **not** a 2019 scheme.
6. **NUHHP** is the National Urban (नगरीय) Housing and Habitat Policy of **2007** (MoHUA).
7. Habitat–scheme chronology: **JNNURM 2005 → NUHHP 2007 → AMRUT 2015 → Jal Jeevan Mission 2019**.
8. **Flora** means plant life of a region. **Fauna** means animal life. Do not swap the two.
9. India has roughly **45,000** plant species and **91,000+** animal species on about **2.4%** of world land.
10. **BSI (1890, Kolkata)** surveys plants. **ZSI (1916, Kolkata)** surveys animals. Both sit in Kolkata; kingdoms differ.
11. Engler is linked to world floral regions. **C.O. Sauer** is linked to planting hearths in cultural geography.
12. A **microhabitat** (सूक्ष्म आवास) is a small subdivision inside a larger habitat.
13. In-situ (स्थान पर) conservation (स्व-स्थाने संरक्षण) protects species in natural habitat; ex-situ (स्थानांतरित) is outside habitat — same fact as biodiversity sheets.
14. Uttar Pradesh (उत्तर प्रदेश)'s richest wildlife habitat belt is the **Terai** (तराई) (Dudhwa (दुधवा), Pilibhit, Kishanpur).
15. UP state animal is **barasingha**. UP state bird is **Sarus crane**.
16. The **Terai Arc Landscape** corridor links UP Terai protected areas toward Nepal.
17. Aquatic fauna facts for UP include the **Gangetic dolphin** in the Ganga (गंगा)–Yamuna (यमुना) system.
18. **Water hyacinth** is a common invasive threat in UP village ponds.
19. Uttar Pradesh has **no mangrove habitat** purposes — it is landlocked for that category.
20. Habitat is an address; niche is a job description. Mixing the two loses marks on definition stems.
21. Wetlands and mangroves are **transitional**, not "purely terrestrial forest" options.
22. MISHTI is mangrove-focused coastal income and protection — not a Terai tiger scheme.
23. Jal Jeevan Mission (**2019**) is the latest fact in the urban/habitat chronology set above; do not date NUHHP as 2019.
24. Corridor questions for UP should start from Terai Arc links, not from Western Ghat mangrove corridors.
25. **Terrestrial habitats** exist on land. Organisms that live on land are called **terrestrial habitants**.
26. **Aquatic habitats** exist in freshwater or marine water. Rivers are lotic and lakes are lentic aquatic habitats.
27. **Transitional habitats** such as wetlands, mangroves, and estuaries are neither purely terrestrial nor purely aquatic.
28. **Abiotic factors** in a habitat include temperature, moisture, light, soil, pH, and salinity.
29. **Biotic factors** include food availability, predators, competitors, and symbiotic partners.
30. **Habitat conservation** protects the natural dwelling place of species so ecological processes can continue.
31. **Habitat loss** completely removes an area's ability to support native species.
32. **Habitat fragmentation** breaks a large habitat into small isolated patches and blocks movement and gene flow.
33. **Habitat degradation** reduces habitat quality without removing it entirely, such as river pollution affecting dolphin habitat.
34. **In-situ conservation** protects species inside their natural habitat. National parks and wildlife sanctuaries are examples.
35. **Wildlife corridors** link fragmented habitats. The **Terai Arc Landscape** connects Dudhwa with Katarniaghat across the Nepal border.
36. **Habitat restoration** rebuilds damaged ecosystems through afforestation, wetland revival, and mangrove planting.
37. **MISHTI** stands for **Mangrove Initiative for Shoreline Habitats and Tangible Income**.
38. **NUHHP** addresses urban habitat quality through housing and green-space planning in cities.
39. **Flora** is the collective **plant** life of a particular region, habitat, or geological period.
40. **Fauna** is the collective **animal** life of a particular region, habitat, or geological period.
41. **Adolf Engler** mapped the world's major **floral regions**.
42. **C.O. Sauer** introduced the concept of **planting hearth**, the region where domesticated plants first originated.
43. **Red Sanders (*Pterocarpus santalinus*)** is a South India dry-deciduous endemic (Palkonda–Seshachalam / southern Eastern Ghats (पूर्वी घाट)) — **not** a tropical rainforest tree.
44. **Taxus** occurs naturally in the Himalaya, is Red Data–listed, and yields **taxol** (anti-cancer; older stems also cite Parkinson's).

📊 Quick Reference Grid

Tags

ITEM	LOCK
Habitat / Niche / Ecosystem	Place / role / functional unit
Habitat classes	Terrestrial · aquatic · transitional
Arboreal / Fossorial / Cursorial / Xericole	Trees / burrows / open plains / desert
#1 threat	Habitat loss (+ fragmentation; corridors reconnect)
Flora / Fauna	Plants / animals — do not swap
India species share	~45k plants; ~91k+ animals on ~2.4% land

Schemes

ITEM	LOCK
MISHTI	Mangrove initiative; Budget **2023–24** (≠ 2019)
NUHHP	Urban Housing & Habitat Policy **2007**
Chronology	JNNURM 2005 → NUHHP 2007 → AMRUT 2015 → JJM 2019

Chapter 06: Protected Areas & Conservation

🎯 Consolidated Must-Score Facts

1. The **Wildlife (Protection) Act, 1972** is India's main protected-area statute.
2. Terrestrial WPA strictness runs **National Park > Wildlife Sanctuary > Conservation/Community Reserve**.
3. A **Wildlife Sanctuary** (Section **18**) allows limited human use with **Chief Wildlife Warden** permission.
4. A **National Park** (Section **35**) is the strictest WPA protected area. Grazing and forestry are largely banned. Boundary change needs a resolution of the **State Legislature** (not Parliament).
5. A **Biosphere Reserve** (जैवमंडल आरक्षित क्षेत्र) follows the UNESCO MAB three-zone model: **core** (no exploitative use) → **buffer** (research, education, regulated tourism/grazing) → **transition** (settlements, cultivation, livelihoods). It has **no WPA statutory basis**.
6. India has **18** biosphere reserves (जैवमंडल आरक्षित क्षेत्र); about **12–13** are in the UNESCO network (count drifts with new WNBR listings). The first was **Nilgiri (नीलगिरि) (1986)**. **Nokrek** is in **Meghalaya** (मेघालय), not Manipur.
7. A **Conservation Reserve** (Section **36A**, 2002) is a government/community forest buffer near a protected area.
8. A **Community Reserve** (Section **36B**, 2002) sits on private or community land with voluntary consent.
9. A **Tiger Reserve** rests on WPA Section **38V** (2006 path) with **core + buffer** under **NTCA**. India has **58** tiger reserves. **Corbett** was first in **1973**.
10. An **Elephant Reserve** comes from **Project Elephant (प्रोजेक्ट एलीफेंट) (1992)** (प्रोजेक्ट एलिफेंट) by executive notification. It has **no WPA statutory basis**. India has about **33** elephant reserves.
11. India's first Marine National Park is **Gulf of Kutch, Gujarat (1980)**. **Gulf of Mannar (TN, 1980)** is also a Marine NP. Uttar Pradesh (उत्तर प्रदेश) has **no marine PA**.
12. **Ramsar (1971)** marks wetlands of international importance. India joined in **1982**. Uttar Pradesh has **12** Ramsar sites.
13. **Rudrasagar** is in **Tripura**. **Sultanpur** is in **Haryana**. Neither is an Uttar Pradesh Ramsar site.
14. An **Eco-Sensitive Zone (ESZ)** (पर्यावरण-संवेदनशील क्षेत्र) is an **EPA 1986** (पर्यावरण संरक्षण 1986) buffer around protected areas that restricts mining and polluting industry. ESZ is **not** a WPA category.
15. Protected-area stock (ENVIS-style ~2023): about **106** National Parks, **573** Wildlife Sanctuaries, **123** Conservation Reserves, **220** Community Reserves. **Madhya Pradesh** leads NP count; **Andaman (अंडमान) & Nicobar (निकोबार)** leads sanctuary count.
16. In a National Park, local people are **not** allowed to collect/use biomass the way limited use is permitted in a sanctuary. BR is for landscape conservation of **all forms of life**, not "a few species only."
17. WPA basis yes: National Park, Sanctuary, Tiger Reserve, Conservation Reserve, Community Reserve. WPA basis no: Biosphere Reserve, Elephant Reserve, Ramsar, ESZ.
18. Uttar Pradesh's only classic National Park fact is **Dudhwa** (दुधवा) (Lakhimpur Kheri; rhino rehabilitation from **1984**, later phases including ~**2018**).
19. UP tiger reserves are **Dudhwa, Pilibhit, and Ranipur**.
20. Important UP sanctuaries include **Katarniaghat (1975), Kishanpur, Chandra Prabha (Chandauli), Nawabganj, and Sur Sarovar**.
21. UP elephant reserves include **UP ER (2009)** and **Terai (तराई) ER (2022)**.
22. UP Ramsar sites include Upper Ganga (गंगा) (Brijghat–Narora), Nawabganj, Parvati Arga, Saman, Samaspur, Sandi, Sarsai Nawar, Sur Sarovar, Haiderpur, Bakhira, Patna (पटना), and Shekha Jheel.
23. River–park traps: **Corbett = Ramganga (रामगंगा) + Kosi (कोसी)** (not Ganga); **Kaziranga (काजीरंगा) = Brahmaputra (ब्रह्मपुत्र) / Diphlu system** (not Manas); **Silent Valley (साइलेंट वैली) = Kunthipuzha** (not Kaveri (कावेरी)).
24. The **Terai Arc** corridor links Dudhwa–Kishanpur–Katarniaghat toward Nepal.
25. Tiger Reserves are statutory under WPA; Elephant Reserves are not — that swap is a high-yield trap.
26. Biosphere Reserve zoning is landscape management under MAB; it is not the same as a National Park notification under Section 35.
27. Community Reserve is private/community land; Conservation Reserve is mainly government forest buffer — do not reverse Sections **36A/36B**.
28. ESZ regulates harmful activity around a PA; it does not itself create a new WPA Schedule animal list.
29. Sanctuary permits regulated use; National Park does not allow the same degree of use — strictness order must stay NP > WLS.
30. Match "firsts" carefully: first BR = **Nilgiri 1986**; first TR = **Corbett 1973**; first Marine NP (प्रथम समुद्री उद्यान) = **Gulf of Kutch 1980**; world first NP = **Yellowstone 1872**.
31. **International Tiger Day = 29 July** (St Petersburg 2010). **M-STrIPES** monitors tiger reserves (patrol + ecology (पारिस्थितिकी) GIS tool).
32. **Sathyamangalam TR (TN)** is a classic ecological link between Eastern and Western Ghats (पश्चिमी घाट).
33. **UP teaching row:** Dudhwa NP / TR, Pilibhit TR, Ranipur TR, **12 Ramsar**, Terai corridors, **no marine PA**, **no biosphere reserve** inside present UP.
34. **Three UNESCO instruments:** World Heritage · Biosphere (MAB / WNBR) · Intangible Cultural Heritage — do not force one site into all three without checking the list.
35. **Nokrek** is in **Meghalaya** (Garo (गारो) Hills; Nokrek peak), not Manipur. **Simlipal** is in **Odisha** (ओडिशा). **Agasthyamalai** spans **Kerala and Tamil Nadu (तमिलनाडु)** (Neyyar, Peppara, Shendurney WLS + Kalakad–Mundanthurai TR).
36. **Asan (आसन) Conservation Reserve** in Uttarakhand is a known example and is also a Ramsar site.
37. **Marine Protected Areas (MPAs)** include **Marine National Parks** and **Marine Wildlife Sanctuaries** notified under WPA.
38. **Gahirmatha Marine Sanctuary** in Odisha is famous for olive ridley turtle nesting.
39. **Uttar Pradesh has no marine protected area** because it is landlocked.
40. **World Wetlands Day** (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).

41. **Uttar Pradesh has 12 Ramsar sites**.
42. **Rudrasagar Lake** is in **Tripura**, not UP.
43. **Sultanpur National Park** is in **Haryana**, not UP.
44. **UNESCO World Heritage Sites** are places of outstanding universal value under the **World Heritage Convention (1972)**.
45. **Natural** World Heritage Sites are recognised for exceptional natural beauty or biodiversity.
46. **Keoladeo** is both a Ramsar wetland and a UNESCO natural World Heritage Site (प्राकृतिक विश्व धरोहर स्थल).
47. **Uttar Pradesh has no natural UNESCO World Heritage Site**. Nanda Devi is in **Uttarakhand** (उत्तराखंड).

📊 Quick Reference Grid

Strictness / categories

ITEM	LOCK
WPA year	**1972**
Strictness (terrestrial)	NP > Sanctuary > Conservation/Community Reserve
NP / Sanctuary	Sec. **35** strictest / Sec. **18** limited use
Biosphere Reserve	UNESCO MAB; core → buffer → transition (**not** WPA)
India BRs	**18** national; first **Nilgiri 1986**
Conservation / Community Res.	Sec. 36A / 36B (**2002**)
Tiger Reserve	Sec. **38V**; NTCA; **58** TRs; Corbett first **1973**
Elephant Reserve	Project Elephant **1992**; ~**33**; **no** WPA statute

Marine / wetland / ESZ

ITEM	LOCK
First Marine NP	Gulf of Kutch, Gujarat **1980**
Ramsar	**1971**; India joined **1982**
Rudrasagar / Sultanpur	Tripura / Haryana (**not** UP)
ESZ	EPA **1986** buffer (≠ WPA category)

Table: Complete Master Matrix of 18 Biosphere Reserves in India

S.NO.	BIOSPHERE RESERVE	STATE / UT	YEAR NOTIFIED	UNESCO WNBR STATUS	KEY FAUNA / UNIQUE ECOLOGICAL FEATURE
1	**Nilgiri**	Tamil Nadu, Kerala, Karnataka	1986	**Yes (2000 — 1st in India)**	Lion-tailed Macaque, Nilgiri Tahr; tri-state junction
2	**Nanda Devi**	Uttarakhand	1988	**Yes (2004)**	Snow Leopard, Himalayan Musk Deer, Valley of Flowers
3	**Nokrek**	Meghalaya	1988	**Yes (2009)**	Red Panda, Asian Elephant, Citrus indica (wild mother orange); Garo Hills
4	**Great Nicobar**	Andaman & Nicobar Islands	1989	**Yes (2013)**	Saltwater Crocodile, Megapode, Leatherback Turtle
5	**Gulf of Mannar**	Tamil Nadu	1989	**Yes (2001)**	Dugong (Sea Cow), Coral Reefs, Seagrass beds
6	**Manas**	Assam	1989	**No** (Removed 2011)	Golden Langur, Pygmy Hog, One-horned Rhino; Bodo territory
7	**Sundarbans**	West Bengal	1989	**Yes (2001)**	Royal Bengal Tiger, Mangrove swamps, Estuarine crocodile
8	**Simlipal**	Odisha	1994	**Yes (2009)**	Tiger, Elephant, Gaur; Mayurbhanj elephant reserve
9	**Dihang-Dibang**	Arunachal Pradesh	1998	**No**	Mishmi Takin, Musk Deer; Eastern Himalayas
10	**Pachmarhi**	Madhya Pradesh	1999	**Yes (2009)**	Giant Squirrel, Flying Squirrel; Satpura Range
11	**Achanakmar-Amarkantak**	Madhya Pradesh, Chhattisgarh	2005	**Yes (2012)**	Four-horned Antelope, Wild Dog; Maikal Hills origin of Narmada/Son
12	**Great Rann of Kutch**	Gujarat	2008	**No** (Largest BR in India)	Indian Wild Ass (Khur), Greater Flamingo; seasonal salt marsh
13	**Cold Desert**	Himachal Pradesh	2009	**No**	Snow Leopard, Tibetan Gazelle; includes Pin Valley NP & Kibber
14	**Khangchendzonga**	Sikkim	2010	**Yes (2018)**	Snow Leopard, Red Panda, Musk Deer; Mixed World Heritage Site
15	**Agasthyamalai**	Kerala, Tamil Nadu	2001	**Yes (2016)**	Nilgiri Tahr, Asian Elephant, Kanikaran indigenous tribe
16	**Panna**	Madhya Pradesh	2011	**Yes (2020 — Newest in WNBR)**	Bengal Tiger, Chital, Chinkara, Sambhar; Ken river valley
17	**Seshachalam Hills**	Andhra Pradesh	2010	**No**	Red Sanders (Pterocarpus santalinus), Slender Loris
18	**Dibru-Saikhowa**	Assam	1997	**No** (Smallest BR in India)	Feral Horses, White-winged Wood Duck; Brahmaputra-Lohit floodplains

Table: Key Ramsar Wetlands of India (Selected High-Yield & State-Wise)

RAMSAR SITE	STATE / UT	AREA / SPECIAL SIGNIFICANCE	EXAM TRAP & DISTINCTION
Chilika Lake	Odisha	First Ramsar Site in India (1981); Largest coastal lagoon	Joined with Keoladeo Ghana in 1981; removed from Montreux Record post-restoration
Keoladeo Ghana National Park	Rajasthan	Wetland & NP; placed on Montreux Record	On Montreux Record due to water shortage & grazing pressure
Loktak Lake	Manipur	Freshwater lake with phumdis (floating mats); Keibul Lamjao NP	Only floating national park; home to Sangai (Dancing Deer); on Montreux Record
Vembanad-Kol Wetland	Kerala	Largest wetland in India by surface area	Spans Alappuzha, Kottayam, Ernakulam; Nehru Trophy Boat Race held here
Sundarban Wetland	West Bengal	Largest Ramsar site by protected area	Mangrove ecosystem; Royal Bengal Tiger; shared with Bangladesh
Renuka Wetland	Himachal Pradesh	Smallest Ramsar site in India (~20 ha)	Natural wetland with freshwater springs; sacred lake
Upper Ganga River	Uttar Pradesh	First Ramsar site of UP (2005; Brijghat to Narora)	Stretch of 82 km; habitat of Gangetic Dolphin, gharial, turtles
Bakhira Wildlife Sanctuary	Uttar Pradesh	10th Ramsar site of UP (Sant Kabir Nagar, 2022)	Natural freshwater marsh; migratory waterbirds
Haiderpur Wetland	Uttar Pradesh	Muzaffarnagar & Bijnor (2021)	Human-made wetland in Hastinapur WLS within Ganga floodplain
Rudrasagar Lake	Tripura	Lowland sedimentation reservoir	Located in Northeast (Tripura) — common trap confusing it with UP or MP
Bhoj Wetland	Madhya Pradesh	Twin lakes: Upper Lake (Bada Talab) and Lower Lake	Created by Raja Bhoj; human-made wetland
Asan Barrage	Uttarakhand	First Ramsar site of Uttarakhand (2020)	Located at confluence of Asan and Yamuna in Dehradun
Lonar Lake	Maharashtra	Meteor impact crater lake; hyper-saline	Turned pink due to haloarchaea microbes in 2020
Tso Kar Wetland Complex	Ladakh	High-altitude saline lake (Startsapuk Tso + Tso Kar)	Breeding ground for Black-necked Crane

Chapter 07: Wildlife Conservation (वन्यजीव संरक्षण)

🎯 Consolidated Must-Score Facts

1. Wildlife conservation protects species and habitats, with **in-situ** (स्थान पर) primary and **ex-situ** (स्थानांतरित) as backup.
2. Major threats include habitat (वास स्थान) loss, poaching, human–wildlife conflict, invasive species (आक्रामक विदेशी प्रजाति), and veterinary **diclofenac** (डाइक्लोफेनाक) for vultures.
3. The **Wildlife (Protection) Act, 1972** followed the Stockholm moment and uses **six Schedules**. **Schedule I** is maximum protection.
4. Schedule I facts include tiger, elephant, rhino, gharial, and vultures among frequently asked species. A **Schedule I tortoise** enjoys the **same protection level as the tiger**.
5. **Schedule VI** plants cannot be cultivated without a **licence** from the Chief Wildlife Warden (or authorised officer) — not "banned under all circumstances."
6. The **Chief Wildlife Warden** is the State implementing officer. **NBWL** is the PM-chaired national advisory body.
7. **NTCA** became statutory through the **2005/2006** WPA amendments and governs **Tiger Reserves only**.
8. **Project Tiger** (प्रोजेक्ट टाइगर) began in **1973**. Corbett was the first tiger reserve. India has **58** tiger reserves. The **2022** census figure often cited is **3,682** tigers.
9. **Project Elephant** (प्रोजेक्ट एलीफेंट) began in **1992** with about **33** elephant reserves and **no WPA statutory basis** — it covers **wild and captive** elephants.
10. The **Crocodile Project (1975)** covers gharial, mugger, and saltwater crocodile. Best natural gharial view: **Chambal** (चंबल) (also Girwa); IUCN (आईयूसीएन) status **Critically Endangered**.
11. **Project Lion (2020)** focuses on Asiatic lions at **Gir (गीर), Gujarat**. Uttar Pradesh (उत्तर प्रदेश) has **no** wild Asiatic lion population (जनसंख्या).
12. **IRV 2020** planned rhino recovery across seven PAs, including **Dudhwa** (दुधवा) in Uttar Pradesh.
13. **Project Snow Leopard (2009)** and **GSLEP (2013)** cover high Himalaya (हिमालय). UP has **no** snow leopard habitat.
14. The **Vulture Action Plan (2006)** and veterinary **diclofenac ban (2006)** address renal failure in Indian vultures. **SAVE** (Nepal–India) targets **vulture** recovery.
15. An **umbrella** species (tiger, elephant) protects wide ecosystems. A **flagship** species mobilises public support. A **keystone** species' loss collapses function (vulture example).
16. World Wildlife Day (विश्व वन्यजीव दिवस) is **3 March**. Wildlife Week in India is **2–8 October**. WWF (डब्ल्यूडब्ल्यूएफ) logo animal is the **Giant Panda**.
17. **Operation Olivia** is an Indian Coast Guard (तटरक्षक बल) nesting patrol on the **Odisha** (ओडिशा) coast.
18. UP tiger reserves are **Dudhwa, Pilibhit, and Ranipur**.
19. UP rhino fact is **Dudhwa National Park** (दुधवा राष्ट्रीय उद्यान) (reintroduced population).
20. UP gharial sites include **Katarniaghat Wildlife Sanctuary** and the **Chambal** sanctuary landscape.
21. UP elephant reserves are **UP ER (2009)** and **Terai (तराई) ER (2022)**.
22. Programme-year chain: WPA **1972** → Project Tiger **1973** → Crocodile Project **1975** → Project Elephant **1992** → Biodiversity Act **2002** → NTCA statutory **2005/2006** → Vulture ban **2006** → Project Snow Leopard **2009** → Project Lion **2020**.
23. **Whale shark** is India's largest fish; seasonal aggregations off **Gujarat** (Mar–May); protection programme from **2004**.
24. **Kharai camel** (Kutch, Gujarat) can swim in seawater and graze mangroves; can be domesticated — **not** "wild only."
25. **Pugmark** technique estimates (प्राकलन समिति) wild animal populations from footprints.
26. **Spider monkey** is Central/South American — **not** naturally found in India (Star tortoise, Monitor lizard, Pygmy hog are).
27. Scientific names: Asiatic wild ass (**Equus hemionus**), Barasingha (**Rucervus/Cervus duvaucelii**), Chinkara (**Gazella bennettii**), Nilgai (**Boselaphus tragocamelus**).
28. **Community Reserve** (सामुदायिक आरक्षित क्षेत्र) (WPA): CWLW authority applies; hunting banned; NTFP collection allowed; **traditional agriculture / jhum not** allowed after notification.
29. **Animal Welfare Board of India** (1962) is under the **Prevention of Cruelty to Animals Act, 1960** — **not** EPA 1986 (पर्यावरण संरक्षण 1986). NTCA is statutory; NGRBA was PM-chaired.
30. Gharial, Indian wild ass, and wild buffalo cannot be hunted except under limited WPA provisions.
31. India holds about **2.4%** of world land area but about **8%** of known biodiversity (~**7.28%** of known animal species in Alfred-type figures).
32. First International Tiger Forum / World Tiger Summit **2010** was at **Saint Petersburg**, Russia.
33. **New York Declaration on Forests (2014)** is voluntary/non-binding; India was **not** an original signatory.
34. **UNCCD (1994)** (मरुस्थलीकरण रोकथाम संधि) is the sole legally binding desertification (मरुस्थलीकरण) convention; bottom-up local participation; no exclusive South Asia finance monopoly claim.
35. Conservation is ecological protection; **Wildlife Protection** language often means the **WPA enforcement** track.
36. NTCA does not run Elephant Reserves; confusing NTCA with Project Elephant administration is a trap.
37. **Wildlife conservation** means protecting wild plants and animals and the habitats they need to survive.
38. **Ex-situ conservation** (बाह्य-स्थाने संरक्षण) uses zoos, captive breeding, and gene banks as backup when wild populations are critically low.
39. **World Wildlife Day** is observed on **3 March**, marking the adoption of CITES (साइट्स) in 1973.
40. **Wildlife Week** in India is observed from **2 to 8 October**.
41. **Operation Olivia** is an **Indian Coast Guard** initiative to protect **Olive Ridley turtles** along the **Odisha coast**.
42. **Wildlife (Protection) Act, 1972** is India's main law for protecting wild animals, birds, and plants.
43. **Schedule I** gives maximum protection. Hunting and trade (पण्याध्यक्ष) are banned for tiger, elephant, rhino, lion, gharial, snow leopard, and Indian vultures.
44. **Schedule V** lists vermin such as the common crow and rat, which may be hunted.

45. **Schedule VI** lists specified plants whose cultivation is **prohibited except with a licence** from the Chief Wildlife Warden or authorised officer (classic pitcher-plant / specified-plant pattern).
46. **Article 48A** directs the State to protect wildlife. **Article 51A(g)** makes it a citizen's duty to protect wildlife.
47. **Animal Welfare Board of India** was established in **1962** under the **Prevention of Cruelty to Animals Act, 1960** — not under EPA 1986.
48. **Project Tiger** was launched in **April 1973** by PM Indira Gandhi (गांधी) at Jim Corbett (जिम कॉर्बेट) National Park.
49. **International Tiger Day** is **29 July** (decided at the **2010 St Petersburg** Tiger Summit).
50. **Nagarjunsagar–Srisailam** holds the largest critical tiger habitat among common options. **Bandipur** (बांदीपुर) pioneered drone monitoring among common stems.

Chronological Sequence of Wildlife Conservation Projects in India

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1972 ⟶ Wildlife (Protection) Act enacted (6 Schedules; rationalized to 4 via 2022 Amendment)
1973 ⟶ Project Tiger launched at Corbett National Park (9 initial reserves; now 55 Tiger Reserves)
1975 ⟶ Crocodile Conservation Project launched with UNDP/FAO support (Mugger, Gharial, Saltwater)
1986 ⟶ Nilgiri Biosphere Reserve designated (India's 1st Biosphere Reserve under UNESCO MAB)
1992 ⟶ Project Elephant launched (Centrally Sponsored Scheme for wild elephant habitat & corridors)
1996 ⟶ Vulture Conservation Action Plan initiated (banning veterinary diclofenac post-crash)
2007 ⟶ Project Snow Leopard launched (covering Ladakh, HP, Uttarakhand, Sikkim, Arunachal)
2020 ⟶ Project Dolphin & Project Lion announced on Independence Day
2022 ⟶ Project Cheetah launched (intercontinental relocation of African cheetahs to Kuno NP, MP)
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📊 Quick Reference Grid

Law / schedules

ITEM	LOCK
WPA Schedules	Six; **Schedule I** = maximum protection
Schedule VI	Plants; licence from CWLW (not absolute (निरपेक्ष) ban)
NBWL / CWLW	PM-chaired advisory / State implementing officer
NTCA	Statutory via 2005–06 path; Tiger Reserves only

Projects

ITEM	LOCK
Project Tiger	**1973**; Corbett first; **58** TRs; ~3682 (2022)
Project Elephant	**1992**; ~33 reserves; wild + captive
Crocodile Project	**1975**; gharial–mugger–saltwater; Chambal
Project Lion	**2020**; Gir only (no wild UP lions)
IRV 2020	Rhino recovery; includes **Dudhwa**
Project Snow Leopard / GSLEP	**2009** / **2013** (no UP habitat)
Vulture plan / diclofenac ban	**2006** / **2006**

Table: Key Tiger Reserves of India & Uttar Pradesh

TIGER RESERVE	STATE	SPECIAL DISTINCTION / STATUS	CORE EXAM HIGH-YIELD NOTES
Jim Corbett	Uttarakhand	First National Park (1936) & First Tiger Reserve (1973)	Highest tiger density in India; Ramganga river traverses
Dudhwa	Uttar Pradesh	Core of Dudhwa TR (Lakhimpur Kheri; 1987)	Only National Park in UP; Swamp Deer (Barasingha) stronghold
Pilibhit	Uttar Pradesh	45th TR of India (2014)	Won international TX2 award in 2020 for doubling tiger population
Amangarh	Uttar Pradesh	Bijnor district (2012)	Buffer zone of Jim Corbett NP left in UP post-bifurcation
Ranipur	Uttar Pradesh	Chitrakoot district (2022; 53rd TR of India)	4th TR of UP; First Tiger Reserve in Bundelkhand region
Bandipur	Karnataka	Western Ghats (1973)	Part of Nilgiri Biosphere Reserve; high tiger and elephant density
Nagarjunsagar Srisailam	Andhra Pradesh	Largest Tiger Reserve in India by total area	Srisailam reservoir, Krishna river; Chenchu tribe
Bor	Maharashtra	Smallest Tiger Reserve in India by area (~138 sq km)	Located in Wardha district
Ranthambore	Rajasthan	Sawai Madhopur	Dry deciduous tiger habitat; famous junction of Aravalli & Vindhya
Kanha	Madhya Pradesh	Satpura Maikal range	Hardground Barasingha mascot 'Bhoorsingh'; inspired Kipling
Bandhavgarh	Madhya Pradesh	Umaria district	Highest density of Bengal tigers in Central India
Guru Ghasidas & Tamor Pingla	Chhattisgarh	54th Tiger Reserve of India (notified 2024)	Inter-state connectivity with Bandhavgarh and Sanjay TRs
Dholpur-Karauli	Rajasthan	55th Tiger Reserve of India (2023)	5th TR of Rajasthan after Ranthambore, Sariska, Mukundra, Ramgarh

Chapter 08: Forests & Forest Management

🎯 Consolidated Must-Score Facts

1. Forest cover (वन आवरण) in ISFR means a patch of at least **1 ha** with canopy **≥10%**, on any land ownership.
2. Canopy classes: **VDF ≥70%**, **MDF 40–70%**, **OF 10–40%**.
3. ISFR **2023** India figures: forest cover about **21.76%**, tree cover (वृक्ष आवरण) about **3.41%**, total green cover about **25.17%**. ISFR **2021** held forest cover at about **21.71%** and forest+tree cover about **24.62%** (still used in older stems).
4. **Madhya Pradesh** has the largest forest **area**. **Mizoram** has the highest **percentage** cover among States. **Arunachal Pradesh** often leads **very dense forest** area.
5. **Forest cover ≠ Recorded Forest Area ≠ tree cover**. Recorded forest area is a legal-notification figure near about **23%**.
6. Under the **Indian Forest Act, 1927**, **Reserved Forest** is strictest (acts banned unless permitted). **Protected Forest** is State-regulated with some customary rights possible.
7. **Champion and Seth (1968)** classify Indian forests into **16 types in 6 groups**. The largest share is **Tropical Dry Deciduous** (about **38%**).
8. **Tropical wet evergreen** (>**200 cm** rain) holds maximum plant diversity / biomass among common options — Western Ghats (पश्चिमी घाट), NE hills, A&N (not Himachal (हिमाचल) as wet evergreen).
9. **Teak** dominates **tropical moist deciduous** forests; **Madhya Pradesh** has the maximum teak area among common options.
10. **Red Sanders (*Pterocarpus santalinus*)** is a dry-deciduous South India endemic (Palkonda–Seshachalam / southern Eastern Ghats (पूर्वी घाट)) — **not** a tropical rainforest tree.
11. **Taxus** (Himalaya (हिमालय); Red Data) yields **taxol** (anti-cancer; also cited vs Parkinson's in older stems).
12. **Amazon rainforest** is often called the **"lungs of the Earth"** (~**20%** of terrestrial photosynthetic oxygen share in coaching figures).
13. UP type facts: Terai (तराई) = moist deciduous; Vindhyan = dry deciduous; Bundelkhand = thorn.
14. Law chronology: **IFA 1927 → FCA 1980 → NFP 1988 → JFM 1990 → FRA 2006 → CAMPA Act 2016**.
15. The **Forest (Conservation) Act, 1980** requires **Central** approval before diverting forest land. Compensatory afforestation is mandatory.
16. **National Forest Policy 1988** targets about **33%** national forest/tree cover and **66%** in hills. **NFP 1952** classes: Protected/Conserved, National, Village Forests, Tree Lands — **not** "National Park" as a forest class.
17. NFP 1988 includes afforestation/wasteland, reforestation, and wood substitutes — **not** wide promotion of insecticides/pesticides.
18. **Chipko (1973, Reni, Uttarakhand (उत्तराखंड))** — Bahuguna, Bhatt, **Gaura Devi**. The **Godavarman (1996)** line broadened "forest."
19. Afforestation tools include **NAP (2000)**, **Green India Mission (2014)**, **Miyawaki** mini-forests, and **CAMPA**.
20. **Social forestry** grows trees on **non-forest** land. **Agroforestry** integrates trees with crops/livestock (**CAFRI Jhansi (झांसी)**).
21. **JFM (1990)** is State–village co-management of **degraded forest**. **FRA 2006** recognises dweller rights (**IFR ~4 ha**).
22. **FCA** controls **diversion**. **FRA** recognises **rights**. **JFM** is executive co-management — three different tools.
23. UP ISFR **2023** forest cover is about **6.24%** of State area.
24. **Sonbhadra** (सोनभद्र) has UP's highest district forest cover. Districts above **20%** include **Chandauli** and **Shravasti**.
25. **Urbanisation is a cause of deforestation**, not an "impact/effect" of deforestation (classic trap).
26. Forests are a **renewable** resource; they also enhance environment (पर्यावरण) quality — but the second fact does **not** alone explain renewability.
27. **Aluminium** is called a **"green metal"** for recyclability/eco profile — not primarily "to save forests by replacing wood."
28. **Living root bridges** are characteristic of **Meghalaya** (मेघालय).
29. ISFR 2021 top absolute (निरपेक्ष) gains: **Andhra Pradesh, Telangana, Odisha**. NE states showed large losses.
30. Mangrove area rank (ISFR 2021 style): **WB > Gujarat > A&N (~616–617 km²) > AP**. **Coringa** = Godavari (गोदावरी) delta (डेल्टा) (AP).
31. **Andaman (अंडमान) & Nicobar (निकोबार)** uniquely combine mangrove + evergreen + deciduous in one option set.
32. International Day of Forests = **21 March** (UN **2012**). Wildlife Week = **2–8 October**. WWF (डब्ल्यूडब्ल्यूएफ) logo = **Giant Panda**.
33. Bhutan constitutionally aims for **≥60%** forest cover (actual cover much higher).
34. Bamboo on **non-forest** land was omitted from "tree" under the **Indian Forest (Amendment) Act, 2017**.
35. **FRI Dehradun (देहरादून) (1906)**; **ICFRE HQ Dehradun**; **IIFM Bhopal**; Survey of India ≠ environment agency.
36. CAMPA manages compensatory afforestation funds after diversion — it is not FRA Gram Sabha (ग्राम सभा) rights recognition (मान्यता).
37. **Global forest types** include tropical rainforest, tropical deciduous, temperate, boreal, Mediterranean scrub, and mangrove.
38. **Reserved Forest** has the strictest protection. Most acts are prohibited unless permitted.
39. **Protected Forest** is state-regulated forest where some grazing, collection, or customary use may continue under rules.
40. **Unclassed Forest** is forest land not yet formally placed in the Reserved or Protected category.
41. **ISFR forest cover** means any patch of **1 hectare or more** with **10% or more** tree canopy, regardless of ownership.
42. **Very Dense Forest** has canopy density **70% or more**. **Moderately Dense Forest** has **40–70%**. **Open Forest** has **10–40%**.
43. **Forest cover** is not the same as **Recorded Forest Area**. Much forest cover exists outside legally notified forest land.
44. **Tree cover** counts scattered trees in orchards and farms that fall below the 1-hectare forest patch rule.
45. **Mangrove and littoral swamp forests** are distinct coastal types with salt-tolerant species such as sundari in the Sundarbans (सुंदरबन).
46. **Champion and Seth (1968)** is India's standard forest type classification with **16 major types** in **6 groups**.

47. **Tropical Wet Evergreen** forest occurs where rainfall exceeds about **200 cm**, mainly on the Western Ghats, Northeast hills (e.g. Arunachal, Mizoram/Khasi (खासी)–Jaintia (जयंतिया) belt), and Andaman & Nicobar — **not** Himachal (pine/temperate belts).
48. **Tropical Semi-Evergreen** forest is a transition type with rainfall around **1500–2500 mm**.
49. **Tropical Moist Deciduous** forest (about **100–200 cm** rain) drops leaves in the dry season; **teak** is the dominant commercial species. **Madhya Pradesh** has the maximum teak area among common state options. Other species: bamboo, sal, shisham, sandalwood, khair, arjun.
50. **Taxus** grows naturally in the Himalaya, is Red Data–listed, and yields **taxol** (anti-cancer; older stems also cite Parkinson's).

📊 Quick Reference Grid

Cover definitions

ITEM	LOCK
Forest cover (ISFR)	≥1 ha; canopy ≥**10%**; any ownership
VDF / MDF / OF	≥70% / 40–70% / 10–40%
ISFR 2023 class	Forest ~**21.76%**; tree ~3.41%; total green ~25.17%
Largest area / highest %	MP / Mizoram
Cover ≠ RFA ≠ tree cover	Legal RFA ~23% class figure

Types / law

ITEM	LOCK
Champion & Seth	**1968**; 16 types / 6 groups; Tropical Dry Deciduous largest
Wet evergreen rain	>**200 cm**; max diversity/biomass
Teak / Red Sanders	Moist deciduous / dry-deciduous S. India endemic
UP belts	Terai moist deciduous · Vindhyan dry · Bundelkhand thorn
Law chain	IFA 1927 → FCA 1980 → NFP 1988 → JFM 1990 → FRA 2006 → CAMPA 2016
Reserved vs Protected Forest	Strictest IFA / State-regulated lighter


India Forest Canopy Architecture & ISFR Forest Types

Chapter 09: Pollution & Waste Management

🎯 Consolidated Must-Score Facts

1. **Pollution** is harmful addition beyond the environment (पर्यावरण)'s absorption capacity. **Degradation** is the broader fall in environmental quality and productivity.
2. Human-caused pollution / human-modified environment = **anthropogenic**. Anthropogenic pollution often raises persistent **non-biodegradable** loads.
3. **Biodegradable** pollutants (sewage, faecal matter, urine, domestic waste, rubber in some stems) break down by microbes. **Non-biodegradable** include pesticides, plastics, mercury, asbestos.
4. **Point sources** are chimneys and drains. **Non-point sources** include farm runoff and roads.
5. **Primary pollutants** are emitted directly (SO_2, PM, CO, NOx, lead). **Secondary pollutants** form in air (O_3, PAN, photochemical smog, acid rain (अम्ल वर्षा)).
6. Law years: **Water Act 1974 → Air Act 1981 → EPA 1986 (पर्यावरण संरक्षण 1986)**. Noise Rules date to **2000**. Air Act amended **1987** to include noise.
7. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) sets ambient air and surface-water standards. **SPCB** enforces in States. **CGWB** handles **groundwater** assessment — not CPCB.
8. India's AQI uses **eight** pollutants: **PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb** — **not CO_2 or CH_4** (those are climate gases). Bands: **0–50 Good** … **401–500 Severe**.
9. London smog is **SO_2 + smoke**. Los Angeles / photochemical smog is **NOx + VOCs + sunlight → O_3 / PAN**. **PAN** is a powerful eye irritant.
10. **CO** binds haemoglobin **~200–300×** more strongly than O_2 (carboxyhaemoglobin). Lead antiknock in petrol harms IQ/brain development.
11. **Fly ash** from coal TPPs: usable in bricks / Portland cement replacement; contains SiO_2, Al_2O_3, CaO **plus toxic metals** — not "oxides only."
12. **Lichens** are SO_2 bio-indicators and die in polluted air. **Radon** is the classic major **indoor** air pollutant (lung cancer risk).
13. **NCAP (2019)** covers non-attainment cities with a **20–30%** PM reduction target. **BS-VI** fuel arrived in **2020**. **WAYU** air-purifying units were installed in **Delhi**.
14. **CCAC** (2012, with UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम)) focuses on **methane, black carbon, HFCs** — not a G20-only club.
15. Water facts: high **BOD** means worse organic pollution; **COD ≥ BOD**; clean DO is often **>6 mg/L**, while **<5 mg/L** / **<4 mg/L** stresses aquatic life.
16. **Eutrophication** (सुपोषण) is nutrient overload (N/P) → algal bloom → dissolved-oxygen crash — a **water** phenomenon, not air pollution.
17. Disease locks: **Minamata–Hg**, **Itai-itai–Cd**, **Knock-knee / fluorosis–F**, **Black Foot–As**, blue-baby–nitrate.
18. **Damodar** (दामोदर) is classically called a **biological desert** from mining/industry pollution. Ganga (गंगा) BOD often highest **Kanpur (कानपुर)–Allahabad (इलाहाबाद)** stretch.
19. Arsenic groundwater hotspots: Ganga–Brahmaputra (ब्रह्मपुत्र) / Bangladesh Padma (पद्मा)–Meghna; Indian states include WB, Bihar, UP, Jharkhand, Assam, Manipur, Chhattisgarh.
20. About **80%** of marine pollution comes from **land**. Ocean acidification (महासागरीय अम्लीकरण) from CO_2 harms calcareous plankton and corals. **IMO** HQ = **London**.
21. Noise is measured in **decibels**. Under **Noise Pollution (Regulation and Control) Rules, 2000**, a person may complain when noise exceeds the ambient standard by **10 dB(A) or more**. A **10 dB** rise also means **10×** intensity on the logarithmic scale. **Green muffler** = roadside tree belts for noise (and dust).
22. Plastic: **PWM Rules 2016**, SUP ban push **2022**, microplastics **<5 mm**; **microbeads** harm marine ecosystems. Polythene = ethylene polymer, non-biodegradable.
23. **Bioremediation** uses microbes to clean pollution; **cannot** readily/completely treat **Cd/Pb** heavy metals. **Oilzapper** (TERI) is for oily sludge/spills. Genetic engineering can create specialised strains (e.g. *Pseudomonas putida*).
24. Waste rules cluster in **2016**: SWM, BMW, Hazardous Waste, and Plastic Waste. **E-waste Rules** updated in **2022**. **EPR** first featured strongly in plastic/e-waste rules **2011+**.
25. **3R** priority is **Reduce > Reuse > Recycle**. **Swachh Bharat** began in **2014**. NAQI launched **17 Oct 2014**.
26. BMW colour codes: **Yellow** incineration; **Red** contaminated recyclables (IV tubes/sets, catheters, urine bags, syringes **without** needles) for disinfection then recycle; **White** sharps (needles); **Blue** glass vials / metal.
27. **Euro / BS** emission norms limit CO, HC, NOx, PM. Euro-II ultra-low sulphur diesel ≈ **≤0.05% S**. India skipped BS-V to **BS-VI (1 Apr 2020)**.
28. Catalytic converters use **Pt / Pd / Rh** to convert CO → CO_2, NOx → N_2, HC → CO_2 + H_2O.
29. UP air NCAP cities include Lucknow (लखनऊ), Kanpur, Ghaziabad, Agra, Varanasi (वाराणसी), Meerut (मेरठ), Noida, Prayagraj (प्रयागराज), Moradabad, and Bareilly.
30. UP issue facts: Kanpur Ganga tanneries; Ghaziabad/Noida air; **Moradabad** noise; **Taj Trapezium** at Agra; Varanasi ghat/plastic pressure.
31. FSI publishes forest cover (वन आवरण), not pollution standards. CWC (केंद्रीय जल आयोग) is surface-water resources, not the groundwater quality report.
32. EPA **1986** is the umbrella for many later rules; it does not replace Water and Air Acts.
33. Secondary ozone at ground level is a pollutant; do not confuse it with stratospheric protective ozone.
34. Hazardous waste follows HW Rules **2016** plus Basel controls — not SWM municipal rules alone.
35. **Asian Brown Cloud** / atmospheric brown cloud is especially noted over **South Asia** (winter aerosols, black carbon, sulphate).
36. Bag filters trap finer particulates (**<~50 µm**); cyclone separators/collectors handle coarser (**>~50 µm**). **Cyclone divider** is not a standard control device name.
37. **Radioactive pollution** is ionizing radiation beyond safe dose. **Becquerel (Bq)** counts disintegrations per second; **Sievert (Sv)** is biological dose; **half-life** is time for **50%** decay — not complete disappearance.
38. **AERB** (Atomic Energy Regulatory Board) regulates nuclear safety in India. **Chernobyl (26 April 1986)** is the classic nuclear-accident teaching case; **Bhopal (1984)** is MIC industrial gas, not radioactive pollution.
39. **Thermal pollution** raises water temperature (mainly power-plant cooling water). Warm water holds **less DO**; even a **2–3°C** rise can stress aquatic life. **Cooling towers/ponds** mitigate discharge under Water Act consent conditions.
40. **Environmental pollution** is the harmful addition of substances or energy when the rate exceeds the environment's **absorption or assimilative capacity**.

41. **Point sources** have identifiable outlets such as factory (फैक्टरी) chimneys and sewage pipes.
42. **Non-point sources** are diffuse, such as fertilizer runoff from farms and urban (नगरीय) stormwater.
43. **Secondary pollutants** form in the atmosphere, including **ground-level ozone**, **PAN**, photochemical smog, and **acid rain**.
44. **CPCB** sets national standards. **SPCBs** enforce pollution control at the state level.
45. **Environmental degradation** is the decline in environmental quality and productivity from pollution, resource depletion, and ecosystem (पारिस्थितिकी तंत्र) damage.
46. **Urbanization** degrades the environment through sewage overload, heat islands, and waste accumulation.
47. **Mining** removes vegetation and can cause acid drainage and toxic tailings. **Sonbhadra (सोनभद्र) (UP)** faces mining-related degradation.
48. **Intensive agriculture** degrades soil through pesticide buildup, salinization, and organic matter loss.
49. **EIA under EPA 1986** aims to prevent degradation before major projects start.
50. **Air pollution** is the presence of harmful gases, dust, and smoke at levels that injure health or visibility.
51. **Lichens** are pollution indicators and do **not** grow well in badly polluted air.
52. **SAFAR** (सफर) is an IITM air-quality and weather forecast system. **WAYU** units were installed in **Delhi**.
53. **CO** forms carboxyhaemoglobin (**~200–300×** affinity vs O_2). **Radon** is the top indoor air-pollutant teaching fact.
54. **BOD** measures organic pollution through microbial oxygen demand. High BOD means worse water quality.
55. **COD** is usually greater than or equal to BOD. Clean water often shows DO above about **6 mg/L** and BOD below about **3 mg/L**.
56. Classic arsenic hotspot teaching points to **Bangladesh** / Gangetic plains. **Knock-knee** = fluoride; **Black Foot** = arsenic; **Minamata** = Hg; **Itai-itai** = Cd.
57. Many pesticides are **not** readily biodegradable — a frequent true statement in papers.
58. Heavy-metal facts include **chromium** around Kanpur plus lead, cadmium, and mercury elsewhere. Mercury sources include **gold mining** and **coal TPPs**; **no known safe exposure level**.
59. **Moradabad** was flagged among the world's noisiest cities in UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) Frontier reporting.
60. India ranks among the world's largest plastic-waste generators (often taught as about **third**).
61. Indoor air pollution is driven by solid cooking fuels, tobacco smoke, poor ventilation, and **radon**.
62. Surface-water pollution includes sewage, industrial effluent, and agricultural runoff into rivers and lakes.
63. Drinking-water quality in India is commonly referenced to **IS 10500** standards.
64. Coastal pollution includes oil, plastics, sewage, and industrial discharge along the shoreline belt.
65. SAMEER-type apps publish AQI for public warning; they do not replace NCAP city action plans.
66. **SWM Rules 2016** give elaborate landfill/site criteria. **R2 Code** = responsible electronics recycling.
67. **H-CNG** reduces CO/HC/CO_2 vs CNG but does **not eliminate** CO; up to ~**1/5** H_2 by volume; not cheaper than CNG.
68. Geoengineering (भू-इंजीनियरिंग) teaching: **cirrus thinning** and **sulphate aerosol** injection aim at cooling / offsetting warming — not cyclone control.
69. Steel industry emits **CO, CO_2, SOx, and NOx**. **Aflatoxins** = mould toxins on poorly stored grains.
70. Excess **nitrogen and phosphorus** fertilisers drive **eutrophication**, raise nitrate in groundwater, and link to **N_2O** climate forcing — Indo-Gangetic wheat–rice systems are high-input geography.
71. Fertiliser pollution is mostly **nutrient overload**, not heavy-metal tanning chemistry.
72. **Air quality** is judged against CPCB **National Ambient Air Quality Standards (NAAQS)**.
73. **Lichens** are sensitive bio-indicators of air pollution, especially SO_2 stress.
74. **NCAP** (राष्ट्रीय स्वच्छ वायु कार्यक्रम) launched in **2019** to improve air quality in non-attainment cities.
75. **Smog** is a secondary air pollution mixture; classical **London smog** is linked to **SO_2 + smoke** in cold, humid air.
76. **Photochemical smog** forms when **NOx and VOCs/hydrocarbons** react in strong sunlight.
77. **PAN (peroxyacetyl nitrate)** is a powerful **eye irritant** (lachrymator) and damages plant chloroplasts.

Chronological Sequence of Global Atmospheric & Pollution Milestones

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1952 ⟶ Great Smog of London ➔ Led to UK Clean Air Act 1956 and modern air pollution epidemiology
1974 ⟶ Molina & Rowland publish CFC-Ozone depletion hypothesis in Nature ➔ Nobel Prize in Chemistry 1995
1979 ⟶ Geneva Convention on Long-Range Transboundary Air Pollution (LRTAP) ➔ First international acid rain treaty
1985 ⟶ Vienna Convention for Protection of the Ozone Layer ➔ Framework convention for global ozone cooperation
1987 ⟶ Montreal Protocol signed (16 September) ➔ World's most successful treaty; phase-out of 99% of ODS
1999 ⟶ Gothenburg Protocol ➔ Multi-pollutant protocol to abate acidification, eutrophication, and ground-level ozone
2013 ⟶ Minamata Convention on Mercury adopted (Japan) ➔ Global legally binding treaty on toxic heavy metal
2016 ⟶ Kigali Amendment to Montreal Protocol ➔ Legally binding phase-down of high-GWP Hydrofluorocarbons (HFCs)
2019 ⟶ National Clean Air Programme (NCAP) launched in India ➔ 20–30% PM reduction target across 131 non-attainment cities
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📊 Quick Reference Grid

Basics

ITEM	LOCK
Primary / Secondary	Emitted direct / formed in air (O_3, PAN, acid rain)
Point / Non-point	Chimney–drain / farm runoff–roads
Law years	Water **1974** → Air **1981** → EPA **1986**; Noise Rules **2000**
AQI 8 pollutants	PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb (**not** CO_2/CH_4)
London / LA smog	SO_2+smoke / NOx+VOCs+sun → O_3+PAN
CO / Lead	Hb affinity ~200–300× / IQ–brain harm
Lichens / Radon	SO_2 bio-indicator / top indoor pollutant
NCAP / BS-VI	**2019** / **2020**

Chapter 10: Climate Change (जलवायु परिवर्तन)

🎯 Consolidated Must-Score Facts

1. **Global warming** is the long-term rise in average surface temperature. It is a subset of broader **climate change**, which also includes rainfall, extremes, and sea-level shifts.
2. IPCC (आईपीसीसी) AR6 facts about **1.1°C** above 1850–1900 globally. India has warmed about **0.7°C** since 1901 (IMD class figure). Past-century surface rise is about **1°C ≈ 1.8°F**.
3. The natural greenhouse effect (प्राकृतिक ग्रीनहाउस प्रभाव) keeps Earth near about **+15°C**. Without it, Earth would be near about **−18°C**.
4. **Joseph Fourier (1820s)** postulated the greenhouse-effect idea. Sun sends **shortwave** (UV + visible + IR); Earth re-emits **longwave IR**. Ozone absorbs harmful UV; GHGs trap outgoing IR.
5. Enhanced greenhouse effect (मानवजनित संवर्धित प्रभाव) means excess anthropogenic GHGs trap outgoing infrared radiation. Main cause of recent warming is **CO_2** (highest concentration among long-lived GHGs).
6. Kyoto-basket majors: **CO_2** (GWP **1**), **CH_4** (GWP about **28–36**), **N_2O** (GWP about **265**), plus HFCs, PFCs, **SF_6** (and **NF_3** in later inventories).
7. Rough **WMO-style relative (सापेक्ष) shares** often used in Indian papers: **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**. Decreasing order: **CO_2 > CH_4 > CFCs > N_2O**.
8. **Water vapour** is Earth's most abundant natural GHG (NASA: ~half of the natural greenhouse effect) but mainly a **feedback**, not the primary policy driver.
9. **Not GHGs: argon, nitrogen (N_2), oxygen, hydrogen, propane**. **SO_2 / NOx** do **not** contribute to warming **directly** (indirect / other roles).
10. **Direct** GHGs include CO_2, CH_4, N_2O, CFCs, SF_6, NF_3. **Indirect** agents include **NOx, CO, NMVOCs, SO_2**.
11. India is about the **third**-largest total emitter (**China** first); **Bhutan** is often cited as **carbon-negative**.
12. A **carbon footprint** totals GHGs in tCO_2e across Scope **1** (direct), **2** (energy), and **3** (supply chain).
13. A **carbon sink** absorbs more CO_2 than it releases (forests, oceans, soils, mangroves / **blue carbon** (नीला कार्बन)). **Sequestration** is the storage process.
14. **Carbon fertilization** = more plant growth from higher atmospheric CO_2 (not the same as ocean acidification (महासागरीय अम्लीकरण) or warming).
15. Pre-industrial CO_2 ~**278–280 ppm (~0.03%)**; mid-20th century ~**315 ppm**; now **415–420+ ppm**.
16. **Methane ("marsh gas")** sources: paddies, cattle, coal mines, landfills, termites; natural wetlands dominate natural CH_4 (~**75–76%** of natural). **Rice** is a key anthropogenic source of **both CH_4 and N_2O**.
17. **Methane hydrates** under Arctic tundra (टुंड्रा) / seafloor can release CH_4 if warmed; atmospheric CH_4 oxidises to CO_2 in roughly a decade or two.
18. Impacts include heat waves, glacier melt, sea-level rise, coral bleaching (प्रवाल विरंजन), disease spread, stronger storms. **Arctic / Greenland ice** is among the most fragile first-hit systems.
19. Rough IPCC-style risk: beyond about **+2°C** pre-industrial → widespread coral mortality; beyond about **+3°C** → terrestrial biosphere tends toward a **net carbon source**.
20. **Adaptation** = coping with impacts. **Mitigation** = cutting emissions and enhancing sinks.
21. **NAPCC (2008)** has **eight** national missions — **not** nuclear power as a listed mission.
22. India's updated **NDC** path includes about **45%** emission-intensity cut and about **50%** non-fossil electricity capacity by **2030** (not "60% energy from renewables" as a false paraphrase).
23. India's **Net Zero** pledge is **2070**. Net zero means emissions balanced by removals — **not** literally zero every emission.
24. **LiFE Mission** launched **June 2022**; idea at **COP26 (2021)**, not COP25.
25. Climate treaty chain: **UNFCCC (यूएनएफसीसीसी) 1992 → Kyoto 1997 (force 2005) → Paris 2015**. **Montreal 1987** = ozone, not climate.
26. **CDM** (Kyoto Art. 12) lets Annex-I parties fund projects in developing countries for **CERs** (1 CER ≈ 1 tCO_2). **Carbon credit** (कार्बन क्रेडिट) concept originates from Kyoto.
27. **Green Climate Fund** was set up at **Cancun COP-16 (2010)**, not Durban. **Earth Hour** (अर्थ आवर) = **WWF** (डब्ल्यूडब्ल्यूएफ), last Saturday of March (lights off ~8:30–9:30 pm).
28. **GHG Protocol** = **WRI + WBCSD** accounting tool. **Blue carbon** = ocean/coastal carbon (mangroves, seagrass, salt (लवणाध्यक्ष) marsh).
29. UP heat vulnerability often cites (साइट्स) Lucknow (लखनऊ), Agra, Prayagraj (प्रयागराज), and NCR.
30. UP drought (सूखा) fact is **Bundelkhand's seven districts**. Flood facts include Ballia, Ghazipur, and Varanasi (वाराणसी) belts.
31. Emissions hotspot language for UP–MP often points to the **Singrauli** coal belt.
32. Terai (तराई) forests at Dudhwa (दुधवा) and Pilibhit are UP **carbon-sink** examples.
33. **Milankovitch** orbital theory (eccentricity, obliquity, precession) — **solar irradiance** is **not** part of that astronomical set.
34. **Ice cores** are the classic **cryogenic** climate archive (Greenland / Antarctica).
35. **Solar flares** are **not** a primary cause of recent climate change in standard NASA/IPCC framing.
36. Farm practices that help soil carbon storage include **contour bunding, relay cropping, and zero tillage**.
37. **Global warming** is the long-term increase in Earth's average near-surface temperature.
38. **Climate change** is broader and includes shifts in rainfall, monsoon timing, cyclones, and extremes.
39. **IPCC AR6** estimates (प्राकलन समिति) warming of about **1.1°C** above the **1850–1900** pre-industrial baseline.
40. **Human activities** — fossil fuels, deforestation, industry, agriculture — are the **dominant cause** of recent warming.
41. **India** has warmed about **0.7°C** since 1901 according to **IMD** data.
42. **Oceans absorb about 90%** of excess heat, causing thermal expansion and marine stress.

43. **Arctic warming** is about **2–3 times faster** than the global average.
44. **Water vapour** is the most abundant natural GHG (about **half** of the natural greenhouse effect in NASA-style framing) but acts mainly as a **feedback**, not a primary emitted driver like fossil CO_2.
45. **Greenhouse gases** absorb and emit infrared radiation in the atmosphere.
46. **CO_2** is the main anthropogenic driver by concentration and total forcing. IPCC-style forcing share is often ~**76%**; older WMO-bulletin style relative shares used in many Indian papers are **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**.
47. **CO_2** has **GWP = 1**. Pre-industrial ~**278–280 ppm (~0.03%)**; ~**315 ppm** around 1950s; now **415–420+ ppm**.
48. **Methane (CH_4)** — also called **marsh gas** — has **GWP around 28–36** (100-yr). Sources: paddies, cattle, coal mines, landfills, termites; natural **wetlands** dominate natural CH_4 (~**75–76%**).
49. **Rice** is a major anthropogenic source of **both methane and nitrous oxide** (anaerobic soils + nitrogen fertiliser).
50. **Nitrous oxide (N_2O)** has **GWP around 265** and links strongly to synthetic fertilisers.
51. IPCC AR6 places warming near **1.1°C** above **1850–1900**, with human influence dominant. Past-century rise ≈ **1°C / 1.8°F**.
52. The **Keeling Curve** records atmospheric **CO_2** at **Mauna Loa** since **1958**.
53. The **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) is an annual **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) product. It is **not** published by the IPCC.
54. WMO-style relative (सापेक्ष) shares often tested: **CO_2 ~64% > CH_4 ~19% > CFCs+others ~11% > N_2O ~6%**. Water vapour dominates the **natural** effect.
55. India ranks near the top in **absolute** (निरपेक्ष) emissions (**China #1**, India ~**3rd**) but stays **low per capita**. **Bhutan** is often carbon-negative.
56. A **carbon credit** equals **1 tonne of CO_2 equivalent**. Carbon trading is the market for credits or allowances (Kyoto (क्योटो) origin).
57. **Green Credit (2023)** (ग्रीन क्रेडिट) and **PAT ESCert** are **not** the same thing as a Kyoto-style carbon credit.
58. **Kyoto** (क्योटो) is climate mitigation (जलवायु शमन). **Montreal** is ozone protection. Do not swap the two treaties.
59. India's **Carbon Credit Trading Scheme (2023)** sits with **BEE** under the Power Ministry.
60. The **IPCC** does science assessment under **WMO + UNEP** parents. It does **not** set national targets or write the Emissions Gap Report.
61. The **carbon cycle** moves carbon among atmosphere, oceans, land, and fossil pools. Sequestration is the storing process.
62. Carbon trading grew from Kyoto-era market tools; Paris-era systems use national NDCs and domestic markets.
63. Do not treat IPCC Assessment Reports as UNFCCC (यूएनएफसीसीसी) treaty text. Science reports and the climate convention are different homes.
64. Adaptation examples include early warning, climate-resilient crops, coastal defence, and heat-action plans — not only emission cuts.
65. **Methane hydrates**, paddy+fertiliser (CH_4+N_2O), and wetland CH_4 are high-yield source traps.
66. **Forest Carbon Partnership Facility** and **BioCarbon Fund ISFL** are **World Bank** programmes — FCPF does **not** fund universities/individuals for research grants.
67. **Climate Action Tracker** = Climate Analytics + New Climate Institute (not IPCC/UNEP wing).
68. **Deep Carbon Observatory** maps deep-Earth carbon (physics, reservoirs, deep energy, deep life).
69. **Common Carbon Metric** (UNEP-linked) measures building-operation carbon footprints.
70. Primary recent warming drivers are fossil fuels, autos, deforestation — **not** solar flares.
71. **Radiative forcing** (W/m^2) measures the net energy imbalance driving warming.
72. **WMO** publishes the **State of the Global Climate** report.
73. **NOAA** and **NASA GISS** maintain key global temperature datasets.
74. **Global Carbon Project** issues the annual **Global Carbon Budget**.
75. **Emissions Gap Report** is published by **UNEP** — it measures the gap between NDCs and 1.5–2°C pathways.
76. **Energy and electricity** (coal, oil, gas) dominate CO_2.
77. **Industry** adds process CO_2 (cement, steel) plus F-gases.
78. **AFOLU** (agriculture, forestry, land use): **CH_4** from livestock/rice; **N_2O** from fertilisers; CO_2 from deforestation.
79. **Transport** and **buildings** add fossil CO_2; **waste** adds landfill methane.
80. **Argon is not a greenhouse gas.**
81. **Fast cycle:** atmosphere ↔ plants ↔ soils ↔ surface ocean (years to decades).
82. **Slow cycle:** rock weathering, sediment burial, fossil carbon (millions of years).
83. **Photosynthesis consumes** CO_2 and releases oxygen. It does **not** add carbon dioxide to the carbon cycle on Earth.
84. **CO_2** has **GWP = 1**. It dominates long-term forcing from fossil fuels, cement, and deforestation.
85. **Methane (CH_4)** has GWP about **28–36**, a shorter lifetime near **12 years**, and strong links to paddies, cattle, landfills, termites, and wetlands (**marsh gas**).
86. **Tropospheric ozone** (क्षोभमंडलीय ओजोन) is a greenhouse gas and pollutant. **Stratospheric ozone** (समतापमंडलीय ओजोन) is the UV shield.
87. **CFCs** are both greenhouse gases and ozone-depleting substances and fall under Montreal controls.
88. **HFCs** are greenhouse gases only (not classic ODS) and are phased down under **Kigali 2016**.
89. **GWP** compares heat-trapping power to CO_2 over a stated horizon, usually **100 years**. Methane's short-horizon GWP is much higher than CO_2 — CH_4 is more potent per molecule.
90. **CO_2e** converts all greenhouse gases into carbon-dioxide equivalent using GWP.
91. Biogas (बायोगैस) is mainly **methane plus CO_2**. Do not treat biogas as CO_2-only.
92. Paddy fields are a classic **methane** source fact in Indian papers.
93. CFC versus HFC is dual-role ODS+GHG versus climate-only substitute under Kigali.

94. N_2O is not the same as N_2. Molecular nitrogen is inert as a greenhouse gas.
95. Water vapour rises as Earth warms; that feedback amplifies warming but does not replace fossil CO_2 as the driver.
96. F-gases (HFCs, PFCs, SF_6) are small by mass but can be huge by GWP.
97. Cement process emissions are a major industrial CO_2 source beyond fuel combustion.
98. Methane's shorter lifetime still matters because near-term warming impact is strong.
99. Ozone's greenhouse role is mainly **tropospheric**; do not call stratospheric ozone the main GHG form.
00. CO_2 is both **useful** (photosynthesis) and **harmful** (enhanced greenhouse effect (मानवजनित संवर्धित प्रभाव)) to life on Earth.
01. **Livestock** enteric fermentation (cattle/buffalo) is a major Indian source.
02. **N_2O** is a long-lived GHG with **GWP ≈ 265** (100-year) and lifetime ~**100+ years**.
03. **Stratospheric ozone** is the UV shield — its ma role is protection, not climate forcing.
04. **CFCs** are human-made compounds once used in refrigeration, aerosols, and foam blowing.
05. **HFCs** replaced many CFCs/HCFCs in air-conditioning and refrigeration.

📊 Quick Reference Grid

Frame

ITEM	LOCK
Warming vs climate change	Temp rise subset / broader incl. rainfall & extremes
Natural GHG effect	~+15°C with; ~−18°C without
Fourier	1820s greenhouse-effect idea
Main recent driver	Anthropogenic **CO_2**
Kyoto basket	CO_2, CH_4, N_2O, HFCs, PFCs, SF_6 (+NF_3 later)
Relative shares (WMO-style)	CO_2~64% > CH_4~19% > CFCs+~11% > N_2O~6%
Not GHGs	Ar, N_2, O_2, H_2, propane
India rank / Bhutan	~3rd absolute (निरपेक्ष) / often carbon-negative
Sink / sequestration	Net CO_2 absorb / storage process
Blue carbon	Ocean–coastal sinks (mangroves etc.)

Chapter 11: Ozone Layer

🎯 Consolidated Must-Score Facts

1. Ozone (O_3) is triatomic oxygen. **Stratospheric ozone** (समतापमंडलीय ओजोन) (about **15–30/35 km**) is protective "good" ozone. **Tropospheric ozone** (क्षोभमंडलीय ओजोन) is a ground-level pollutant.
2. About **90%** of ozone is stratospheric; about **10%** is tropospheric. Stratospheric ozone is regulated in part by **NO_2**.
3. The **Chapman cycle** describes natural formation and destruction balance of stratospheric ozone (UV acting on O_2).
4. **UV-A (315–400 nm)** reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is fully blocked. Ozone blocks ~**93–99%** of the most damaging UV.
5. Ozone-depleting substances release chlorine and bromine radicals that catalytically destroy ozone. One chlorine atom can destroy on the order of **100,000** ozone molecules.
6. Main ODS include **CFCs, Halon-1211/1301, carbon tetrachloride, methyl chloroform, methyl bromide, and transitional HCFCs**. Fridge coolants often sold as **Mafron**; large plants may use **ammonia**.
7. An **ozone hole** is not a physical hole. It means column ozone below **220 Dobson Units**. **1 DU ≈ 0.01 mm** pure ozone at 0°C, 1 atm.
8. The Antarctic ozone hole peaks in Southern Hemisphere spring (**September–October**) under a polar vortex with polar stratospheric clouds + CFCs.
9. The Antarctic ozone hole was discovered in **1985** by **Farman (फ़रमान), Gardiner, Shanklin** (British Antarctic Survey). **Kent Moore** reported Tibetan Plateau ozone feature in **2005**.
10. Treaty chain: **Vienna Convention 1985 → Montreal Protocol 1987 (force 1 Jan 1989) → Kigali Amendment 2016 (force 1 Jan 2019)**.
11. Montreal is the binding ODS phase-out treaty. International Ozone Day is **16 September** (UN observance from **1994**).
12. **Kigali (2016)** phases down **HFCs**. HFCs are not classic ODS, but they have high global-warming potential.
13. Montreal achieved near-universal ratification and cut ODS emissions by about **99%** from peak levels.
14. Treaty trap: **Montreal = ozone**; **Kyoto/Paris = climate**. Never swap the issue tags. Carbon trading ≠ Montreal.
15. Global average ozone column is often near about **300 DU**; the hole threshold is **<220 DU**.
16. India ratified Montreal in **1992**. The nodal unit is the **Ozone Cell, MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय)**.
17. India CFC phase-out fact is **2010**. HCFC freeze is **2020**; complete HCFC phase-out aims at **2030**.
18. India's Kigali HFC phase-down commitment begins from **2028** in standard coaching facts.
19. Stratospheric ozone protects life; tropospheric ozone harms lungs and crops — same molecule, opposite verdict by layer.
20. Vienna is the framework convention; Montreal is the binding control protocol — parallel to UNFCCC (यूएनएफसीसीसी)/Kyoto structure, but for ozone.
21. Dobson Unit measures total ozone column, not ground-level ppm of pollution ozone.
22. Polar stratospheric clouds enable chlorine activation — that is why the deep hole is polar-spring specific.
23. HCFCs were transitional substitutes on the way out of CFCs; they are still ODS, just weaker/longer phase-out.
24. Kigali sits under the Montreal family, not under UNFCCC as a climate protocol.
25. Ozone Day is **16 September**; do not confuse it with World Environment (पर्यावरण) Day (विश्व पर्यावरण दिवस) (**5 June**) or Wildlife Day (**3 March**).
26. CFCs are both ODS and GHGs; CFC source is not primarily automobiles; CFC use is not tubeless tyres.
27. Extra UV mainly raises **skin cancer** risk (classic stem).
28. Homosphere (सममंडल) ≈ troposphere (क्षोभमंडल) + stratosphere (समतापमंडल) + mesosphere (मध्यमंडल); **ionosphere not** in that set.
29. Column thickness often highest in **spring**, lowest in **rainy season** (general seasonal pattern).
30. Full teaching depth for this chapter also lives in Topic **36**.
31. **Ozone (O_3)** is a molecule of three oxygen atoms that absorbs harmful ultraviolet radiation.
32. **UV-A** largely reaches the surface. **UV-B** is mostly blocked by ozone and causes sunburn and skin cancer if exposure rises.
33. **UV-C** is fully absorbed by ozone and oxygen under normal conditions.
34. **Column ozone** is measured in **Dobson Units (DU)**. Global average is about **300 DU**.
35. **Stratospheric ozone** is protective. **Tropospheric ozone** is a harmful air pollutant and greenhouse gas.
36. **Ozone depletion** is the sustained reduction of stratospheric ozone below natural levels.
37. **Ozone-Depleting Substances (ODS)** contain chlorine and/or bromine that destroy ozone in the stratosphere.
38. **CFCs** were used in refrigerants, aerosols, foam blowing, and solvents.
39. **One chlorine atom can destroy up to about 100,000 ozone molecules** before removal.
40. **Halons** used in fire suppressants release **bromine**, which is even more destructive per atom.
41. **Polar Stratospheric Clouds (PSCs)** (ध्रुवीय समतापमंडलीय बादल) form at very low temperatures and enable chlorine activation.
42. **Arctic** ozone loss can occur in cold winters but is **less severe and less regular** than Antarctic depletion.
43. **International Ozone Day** is **16 September**, the date the Montreal Protocol was signed.
44. **India ratified in 1992**. The **Ozone Cell under MoEFCC** coordinates compliance.
45. **Montreal controls ozone**. **Kyoto controls climate**. Do not swap them.
46. Stratospheric ozone (समतापमंडलीय ओजोन) forms when **UV** acts on **O_2** (Chapman cycle). Ozone absorbs **ultraviolet**, not IR as its main protective role.
47. Ozone depletion is driven by **ODS** such as CFCs through chlorine and bromine catalytic cycles.

48. Refrigerant trade (पण्याध्यक्ष) name **Mafron** (halocarbon coolants) is common in domestic refrigerators; **ammonia** is used in many large plants. CFCs are **not** primarily "automobile exhaust" gases.
49. **HFCs** are high-GWP climate gases under **Kigali**. They are **not** classic ozone-depleting substances.
50. ODP measures ozone-damage potential. GWP measures heat-trapping potential.
51. CFCs are both ODS and strong greenhouse gases — "CFC not a GHG" is false.
52. Polar spring timing matters: the deepest Antarctic loss is not midsummer.
53. Vienna does not itself list every control measure; Montreal carries the binding phase-out schedules.
54. Troposphere is deepest in the tropics (~**18–20 km**) and shallower near the poles.
55. Dobson Unit is a column measure, not a surface concentration in ppm.
56. A stem pairing Montreal with climate mitigation (जलवायु शमन) is usually the Kyoto/Paris trap.
57. Ozone as a gas near the surface is poisonous to lungs/respiratory systems; as a stratospheric layer it is essential for biotic life via UV shielding.
58. Formation of ozone is slower over polar regions, so depletion effects are strongest there.
59. Acid rain (अम्ल वर्षा) is driven by **SO_2 / NOx**, not "nitric acid alone" as a matched pair with ozone–CFC style keys.
60. **Rowland–Molina (1974)** linked CFCs to ozone loss; Nobel Prize in Chemistry **1995**.
61. **ODS** are chemicals controlled under the Montreal Protocol because they destroy stratospheric ozone.
62. **Halons (e.g. Halon-1211, Halon-1301):** fire suppressants — bromine; often higher ozone-depleting impact per molecule.
63. **ODP (Ozone Depleting Potential)** ranks destructiveness relative (सापेक्ष) to **CFC-11 = 1**.
64. **Nitrous oxide (N_2O)** also contributes to ozone chemistry / long-term ozone loss discussions, but CFCs/halons dominate classic "ozone hole" stems.

📊 Quick Reference Grid

Science

ITEM	LOCK
Good / bad ozone	Stratosphere (~15–35 km) / tropospheric pollutant
Share	~90% stratospheric; ~10% tropospheric
UV-A / B / C	Reaches surface / mostly blocked / fully blocked
Ozone hole	Column <**220 DU** (not a physical hole)
Antarctic peak	SH spring Sep–Oct; PSCs + vortex + CFCs
Discovery	**1985** Farman–Gardiner–Shanklin (BAS)

Treaties

ITEM	LOCK
Chain	Vienna **1985** → Montreal **1987** → Kigali **2016**
Montreal vs Kyoto/Paris	Ozone ODS / climate GHGs — never swap
Kigali	Phases down **HFCs** (high GWP; not classic ODS)
Ozone Day	**16 September**


Chapman Ozone Cycle & CFC Catalytic Depletion Chain

Chapter 12: Acid Rain (अम्ल वर्षा)

🎯 Consolidated Must-Score Facts

1. **Acid rain** means wet or dry acidic deposition with pH below **5.6**.
2. Normal unpolluted rain is already about **pH 5.6** because dissolved CO_2 forms weak carbonic acid — it is **not** pH 7.
3. **Wet deposition** is acids in rain, snow, or fog. **Dry deposition** is acidic gases and particles settling without rain.
4. **SO_2** oxidises to **H_2SO_4** (sulphuric acid), mainly from coal TPPs, smelters, and refineries.
5. **NOx** oxidises to **HNO_3** (nitric acid), mainly from vehicles and high-temperature combustion.
6. CO_2 forms only weak H_2CO_3 and sets the **5.6 baseline**; it is **not** the main culprit of severe acid rain.
7. On the pH scale, acidic is **<7**, basic is **>7**, neutral is **7**. Each step down multiplies H^+ by **10**.
8. Freshwater below about **pH 5** stresses fish and can release toxic **Al^{3+}**.
9. Acid deposition leaches soil **Ca^{2+} and Mg^{2+}** and can free toxic aluminium.
10. Conifers are especially sensitive forest receptors. Marble and limestone monuments form a **gypsum crust** ($CaSO_4$) under sulphuric attack — the Taj story.
11. **Lichens** die in high SO_2 and act as pollution bio-indicators.
12. Prevention tools include **FGD scrubbers** (lime/limestone), catalytic converters, low-sulphur coal, CNG/LPG switches, and **BS-VI** fuel (about **10 ppm** sulphur).
13. The **Gothenburg Protocol (1999)** (गोथेनबर्ग प्रोटोकॉल) under **CLRTAP (1979)** cuts transboundary acidifying emissions.
14. Treaty trap: Gothenburg/CLRTAP = acidifying air pollution; Montreal = ozone; Kyoto (क्योटो) = climate GHGs.
15. The **Taj Trapezium Zone** is about **10,400 km^2** around the Taj Mahal (ताज महल), strengthened by Supreme Court orders from **1996**.
16. **Mathura (मथुरा) refinery** is a classic SO_2 source in the Taj Trapezium narrative.
17. The **Singrauli–Sonbhadra (सोनभद्र)** coal TPP belt is a major UP–MP SO_2 source region.
18. UP NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) cities linked to acidifying/urban (नगरीय) air pressure include Kanpur (कानपुर), Ghaziabad, Lucknow (लखनऊ), and Noida.
19. SO_2 is denser than air — a physical property sometimes used in distractors.
20. Acid rain is **not** "rain only"; dry deposition counts too.
21. Secondary pollutants H_2SO_4 and HNO_3 form in the atmosphere from primary SO_2 and NOx.
22. FGD is flue-gas desulphurisation at power-plant chimneys — a mitigation hardware fact, not a treaty name.
23. Calling CO_2 the chief severe acid-rain driver is wrong; SO_2 and NOx are the pair.
24. Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) does **not** control acid rain. Keep ozone chemistry out of acid-rain treaty options.
25. **Acid rain** is precipitation or atmospheric deposition with **pH below 5.6**, mainly from **sulphuric acid (H_2SO_4)** and **nitric acid (HNO_3)** (नाइट्रिक अम्ल).
26. **Normal unpolluted rain** has pH about **5.6**, not 7, because atmospheric **CO_2** forms weak **carbonic acid (H_2CO_3)**.
27. **Wet deposition** carries acids in rain, snow, fog, or dew.
28. **Dry deposition** means **SO_2**, **NO_2**, and sulphate/nitrate particles settle on soil, leaves, and buildings **without rainfall**.
29. **SO_2** and **NOx** are **primary pollutants**. Strong acids form in the atmosphere, so acid rain is a **secondary pollutant** phenomenon.
30. **Sulphur dioxide (SO_2)** is the main source of **sulphuric acid rain**.
31. **Nitrogen oxides (NOx)** form at **high combustion temperatures** when nitrogen in air reacts with oxygen.
32. **CO_2** dissolves as weak carbonic acid and sets the **pH 5.6 baseline**. It is **not** the main driver of severe acid rain.
33. **Coal thermal power plants** are India's largest **SO_2** point sources.
34. **Vehicles** are a major **NOx** source in cities.
35. **SO_2 is heavier than air** and tends to accumulate near ground-level emission sources.
36. **Aquatic ecosystems** fail when lake/river pH drops **below about 5.0** — fish eggs do not hatch and adult fish die.
37. **Aluminium (Al^{3+})** leached from acidified soil is **toxic to fish gills** and plant roots.
38. **Soil acidification** leaches essential **calcium (Ca^{2+})** and **magnesium (Mg^{2+})** and releases toxic **Al^{3+}**.
39. **Coniferous forests** (pine, spruce, fir) are **most sensitive** to acid rain and acid fog.
40. **Crops** suffer from direct leaf damage and nutrient loss in soil.
41. **Marble and limestone ($CaCO_3$)** react with sulphuric acid to form **gypsum ($CaSO_4$)**, causing yellow crust and pitting.
42. **Metal structures** rust faster under acid deposition.
43. **SO_2 and NOx** cause **respiratory illness** — asthma, bronchitis, and lung irritation from acid aerosols and $PM_{2.5}$.
44. **Lichens** are **bio-indicators of clean air**. They **die in high SO_2** and do **not** thrive in polluted areas.
45. **Flue Gas Desulphurisation (FGD) scrubbers** pass chimney gas through **lime or limestone slurry** to absorb **SO_2** and can remove **90%+** of it.
46. **Coal washing** and **low-sulphur coal** cut sulphur before combustion.
47. **Catalytic converters** in vehicles convert **NO to N_2** and reduce **NOx** that forms nitric acid.
48. **BS-VI norms (2020)** brought **10 ppm sulphur** in petrol and diesel, cutting both SO_2 and NOx precursors.
49. **CNG, LPG, and natural gas** produce **negligible SO_2** and lower NOx than coal.
50. **CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) emission standards** under the Air Act cap SO_2 and NOx from TPPs, cement, steel, and refineries.
51. The main strong acid pathway is **SO_2 → H_2SO_4**. **NOx → HNO_3** is the other major path.

52. Reversed statements that call normal rain alkaline or acid rain alkaline are both false.
53. Acid deposition harms conifers and weathers marble and limestone. The Taj marble → **gypsum** story is the classic Indian fact.
54. The **Gothenburg Protocol (1999)** under **CLRTAP (1979)** targets acidifying air pollutants. **Montreal** targets ozone ODS — different problem.
55. The **Singrauli** thermal belt and industrial NCR/Kanpur (कानपुर) airsheds are Indian acidifying-emission hotspots.
56. SO_2 is denser than air and can travel long distances before deposition.
57. Coal power and some industries are major SO_2 sources. Vehicles and combustion add NOx.
58. Liming lakes is an emergency response, not a substitute for cutting emissions at source.
59. Do not blame acid rain mainly on CO_2 just because normal rain is weakly acidic.
60. Dry deposition can damage materials even on rainless days.
61. Acid mist and acid fog are wet forms even when they are not classic rainfall.
62. Cultural-heritage damage and forest dieback are both valid effect facts alongside aquatic harm.
63. India's focus pairs chemistry (SO_2/NOx) with Taj protection geography.
64. **Conifers** (pine, spruce) are especially sensitive to acid fog and soil change.
65. **Taj Trapezium Zone (~10,400 km²)** — Supreme Court-driven controls (1996 onward) forced gas switch/relocation of polluting units around Agra.

📊 Quick Reference Grid

Chemistry

ITEM	LOCK
Acid rain pH	Below **5.6**
Normal rain (सामान्य वर्षा)	~pH **5.6** from CO_2 → H_2CO_3 (≠ 7)
Wet / Dry	Rain–snow–fog / gases–particles without rain
SO_2 / NOx paths	→ H_2SO_4 (TPPs) / → HNO_3 (vehicles)
CO_2 role	Sets 5.6 baseline — **not** main severe-acid culprit
Receptors	Fish <~pH 5; Al^{3+} release; conifers; marble → gypsum
Lichens	SO_2 bio-indicators
Gothenburg / CLRTAP	**1999** / **1979** acidifying emissions


Acid Rain Pathways Landscape Profile & Taj Mahal Stone Cancer

Chapter 13: Desertification (मरुस्थलीकरण) & Land Degradation (भू-क्षरण)

🎯 Consolidated Must-Score Facts

1. **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** areas from climate variation plus human pressure.
2. Desertification is **not** only "desert expanding." Any dryland losing productivity can count.
3. **Land degradation** is broader: decline in productive capacity on any land through erosion, salinization, waterlogging, mining, or pollution.
4. India has about **96.4 million ha** degraded land — roughly **29–30%** (ISRO Atlas **2016** class figure).
5. **Normal geological erosion** is slow and roughly balanced by soil formation. **Accelerated erosion** is human-driven and faster than formation — the focus.
6. Water-erosion sequence: **sheet** → **rill** → **gully** → **ravine**. Chambal (चंबल) badlands are the ravine end-member.
7. Wind erosion works by **deflation** (lifting) and **abrasion** (sandblasting).
8. Leading causes include overgrazing in drylands, deforestation, unsustainable farming, excessive irrigation (salinization/waterlogging), and mining.
9. India hotspots: **Thar** (थार) wind erosion; **Chambal** (चंबल) ravines (MP–UP–Rajasthan); **Bundelkhand** drought (सूखा)–degradation cycle; Punjab–Haryana salinization belts.
10. Conservation tools include contour bunding, terraces, shelter belts, afforestation, gully plugging, check dams, and watershed management (वाटरशेड (जलग्रहण) प्रबंधन).
11. Earthworms are soil-fertility allies in restoration narratives.
12. **UNCCD** was adopted (अंगीकृत) in **1994**. Desertification and Drought Day is **17 June**.
13. **LDN** (Land Degradation Neutrality) under **SDG 15.3** targets no net loss of healthy land by **2030**.
14. **DPAP** launched in **1973**. **DDP** is **1977–78**. **IWMP** is **2009**. **MGNREGA (2005)** funds bunds and check dams.
15. Treaty trap: **UNCCD ≠ UNFCCC (यूएनएफसीसीसी)**. Desertification is not the climate convention.
16. Afforestation helps **prevent** degradation; it is not a cause of desertification.
17. Gully erosion (अवनलिका अपरदन) is not the same as sheet erosion (परत (चादर) अपरदन) — sheet is the first thin uniform stage.
18. Chambal ravine area is about **3 million ha** in standard coaching numbers.
19. UP Chambal ravine districts include **Etawah (इटावा) and Agra** (with Morena–Bhind on the MP side).
20. Bundelkhand's **seven UP districts** are the drought-plus-degradation fact.
21. Thar-margin UP questions emphasise wind erosion and dune encroachment.
22. **Sonbhadra** (सोनभद्र) is a UP mining-related land-degradation hotspot.
23. Wasteland means degraded land poorly suited to cultivation — related to, but not identical with, desertification's ASDM definition.
24. LDN by 2030 is an SDG land target; do not date UNCCD itself as 2030.
25. **Desertification** is **land degradation in arid, semi-arid, and dry sub-humid (ASDM) areas** caused by climate variation and human activities.
26. **Overgrazing** is the **main human cause** in drylands. Livestock remove vegetation, compact soil, and expose topsoil to wind and water erosion.
27. **Deforestation**, **fuelwood collection**, and **unsustainable cropping** remove root binding and reduce local moisture retention.
28. **Improper irrigation** can cause **salinization** and **waterlogging**, turning fertile irrigated land barren — common in parts of Rajasthan canal commands and the Indo-Gangetic fringe.
29. **Drought and climate change (जलवायु परिवर्तन)** kill vegetation and trigger flash erosion on bare soil, creating a feedback loop of further degradation.
30. **Wind erosion** dominates true arid zones like the **Thar Desert**, where dunes migrate and bury fields.
31. **17 June** is observed as **World Desertification and Drought Day**.
32. **SDG 15.3** targets **Land Degradation Neutrality (LDN) (भूमि क्षरण तटस्थता) by 2030** — avoid new degradation and restore degraded land.
33. **Land degradation** is the **broader decline in a land's productive capacity** from natural or human factors.
34. **Desertification** is one **subset** limited to ASDM drylands. **Soil erosion** is one **major process** within degradation.
35. **Human causes dominate** India's degradation statistics: overgrazing, deforestation, unsustainable agriculture, excessive fertilizers, mining, industrial pollution, and urban (नगरीय) sprawl.
36. **Salinization** from excessive canal irrigation affects **Punjab, Haryana, and western UP**.
37. **Mining degradation** is severe in **Sonbhadra (UP), Jharkhand, and Odisha**.
38. **Salinization and waterlogging** can often be reversed through drainage, gypsum, and drip irrigation. **Deep ravines** are hardest to restore.
39. **Afforestation prevents degradation** — it is **not** a cause. Pick it when asked "NOT a cause."
40. **Topsoil (A-horizon)** is the fertile organic-rich layer, usually **15–30 cm deep**. It can take **centuries to form** but may be lost in **one heavy monsoon** on bare slopes.
41. **Normal (geological) erosion** is slow and balanced by soil formation. **Accelerated erosion** from human activity is **10–100× faster** than formation — this is the usual default meaning of "soil erosion."
42. **Water erosion** dominates humid and monsoon India. Raindrop impact dislodges soil and runoff carries it downslope.
43. **Water erosion sequence:Sheet erosion** (thin uniform layer) → **rill erosion** (क्षुद्र सरिता (रिल) अपरदन) (small channels) → **gully erosion** (deep cuts) → **ravines** (extensive badland networks).
44. **Chambal ravines** (चंबल) cover about **3 million hectares** across **MP (Morena, Bhind), UP (Etawah, Agra), and Rajasthan (Dholpur, Karauli)**. They are India's classic **gully/ravine erosion** landscape.

45. **Wind erosion** dominates arid zones through **deflation** (lifting fine particles) and **abrasion** (sandblasting surfaces). It affects the **Thar**, Rann of Kutch, and western Rajasthan-UP margins.
46. **Gravity erosion** (landslides, slumping) matters in **Himalayan and NE hill states**, especially after deforestation and road cutting.
47. **Vegetation cover** is the best natural erosion shield. Deforestation and overgrazing are the top acceleration factors.
48. **Conservation measures:contour bunding/ploughing**, **terrace farming**, **shelter belts/wind breaks**, **afforestation**, **gully plugging**, **check dams**, **strip cropping**, and **cover crops**.
49. **Earthworms** aerate soil, improve structure, and boost fertility — healthy soil resists erosion.
50. Drivers combine climate variation with human pressures such as overgrazing, deforestation, and bad irrigation.
51. **Land degradation** is the broader productivity decline and can occur in any climate zone.
52. Drought types run **meteorological → hydrological → agricultural**. Meteorological drought (मौसमी (मौसम विज्ञान) सूखा) is often linked to rainfall below about **75%** of long-period average.
53. **DPAP** began in **1973**. Drought and desertification reinforce each other in a feedback loop.
54. Salinity from poor irrigation is a human-made degradation path even where rainfall is not zero.
55. Desertification can reverse with sustainable land management; it is not an irreversible death sentence in every plot.
56. Do not define desertification as any bare land in a humid forest belt.
57. IMD drought monitoring and DPAP geography often appear beside desertification stems.
58. SDG 15.3 LDN is the policy end-state; UNCCD is the global legal home.
59. Thar wind processes and Chambal water ravines are twin Indian landform facts.
60. Deforestation removes cover and raises runoff and erosion risk in drylands and beyond.
61. **Desertification** is the ASDM subset; erosion, salinity, waterlogging, mining, and pollution are processes under the broader term.
62. **Hydrological drought (जल-वैज्ञानिक सूखा):** rivers, reservoirs, and groundwater fall.
63. **Agricultural drought (कृषि सूखा):** soil moisture fails crops even if some rain fell.
64. **Drought Prone Areas Programme (DPAP)** launched **1973–74** for dryland watershed-type works.

📊 Quick Reference Grid

Concepts

ITEM	LOCK
Desertification	Degradation in arid–semi-arid–dry sub-humid (ASDM)
≠ expanding desert only	Any dryland productivity loss can count
India degraded land	~**96.4 mha**; ~**29–30%** (ISRO Atlas 2016)
Water erosion sequence	Sheet → rill → gully → ravine (Chambal)
Wind tools	Deflation + abrasion
UNCCD / Day	**1994** / **17 June**
LDN	SDG **15.3**; no net loss by **2030**
DPAP / DDP / IWMP	**1973** / **1977–78** / **2009**

Chapter 14: Environmental Impact Assessment

🎯 Consolidated Must-Score Facts

1. **EIA** studies a project's likely environmental effects **before** approval. The aim is anticipatory prevention, not reactive cleanup.
2. Legal base is the **Environment (पर्यावरण) (Protection) Act, 1986** (Section **3**). First EIA Notification was **1994**; the current framework is **EIA Notification 2006 (14 September)**.
3. **Environmental Clearance** is mandatory prior clearance for listed projects.
4. **Category A** is appraised centrally by **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) through the **EAC**.
5. **Category B** is appraised by **SEIAA** through **SEAC**. **B1** needs full EIA (usually with public hearing); **B2** is simplified.
6. Under a **General Condition**, a Category B project in a sensitive area is treated as Category A.
7. The four (चातुर्याम) classic stages are **screening → scoping → public consultation → appraisal**. Screening applies to Category B only.
8. **Scoping** fixes the **Terms of Reference (ToR)** (संदर्भ की शर्तें) for the EIA study.
9. An EIA report covers baseline, impact prediction, mitigation, and an **Environmental Management Plan (EMP)** (पर्यावरण प्रबंधन योजना).
10. **Rapid EIA** uses one season; **comprehensive EIA** covers all seasons.
11. MoEFCC/SEIAA **grant** EC. EAC/SEAC only **recommend** (सिफारिश). CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड)/SPCB set standards and monitor — they do **not** grant EC.
12. **NGT (Act 2010)** hears EC and pollution appeals. **PARIVESH** is MoEFCC's online EC portal.
13. **SEA** assesses policies, plans, and programmes upstream of projects. India has project-level EIA law but **no standalone SEA statute**.
14. SEA references include the Kiev SEA Protocol **2003** (Espoo family) and the EU SEA Directive.
15. Rio Principle **17** supports EIA for projects with significant impacts.
16. EC must come **before** construction. Ex-post facto clearance is illegal in the standard line.
17. Timeline: Bhopal **1984** → EPA **1986** → EIA Notification **1994** → EIA Notification **2006** → NGT Act **2010** → draft EIA amendment debates **2020** (not a full replacement of 2006).
18. UP examples: Ganga (गंगा) Expressway as Category A; Singrauli TPP expansions; Upper Ganga ESZ plan-level restrictions before project EC.
19. EIA is project-level; SEA is plan/policy-level — do not merge the acronyms.
20. Water Act **1974**/Air Act **1981** create consent regimes; they are **not** the EIA enabling Act (**EPA 1986** (पर्यावरण संरक्षण 1986) is).
21. DEIAA appears in the minor-minerals / B2 district track — not as the Category A central authority.
22. Public consultation includes hearing plus written comments; skipping it on B1 where required is a process defect.
23. EMP is mitigation and monitoring inside the EIA package — not a substitute for the EC grant itself.
24. PARIVESH tracks applications; it does not replace NGT's appellate role.
25. **Environmental Impact Assessment (EIA)** is a systematic study of a proposed project's likely environmental, social, and health effects **before** major decisions are taken.
26. **Environmental Clearance (EC)** is mandatory for **listed projects** in the Schedule before construction or land preparation.
27. **Category A** projects are large, high-impact projects appraised centrally by **MoEFCC** through the **Expert Appraisal Committee (EAC)**.
28. **Category B** projects are appraised by the **State Environment Impact Assessment Authority (SEIAA)** through the **State Expert Appraisal Committee (SEAC)**.
29. **Screening** applies to **Category B only** and splits projects into **B1** (full EIA required) or **B2** (no full EIA report).
30. **Rapid EIA** uses one season of baseline data. **Comprehensive EIA** covers all seasons and is more reliable for monsoon-dry variation.
31. **CPCB and SPCBs** set pollution standards and monitor compliance. They **do not grant EC**.
32. **PARIVESH** is MoEFCC's online single-window portal for EC applications and tracking.
33. **NGT (National Green Tribunal)**, under the **NGT Act 2010**, hears appeals against EC decisions.
34. **Strategic Environmental Assessment (SEA)** evaluates the environmental consequences of **policies, plans, and programmes (PPP)** at the earliest planning stage.
35. **SEA sits upstream of EIA**. It asks whether a development path is wise **before** individual projects are designed.
36. **EIA** can only accept, modify, or reject **one pre-selected project**. It cannot question whether an entire sector plan — such as many dams in one basin — is environmentally sound.
37. **SEA assesses cumulative impacts** when multiple projects together may overwhelm an ecosystem (पारिस्थितिकी तंत्र) even if each passes individual EIA.
38. **Public participation** in SEA happens when policy choices are still flexible — earlier and cheaper than late-stage project hearings.
39. **Rio Declaration (1992) Principle 17** calls for **EIA** for proposed activities with significant adverse impact. It does **not** mandate SEA by name.
40. **Espoo Convention (1991)** covers **transboundary EIA**. The **SEA Protocol (2003)** signed at **Kiev** extends assessment to plans and programmes.
41. **India has robust project-level EIA** under the 2006 Notification but **no standalone SEA legislation** as of current law.

📊 Quick Reference Grid

Framework

ITEM	LOCK
Legal base	EPA **1986**; Notification **1994** → **2006** frame
Category A / B	MoEFCC–EAC central / SEIAA–SEAC state
B1 / B2	Full EIA (+ hearing usual) / simplified
General Condition	Sensitive-area B treated as A
Four stages	Screening → scoping → public consultation → appraisal
Screening	Category **B** only
Rapid vs comprehensive	One season / all seasons
EC granter	MoEFCC/SEIAA (EAC/SEAC recommend only)
NGT / PARIVESH	Act **2010** appeals / MoEFCC online portal
SEA	Policies–plans–programmes; India has no standalone SEA statute


EIA Notification 2006 Statutory Stages & Categorization Architecture

Chapter 15: Sustainable Development & Environmental Governance

🎯 Consolidated Must-Score Facts

1. **Sustainable development** (Brundtland **1987**, *Our Common Future*) means meeting present needs without compromising future generations' ability to meet theirs.
2. Brundtland is **not** *Limits to Growth* (Club of Rome, **1972**).
3. SD rests on **economic, social, and environmental** pillars (plus institutional capacity). It is not environment-only.
4. Core objectives include basic needs, intra- and inter-generational equity, ecological limits, participation, and integrated planning.
5. Priority objectives also include **poverty reduction**, **biodiversity conservation**, and **pollution control**.
6. Indicators include **HDI (UNDP (यूएनडीपी))**, ecological footprint (पारिस्थितिक पदचिह्न), NITI (नीति) Aayog (नीति आयोग) SDG Index, global SDG indicator sets, and EPI.
7. **Natural capital** means renewable and non-renewable environmental resources (air, water, land, minerals, forests). **Roads** are not natural capital.
8. **SDGs** were adopted (अंगीकृत) in **September 2015**, run to **2030**, and include **17 goals** and **169 targets**.
9. SDGs replaced **MDGs** (8 goals, **2000–2015**) and apply universally to all countries.
10. Goal facts worth ratta include SDG **6** water, **7** energy, **13** climate, **14** life below water, and **15** life on land.
11. Global SDG Index ranks often place **Finland** first. India was about **116th (2017)** and about **112th (2023)**.
12. NITI Aayog **SDG India Index** has repeatedly placed **Kerala** first; Himachal (हिमाचल) Pradesh and Tamil Nadu (तमिलनाडु) sit near the top. **Gujarat** was **not** in the 2019–20 top-five set often tested.
13. Conference chain: **Stockholm 1972 (स्टॉकहोम सम्मेलन) → Brundtland 1987 → Rio and Agenda 21 in 1992 → SDGs 2015**.
14. **Agenda 21** is Rio **1992**, not 1995.
15. **LiFE Mission** launched in **June 2022**; the idea was floated at **COP26 Glasgow 2021**, not COP25.
16. MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is the nodal environment (पर्यावरण) ministry. **NGT Act 2010** creates the green tribunal.
17. Rio principles frequently tested are **Precautionary Principle, Polluter Pays, and Public Trust**.
18. Constitutional hooks are **Article 48A**, **Article 51A(g)**, and **Article 21** (अनुच्छेद 21) read with a healthy environment.
19. UP SDG reporting typically sits with the **Planning Department**.
20. Environmental justice means fair distribution of environmental benefits and burdens plus fair participation in decisions.
21. Bhopal **1984** led to EPA **1986**. UP justice/pollution examples include Kanpur (कानपुर) tanneries and the Singrauli coal belt.
22. Ethics labels: **anthropocentrism** (human-centred), **biocentrism** (all life), **ecocentrism** (ecosystems). Leopold's Land Ethic is **1949**; deep ecology (पारिस्थितिकी) links to **Arne Næss**.
23. Gandhi (गांधी)'s **trusteeship** is a frequent Indian environmental-ethics fact.
24. **Green economy** (UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) **2011** line) means low-carbon, resource-efficient, socially inclusive growth.
25. **Circular economy** redesigns take-make-dispose into reduce-reuse-recycle loops, often with **EPR** rules. Circular sits inside, but is not identical to, green economy.
26. **Sustainable agriculture** means using land so its **quality remains intact**, with mixed cropping, organic manures, nitrogen-fixing plants, and pest-resistant varieties as the environment-friendly strategy set.
27. **NMSA** (National Mission for Sustainable Agriculture) is a **NAPCC** mission on productivity, water-use efficiency, nutrient management, and livelihood diversification.
28. Example scheme–SDG matches: Jal Jeevan / Swachh Bharat → SDG **6**; SAUBHAGYA / UJALA → SDG **7**; NAPCC / Net Zero 2070 → SDG **13**.
29. MDGs had 8 goals; SDGs have 17 — do not reverse the counts.
30. HDI is an UNDP development indicator; it is not itself an MoEFCC clearance tool.
31. Polluter Pays allocates cleanup cost to the polluter; Precautionary Principle acts under scientific uncertainty — different tools.
32. Public Trust treats certain natural resources as held for public benefit, not absolute (निरपेक्ष) private exhaustion.
33. **Sustainable development** means meeting the needs of the present **without compromising** the ability of future generations to meet their own needs.
34. **Inter-generational equity** protects future generations. **Intra-generational equity** ensures fair sharing among people alive today.
35. **Objectives** include fulfilling **basic needs**, respecting **ecological limits** and carrying capacity, integrating environment into economic planning, and ensuring **public participation**.
36. **Natural capital** includes all renewable and non-renewable environmental resources such as air, water, land, minerals, and forests. **Roads** and other purely human-made assets are **not** natural capital.
37. **Indicators** include **HDI** (UNDP), **ecological footprint** (पारिस्थितिक पदचिह्न), **SDG Index**, **Environmental Performance Index (EPI)**, and **231 SDG indicators**.
38. **NITI Aayog** (नीति आयोग) publishes the **SDG India Index** ranking states on SDG progress.
39. **SDGs replaced MDGs**, which had **8 goals (2000–2015)** and focused mainly on developing countries.
40. **SDGs are universal** — they apply to all countries, including developed ones.
41. **Leave No One Behind** requires progress to reach the poorest and most vulnerable groups.
42. **NITI Aayog** coordinates SDG localisation in India. **MoSPI** maintains the National Indicator Framework.
43. **NITI Aayog's SDG India Index** ranks states and UTs. **Kerala** has repeatedly held the **first** position (including the 2019–20 and 2020–21 editions).
44. **Environmental governance** is the system of rules, institutions, and processes through which environmental decisions are made and enforced.

45. **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is India's nodal ministry for environment, forest, and climate policy.
46. **Rio principles** include **Precautionary principle**, **Polluter Pays**, **Public Trust doctrine**, and **inter-generational equity** — adopted in Indian judgments like *Vellore Citizens' Welfare Forum*.
47. **NGT (2010)** is the specialised tribunal for environmental cases and EC appeals.
48. **Agenda 21 (1992)** is a Rio blueprint for sustainable development planning, including local-level action.
49. **Environmental justice** means fair treatment of all in environmental benefits and burdens — both **distributive** (who gets pollution) and **procedural** (who participates in decisions).
50. **Environmental ethics** asks what moral duties humans owe to nature.

📊 Quick Reference Grid

Spine

ITEM	LOCK
Brundtland	**1987** *Our Common Future* (≠ Limits to Growth 1972)
Three pillars	Economic · social · environmental
SDGs	**Sep 2015** → **2030**; **17** goals / **169** targets
MDGs → SDGs	8 goals 2000–15 / universal to all countries
Key goals	6 water · 7 energy · 13 climate · 14 ocean · 15 land
Conference chain	Stockholm 72 → Brundtland 87 → Rio 92 → SDGs 15
Agenda 21	Rio **1992** (≠ 1995)
NITI SDG Index top	Kerala repeatedly first; HP/TN near top

Chapter 16: Environmental Organizations (India)

🎯 Consolidated Must-Score Facts

1. **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is India's nodal ministry for environment (पर्यावरण), forests, wildlife, and climate change (जलवायु परिवर्तन). It began as MoEF in **1985** and was renamed MoEFCC in **2014**. Headquarters is **New Delhi**.
2. MoEFCC grants **Category A Environmental Clearance**, runs the **PARIVESH** portal, and administers EPA, Wildlife Act, Forest Conservation Act, and Biodiversity Act. It does **not** set NAAQS pollution standards.
3. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) was created under the **Water Act, 1974**, sits in **New Delhi**, sets **NAAQS**, and coordinates State Pollution Control Boards. The Air Act **1981** widened its air-pollution role.
4. CPCB does **not** grant Environmental Clearance, does **not** publish ISFR, and does **not** prepare the Annual Groundwater Quality Report — that report belongs to **CGWB**.
5. Every state has an **SPCB**. SPCBs issue **Consent to Establish** and **Consent to Operate** under the Water and Air Acts. They do **not** grant Category B EC — that is **SEIAA**.
6. **UPPCB** is Uttar Pradesh (उत्तर प्रदेश)'s board with headquarters at **Lucknow** (लखनऊ). It enforces norms on Kanpur (कानपुर) tanneries, Noida–Ghaziabad industry, and Ganga (गंगा)–Yamuna (यमुना) discharges.
7. **NGT** was created by the **NGT Act, 2010** (operational **2011**). Headquarters is New Delhi, with regional benches at Bhopal, Pune (पुणे), Kolkata, and Chennai.
8. NGT hears environmental appeals, including EC appeals, and can award compensation under Polluter Pays. The Act aims for disposal within **six months**. NGT does **not** itself grant EC.
9. The NGT Chairperson must be a retired **Supreme Court** judge or a High Court Chief Justice.
10. **WII** (**1982**, Dehradun (देहरादून)) does wildlife research and trains forest officers. It helped frame India's **ten** biogeographic zones and tiger-census methods. It is **not** the plant or animal survey body.
11. **BSI** (**1890**, Kolkata) surveys **plants** and publishes *Flora (वनस्पति) of India* and the plant Red Data Book (रेड डेटा बुक). **ZSI** (**1916**, Kolkata) surveys **animals** and publishes *Fauna (प्राणीजात) of India* and the animal Red Data Book.
12. **FSI** (**1981**, Dehradun) maps forest cover (वन आवरण) and publishes the **India State of Forest Report** every **two years**. ISFR is **not** an annual CPCB product.
13. **NEERI** (**1958**, **Nagpur**) began as **CPHERI** and is a CSIR laboratory for environmental science and engineering under the Ministry of Science and Technology.
14. Headquarters fact: **Delhi** — MoEFCC, CPCB, NGT; **Dehradun** (देहरादून) — WII, FSI; **Kolkata** — BSI, ZSI; **Nagpur** — NEERI; **Lucknow** — UPPCB.
15. Category A EC → **MoEFCC**. Category B EC → **SEIAA**. Industry consent → **SPCB**. EC appeal → **NGT**.
16. Pollution standards → **CPCB**. Forest cover report → **FSI**. Plant taxonomy → **BSI**. Animal taxonomy → **ZSI**. Wildlife training → **WII**. Environmental engineering research → **NEERI**.
17. MoEFCC was **not** created in **1972**. That year marks the Stockholm Conference and the Wildlife (Protection) Act.
18. Consent under Water/Air Acts is **not** the same as Environmental Clearance under EPA/EIA rules. Confusing CTE/CTO with EC is a classic trap.
19. Forest clearance for diversion of forest land sits with **MoEFCC** under the Forest (Conservation) Act — not with CPCB or NGT.
20. BSI and ZSI both sit in Kolkata, but BSI is plants only and ZSI is animals only. Swapping kingdoms is a frequent wrong match.
21. WII and FSI both sit in Dehradun, but WII is wildlife science/training and FSI is forest-cover assessment. Do not give ISFR to WII.
22. NGT adjudicates; it does **not** replace CPCB on standards or MoEFCC on clearances. The older **NEAA (1997)** was dissolved after NGT was set up.
23. SPCBs work under CPCB technical guidance and state administrative control. They enforce standards; they do not write national NAAQS.
24. PARIVESH is MoEFCC's online EC application system — not a CPCB monitoring portal.
25. Match report to publisher: ISFR → **FSI**; Groundwater Quality Report → **CGWB**; NAAQS bulletins → **CPCB**; *Flora* → **BSI**; *Fauna* → **ZSI**.
26. NEERI is **Nagpur 1958**, not New Delhi 1970 and not a pollution-standards board.
27. **CAZRI = Jodhpur**; **IIFM = Bhopal**; operational Antarctica (भारती) stations = **Maitri + Bharati** (मैत्री).
28. **INCOIS = Hyderabad (हैदराबाद)**; ENVIS population (जनसंख्या)–environment centre = **IIPS Mumbai**.
29. **MoEFCC** is India's **nodal ministry** for environment, forests, wildlife, and climate change policy.
30. **Headquarters** is **New Delhi**.
31. **CPCB** was established in **1974** under the **Water (Prevention and Control of Pollution) Act, 1974**.
32. **UPPCB** is Uttar Pradesh's board with HQ at **Lucknow**.
33. **NGT** was established under the **National Green Tribunal Act, 2010** and became operational in **2011**.
34. **Headquarters** is **New Delhi** with regional benches at **Bhopal, Pune, Kolkata, and Chennai**.
35. **WII** was established in **1982** at **Dehradun** under **MoEFCC**.
36. **BSI** was established in **1890** and is headquartered at **Kolkata**.
37. **ZSI** was established in **1916** and is headquartered at **Kolkata**.
38. **FSI** was established in **1981** at **Dehradun** under **MoEFCC**.
39. **NEERI** is India's pioneering CSIR laboratory for **environmental science and engineering**.
40. **Wildlife Institute of India** — **Dehradun**. **Forest Research Institute** — **Dehradun**. **Indian Institute of Forest Management** — **Bhopal** (not Kolkata).
41. **Central Arid Zone Research Institute (CAZRI)** — **Jodhpur** (ICAR; arid-zone research).
42. **Indian Grassland and Fodder Research Institute** — **Jhansi** (झांसी). **IARI** — **New Delhi**.
43. **National Institute of Nutrition** — **Hyderabad**. **National Institute of Ayurveda** — **Jaipur**. **National Institute of Naturopathy** — **Pune**.

44. **Indian Institute of Ecology (पारिस्थितिकी) and Environment** — **New Delhi** (founded **5 June 1980** teaching).
45. **ENVIS Centre on Population and Environment** — **IIPS Mumbai**.
46. **Indian National Centre for Ocean Information Services (INCOIS)** — **Hyderabad** (1999; Ministry of Earth Sciences).
47. **G.B. Pant Institute of Himalayan Environment and Development** — **Almora** (अल्मोड़ा). **Centre for Ecological Sciences** — **Bengaluru**.
48. **Kailash Sacred Landscape Conservation and Development Initiative (KSLCDI)** partners: **India, China, Nepal**.

📊 Quick Reference Grid

HQ map

ITEM	LOCK
MoEF → MoEFCC	**1985** → **2014**; Delhi
CPCB	Water Act **1974**; Delhi; NAAQS; coordinates SPCBs
UPPCB	Lucknow
NGT	Act **2010** (ops 2011); Delhi + Bhopal/Pune/Kolkata/Chennai
NGT Chair	Retired SC judge **or** HC Chief Justice
WII / FSI	**1982** / **1981**; both **Dehradun**
BSI / ZSI	**1890** / **1916**; both **Kolkata**
NEERI	**1958** Nagpur; CSIR (≠ MoEFCC)

Chapter 17: Environmental Laws & Policies

🎯 Consolidated Must-Score Facts

1. Memorise the act-year chain: **WPA 1972 → Water 1974 → FCA 1980 → Air 1981 → EPA 1986 → BDA 2002 → FRA 2006 → NGT Act 2010**.
2. **Article 48A** (DPSP (राज्य के नीति निदेशक तत्व)) directs the State to protect the environment (पर्यावरण). **Article 51A(g)** is the citizen's Fundamental Duty to protect nature. Do not swap them.
3. **Article 21** (अनुच्छेद 21) has been read to include a healthy environment. **Article 48** is about agriculture and cattle — it is **not** the environment article (**48A** is).
4. The **Environment (Protection) Act, 1986** is the post-Bhopal umbrella law under **Article 253**. Section **3** powers support EIA, CRZ, ESZ, **GEAC**, and **CGWA**. It **supplements** Water and Air Acts; it does not repeal them.
5. EIA Notification and related clearances flow from **EPA**, not from the Water Act.
6. The **Wildlife (Protection) Act, 1972** uses **six** Schedules. Schedules **I–IV** protect animals; Schedule **V** is vermin; Schedule **VI** covers plants. **Schedule I** is the highest protection; Schedule **V** animals may be hunted as vermin.
7. Under WPA, a **National Park** (Section **35**) is stricter than a **Sanctuary** (Section **18**). Tiger Reserves rest on Section **38V** (2006 amendment path).
8. The **Forest (Conservation) Act, 1980** requires prior **Central** approval before diverting forest land. CAMPA and NPV sit in this clearance world.
9. The **Forest Rights Act, 2006** recognises rights of Scheduled Tribes and Other Traditional Forest Dwellers. Individual forest rights are capped around **4 hectares**, with a cutoff of **13 December 2005**, and recognition (मान्यता) begins at the **Gram Sabha** (ग्राम सभा).
10. **FCA** controls **diversion** of forest land. **FRA** recognises **rights** of forest dwellers. Do not treat them as one Act.
11. The **Biological Diversity Act, 2002** implements the CBD through **NBA** (national), **SBB** (state), and **BMC** (local) with People's Biodiversity Registers.
12. The **Water Act, 1974** created **CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड)/SPCB** and the consent regime. The **Air Act, 1981** covers air pollution and NAAQS. Water is **1974**; Air is **1981** — do not reverse the years.
13. Industry **consent** (CTE/CTO) is under Water/Air Acts via **SPCB**. **Environmental Clearance** is under EPA/EIA via **MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय)/SEIAA**. Forest clearance is under **FCA** via MoEFCC.
14. The **NGT Act, 2010** creates the green tribunal for environmental appeals and compensation, with a six-month disposal target.
15. National Forest Policy **1952** was revenue-leaning. Later practice follows National Forest Policy **1988** targets of about **33%** forest cover (वन आवरण) nationally and **66%** in hills.
16. EPA is **1986**, not 1972. WPA is **1972**, not 1986. Mixing those years is a standard trap.
17. Schedule **V** under WPA is **not** the highest protection tier — Schedule **I** is.
18. Consent is not EC, and EC is not forest clearance. Keep the three tools on separate legal tracks.
19. FRA's nodal line is tribal (आदिवासी) forest rights, not MoEFCC forest diversion under FCA.
20. BDA is about biodiversity access and benefit-sharing institutions — not the same as WPA schedules or NGT appeals.
21. Article **48A** binds the State as a Directive Principle; **51A(g)** binds the citizen as a duty. Both environment hooks sit beside Article **21** case law.
22. Water Act creates the pollution-board architecture; EPA is the later umbrella for broader environmental regulation including EIA.
23. Air Act year fact is **1981**. Pairing Air with **1974** or Water with **1981** is wrong.
24. NGT hears appeals; it does not rewrite the year of EPA or replace CPCB's standards role.
25. CAMPA/NPV issues arise after forest diversion under FCA — they are not FRA Gram Sabha rights tools.
26. Match clearance labels carefully: **SPCB consent**, **MoEFCC/SEIAA EC**, **MoEFCC forest clearance**.
27. Wildlife and forests sit on the **Concurrent List** (समवर्ती सूची) — Centre and states both legislate.
28. Indian courts have adopted Rio principles — **Precautionary**, **Polluter Pays**, and **Public Trust** (*Vellore Citizens' Welfare Forum*).
29. **Environmental law** in India combines **constitutional duties**, **statutory Acts**, and **Supreme Court/NGT jurisprudence**.
30. **Article 48A** (DPSP) directs the **State** to protect and improve the environment and safeguard forests and wildlife.
31. **Article 51A(g)** (Fundamental Duty) requires every **citizen** to protect and improve the natural environment.
32. **Article 21** (Right to Life) was expanded by the Supreme Court to include the **right to a wholesome environment** (*M.C. Mehta*, *Subhash Kumar*).
33. **Article 48** deals with **agriculture and animal husbandry** — it is **not** the environment article. Do not confuse with **48A**.
34. **Section 6** empowers the Centre to make **rules** for environmental protection.
35. **Environmental Clearance** under EIA is distinct from SPCB **consent** under Water/Air Acts. Large projects may need **both**.
36. **CGWA** (regulatory authority under EPA) is **not** the same as **CGWB** (assessment board under Jal Shakti (शक्ति)).
37. **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is the nodal ministry for EPA implementation.
38. **Six Schedules** classify species by protection level.
39. **Hunting** is banned for Schedule **I–IV** species with narrow exceptions for research or public safety with permission.
40. **National Park (Section 35)** has the **strictest** protection — no grazing or exploitation.
41. **Wildlife Sanctuary (Section 18)** allows more regulated human use under the Chief Wildlife Warden.
42. **Tiger Reserves** are notified under **Section 38V** (2006 amendment) with **NTCA** oversight.
43. **Prior Central Government approval** is mandatory before using **forest land** for **non-forest purposes** like mining, dams, or industry.
44. **Community Forest Resource (CFR)** (सामुदायिक वन संसाधन अधिकार) rights allow communities to manage and protect forest resources.
45. **State Biodiversity Boards (SBB)** operate at state level.
46. **Biodiversity Management Committees (BMC)** function at local panchayat level.

47. **People's Biodiversity Registers (PBR)** document local biodiversity.

Chronological Sequence of Environmental Legislation & Governance in India

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1972 ⟶ Wildlife (Protection) Act (WPA) ➔ Apex PA statute, hunting prohibition, statutory schedules
1974 ⟶ Water (Prevention & Control of Pollution) Act ➔ Established Central & State Pollution Control Boards (CPCB/SPCB)
1980 ⟶ Forest (Conservation) Act (FCA) ➔ Mandatory prior central approval for diversion of forest land
1981 ⟶ Air (Prevention & Control of Pollution) Act ➔ Air pollution control; noise included via 1987 amendment
1986 ⟶ Environment (Protection) Act (EPA) ➔ Post-Bhopal umbrella statute under Art 253; EIA, CRZ, ESZ notifications
1991 ⟶ Public Liability Insurance Act ➔ Mandatory insurance for handling hazardous substances
2002 ⟶ Biological Diversity Act (BDA) ➔ 3-tier decentralized framework: NBA (Chennai) -> SBB -> BMC
2006 ⟶ Scheduled Tribes & Other Traditional Forest Dwellers (FRA) Act ➔ Individual & community forest rights
2010 ⟶ National Green Tribunal (NGT) Act ➔ Specialized judicial body for environmental disputes under Article 21
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📊 Quick Reference Grid

Act chain

ITEM	LOCK
Spine	WPA 72 → Water 74 → FCA 80 → Air 81 → EPA 86 → BDA 02 → FRA 06 → NGT 10
48A / 51A(g) / 48	State DPSP env / duty / agri–cattle (**not** env)
EPA umbrella	Post-Bhopal; Art. 253; EIA/CRZ/ESZ/GEAC power
WPA Schedules	I–IV animals; V vermin; VI plants
NP vs Sanctuary	Sec. 35 stricter / Sec. 18
FCA vs FRA	Central diversion approval / forest-dweller rights
BDA ladder	NBA → SBB → BMC + PBR
Consent vs EC vs forest	SPCB Water/Air / MoEFCC–SEIAA EPA / FCA Central


India Environmental Legal Architecture & Statutory Chronology

Chapter 18: International Environmental Agreements & Conferences

🎯 Consolidated Must-Score Facts

1. Conference spine: **Stockholm 1972 → Brundtland 1987 → Rio 1992 → Johannesburg 2002 → Rio+20 2012 → SDGs and Paris 2015**.
2. **Stockholm 1972** (स्टॉकहोम सम्मेलन) was the first UN Conference on the Human Environment (पर्यावरण). It led to **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) (Nairobi). World Environment Day (विश्व पर्यावरण दिवस) is **5 June**. The slogan fact is "Only One Earth."
3. The **Brundtland Report** (**1987**, *Our Common Future*) popularised **sustainable development**. It is **not** *Limits to Growth* (**1972**, Club of Rome).
4. **Rio 1992** (UNCED) produced the Rio Declaration, **Agenda 21**, Forest Principles, and opened **UNFCCC** (यूएनएफसीसीसी) and **CBD** for signature. **CITES** is **not** a Rio product — CITES is **1973**.
5. **Agenda 21** is Rio's non-binding sustainable-development action plan for the twenty-first century. It was **not** signed in 1995.
6. Climate chain: **UNFCCC 1992 → Kyoto Protocol 1997 → Paris Agreement (पेरिस समझौता) 2015**. Paris uses universal **NDCs** and the **1.5°C** ambition language.
7. Ozone chain: **Vienna Convention 1985 → Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) 1987 → Kigali Amendment 2016** (HFCs). Montreal is **ozone**, not climate.
8. Treaty–issue fact: Montreal/Vienna = ozone; Kyoto/Paris/UNFCCC = climate; CBD = biodiversity; Ramsar = wetlands; CITES = wildlife **trade** (पण्याध्यक्ष).
9. **CMS (Bonn)** covers migratory species. **Basel** covers hazardous **waste**. **Rotterdam** covers hazardous **chemicals** (PIC). **Stockholm POPs** covers persistent organic pollutants. **Minamata** covers **mercury**.
10. **CBD** has three pillars: conserve biodiversity, sustainable use, and fair benefit-sharing. **Nagoya Protocol (2010)** details access and benefit-sharing.
11. **Ramsar** (रामसर) (**1971**) is the wetlands treaty under the Wise Use idea. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).
12. **CITES** (**1973**, Washington) regulates trade through Appendices **I–III**. IUCN (आईयूसीएन)'s Red List assesses extinction risk; CITES regulates trade — different jobs.
13. **GEF** was established in **1991** on the eve of Rio as a finance mechanism for major environmental conventions.
14. **UNCCD (1994)** (मरुस्थलीकरण रोकथाम संधि) is the sole legally binding convention linking environment, development, and sustainable land management; it stresses bottom-up local participation against desertification (मरुस्थलीकरण).
15. The **New York Declaration on Forests (2014)** is a voluntary, non-legally binding political declaration (halve forest loss by 2020 / end by 2030 ambition language); India was **not** an original signatory.
16. Kyoto set binding targets mainly for developed parties. Paris is a universal NDC architecture. Do not treat them as identical instruments.
17. Kigali amends Montreal to phase down **HFCs**. It is an ozone-protocol amendment path, not a UNFCCC protocol.
18. Basel is waste shipment control. Rotterdam is prior informed consent for chemicals. Stockholm POPs is about persistent toxics such as listed DDT/PCB classes. Minamata is mercury.
19. CMS is migratory species; CITES is trade; Ramsar is wetlands. Mixing those three is a classic match trap.
20. UNEP was born from Stockholm, not from Rio. Rio opened UNFCCC and CBD.
21. Paris is **2015**; Kyoto is **1997**; UNFCCC framework is **1992**. Keep the climate chronology straight.
22. Montreal successful ODS phase-out is often contrasted with slower climate politics — but stems still demand the correct issue tag, not slogans.
23. Rudrasagar is a **Tripura** Ramsar site. Sultanpur is **Haryana**. Neither is an Uttar Pradesh (उत्तर प्रदेश) site.
24. Uttar Pradesh Ramsar examples include sites such as Nawabganj, Samaspur, Sarsai Nawar, Upper Ganga (गंगा), Bakhira, and Haiderpur — verify the live list before the paper, but never park Rudrasagar or Sultanpur in UP.
25. CBD opened at Rio **1992**; Nagoya ABS is **2010**. Do not date CBD as 2010.
26. Forest Principles at Rio are non-binding soft law — not the same as CITES appendices.
27. Johannesburg **2002** and Rio+20 **2012** continue the sustainable-development conference line after Rio; they do not reopen CITES as a 1992 invention.
28. "Only One Earth" and UNEP creation fact to Stockholm **1972**, not to Paris **2015**.
29. Wise Use is the Ramsar philosophy. It is not the CBD's three objectives word-for-word.
30. Master year table worth ratta: Ramsar 1971, CITES 1973, Vienna 1985, Montreal 1987, Basel 1989, UNFCCC/CBD 1992, UNCCD 1994, Kyoto 1997, Rotterdam 1998, Stockholm POPs 2001, Nagoya 2010, Minamata 2013, New York Forests Declaration 2014, Paris 2015, Kigali 2016.
31. **World Environment Day** is celebrated on **5 June** every year, marking the Stockholm opening day.
32. **Agenda 21** is a non-binding action plan for **sustainable development in the 21st century** — adopted at **Rio 1992**, not 1995.
33. **Local Agenda 21** encourages cities and local bodies to prepare their own SD plans.
34. **Earth Summit +5** was the UN General Assembly special session (पंगु सत्र) held in **New York in June 1997** to review Rio progress.
35. **Johannesburg Summit (2002)** was Rio+10 on sustainable development and reaffirmed Agenda 21.
36. **Rio+20 (2012)** — the United Nations Conference on Sustainable Development — reviewed SD progress and produced the outcome document ***The Future We Want***.
37. **PAGE (Partnership for Action on Green Economy (हरित अर्थव्यवस्था))** was created as the UN response to Rio+20's call for green-economy transitions.
38. **SDGs were adopted in 2015** at the UN General Assembly.
39. **Earth Hour** (अर्थ आवर) is a **WWF** (डब्ल्यूडब्ल्यूएफ) climate-awareness event (usually last Saturday of March). **GHG Protocol** = **WRI + WBCSD**.
40. **International Ozone Day** is **16 September** (Montreal signing date).
41. **TRAFFIC** (WWF–IUCN alliance) monitors wildlife trade for sustainability — it is **not** a UNEP bureau.
42. **BirdLife International** identifies Important Bird and Biodiversity Areas; biodiversity **hotspots** were framed by **Norman Myers**, not BirdLife.

43. **CITES (1973, Washington)** regulates **international trade** in endangered wild fauna (प्राणीजात) and flora (वनस्पति).
44. **Appendix I** — trade banned except in exceptional cases.
45. **Appendix II** — regulated trade with permits.
46. **Appendix III** — species listed by a member country requesting cooperation.
47. **World Wetlands Day** is **2 February**.
48. **CBD (1992)** is the biodiversity convention. **Cartagena** is the biosafety protocol under CBD.
49. **Stockholm Convention (2001)** (स्टॉकहोम संधि) covers **POPs** such as DDT and PCBs. It is **not** the Stockholm Conference of **1972**.
50. UNFCCC was opened at **Rio 1992**. Its secretariat sits in **Bonn**. Annual COP meetings drive negotiations.
51. Ramsar "wise use" allows sustainable human use of wetlands — it is not a total ban slogan.
52. A stem that says Kyoto is about "saving water" is false. Kyoto is climate.
53. Stockholm **1972** created the UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) moment. Stockholm **2001** is the POPs treaty. Keep the two Stockholms apart.
54. CBD's three objectives are conservation, sustainable use, and fair benefit-sharing from genetic resources.
55. CMS protects migratory animals across borders. CITES controls international commercial trade in listed species.
56. Minamata's mercury focus includes artisanal gold mining and product phase-downs in teaching notes.
57. Do not put Ramsar, CITES, or CBD into the climate-mitigation answer slot when Kyoto is present.
58. UP Ramsar geography traps belong beside Ramsar treaty facts but do not change the 1971 identity.
59. Bonn Convention means **CMS**, not the UNFCCC Bonn secretariat alone.
60. **Montreux Record** = Ramsar sites with adverse / likely adverse ecological change from human interference.
61. **World Wildlife Day (विश्व वन्यजीव दिवस) = 3 March** (CITES adoption date).

Chronological Sequence of Global Climate Summits & Treaties

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1972 ⟶ UN Conference on Human Environment (Stockholm) ➔ Creation of UNEP (Nairobi) & World Environment Day (5 June)
1979 ⟶ First World Climate Conference (Geneva) ➔ Established World Climate Programme
1987 ⟶ Montreal Protocol on Substances that Deplete the Ozone Layer (Signed 16 Sep; ODS phase-out)
1988 ⟶ IPCC established by UNEP and WMO (Co-founders: Bert Bolin; Headquarters: Geneva)
1992 ⟶ UN Conference on Environment & Development (Rio Earth Summit) ➔ Rio Declaration, Agenda 21, UNFCCC, CBD, UNCCD
1997 ⟶ Kyoto Protocol (COP 3) ➔ Legally binding Annex-I emission reduction targets (Common But Differentiated Responsibilities)
2002 ⟶ World Summit on Sustainable Development (Johannesburg Earth Summit) ➔ Focus on water, energy, health, agriculture
2012 ⟶ Rio+20 Conference (UNCSD, Rio de Janeiro) ➔ "The Future We Want", process to launch SDGs
2015 ⟶ Paris Agreement (COP 21) ➔ Universal pledge-and-review; limit warming to "well below 2°C, pursue 1.5°C"; NDCs
2016 ⟶ Kigali Amendment to Montreal Protocol ➔ Phase-down of climate-warming Hydrofluorocarbons (HFCs)
2021 ⟶ Glasgow Climate Pact (COP 26) ➔ "Phase-down" of unabated coal; launch of LiFE (Lifestyle for Environment) by India
2022 ⟶ Sharm el-Sheikh Implementation Plan (COP 27) ➔ Landmark decision to establish a Loss and Damage Fund
2022 ⟶ Kunming-Montreal Global Biodiversity Framework (CBD COP 15) ➔ 30×30 conservation target by 2030
2023 ⟶ COP 28 (Dubai, UAE) ➔ "UAE Consensus", First Global Stocktake, agreement to "transition away from fossil fuels"
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📊 Quick Reference Grid

Conferences

ITEM	LOCK
Spine	Stockholm 72 → Brundtland 87 → Rio 92 → Joburg 02 → Rio+20 12 → SDGs/Paris 15
Stockholm → UNEP	**1972**; UNEP Nairobi; WED **5 June**
Rio products	Agenda 21 · UNFCCC · CBD (CITES **not** Rio product)
Climate ladder	UNFCCC 92 → Kyoto 97 → Paris 15
Ozone ladder	Vienna 85 → Montreal 87 → Kigali 16

Issue tags

ITEM	LOCK
Montreal / Kyoto–Paris	Ozone / climate
CBD / Ramsar / CITES	Biodiversity / wetlands / wildlife trade
Basel / Rotterdam / Stockholm POPs	Haz. waste / PIC chemicals / POPs
CMS (Bonn)	Migratory species
GEF / UNCCD	**1991** finance / **1994** desertification



Chapter 19: Climate & Environmental Institutions

🎯 Consolidated Must-Score Facts

1. **IPCC** sits in **Geneva**, issues Assessment Reports, and synthesises climate science. It does **not** set NDCs and is **not** the UNFCCC treaty body.
2. IPCC was established in **1988** by **WMO and UNEP** and shared the Nobel Peace Prize in **2007**.
3. **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) headquarters is **Nairobi**. Flagship products include the **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) and GEO. UNEP was created after Stockholm **1972**.
4. UNEP hosts several MEA secretariats (including Montreal, Basel, and Minamata lines) and works with GEF-linked convention finance.
5. **UNDP** headquarters is **New York**. It publishes the **Human Development Report** (from **1990**) and the **HDI**. HDR is UNDP, not UNEP.
6. HDI is linked to **Mahbub ul Haq** and **Amartya Sen**. Do not invent a wrong founding-name trap for HDR.
7. **IUCN** (आईयूसीएन) headquarters is **Gland**. It publishes the **Red List** of extinction risk. IUCN assesses risk; it does **not** regulate wildlife trade (पण्याध्यक्ष) — that is **CITES** (साइट्स).
8. On the Red List, **Threatened** means **Vulnerable + Endangered + Critically Endangered** (VU + EN + CR). "Endangered alone" is not the full threatened set.
9. **WWF** is a conservation NGO also linked to Gland, known for the panda logo and the **Living Planet Report / Living Planet Index**.
10. **FAO** headquarters is **Rome**. Environment-facing outputs include the **State of the World's Forests (SOFO)** and agriculture–forest links. FAO partners UNEP on some chemical-convention work such as Rotterdam.
11. HQ memory: Nairobi = UNEP; New York = UNDP; Geneva = IPCC; Gland = IUCN and WWF; Rome = FAO.
12. IPCC ≠ UNFCCC. Science assessment is not the same as the climate treaty and COP process.
13. UNEP ≠ UNDP. Environment (पर्यावरण) programme versus development agency — swap reports and you lose marks.
14. Emissions Gap Report → **UNEP**. Human Development Report → **UNDP**. Red List → **IUCN**. Living Planet → **WWF**. Assessment Reports → **IPCC**.
15. IUCN ≠ CITES. Risk categories versus trade appendices.
16. WWF ≠ IUCN. NGO Living Planet work versus Red List authority.
17. FAO is food and agriculture first; it is not the Red List body and not the Emissions Gap publisher.
18. GEF (**1991**) finances global environmental conventions; do not confuse GEF with IPCC authorship of Assessment Reports.
19. Red List codes to know: CR, EN, VU as the threatened band; also EX, EW, NT, LC, DD, NE in the wider ladder.
20. UNEP's Green Economy (हरित अर्थव्यवस्था) work and GEO sit in the environment-programme lane — not in UNDP's HDI lane.
21. IPCC reports inform negotiators; parties still adopt NDCs under the UNFCCC/Paris track.
22. Living Planet Index tracks vertebrate abundance trends; it is not a legal trade permit system.
23. SOFO is FAO's forest flagship — not FSI's India State of Forest Report.
24. Match org → city → product before matching slogans. Geography of headquarters is a frequent prelims filter.
25. **Headquarters** is in **Geneva, Switzerland**.
26. **Working Group I** covers physical science. **WG II** covers impacts and adaptation. **WG III** covers mitigation.
27. **UNFCCC** (यूएनएफसीसीसी) is the **treaty body** where countries negotiate **NDCs** at **COP** meetings. IPCC informs but does not decide policy.
28. **UNEP** was created by the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) as the UN's principal environmental authority.
29. **Headquarters** is in **Nairobi, Kenya** — the only major UN environment HQ in the Global South.
30. **GEO (Global Environment Outlook)** is UNEP's comprehensive state-of-environment assessment.
31. **UNDP** is the UN's global **development** network working on poverty, governance, and SDG implementation.
32. **Headquarters** is in **New York**.
33. **HDI** combines **life expectancy**, **education**, and **GNI per capita (PPP)**.
34. **IUCN** is a global conservation union of states and NGOs, founded in **1948**.
35. **Headquarters** is in **Gland, Switzerland**.
36. **Threatened species** (संकटग्रस्त प्रजातियां) means **Vulnerable + Endangered + Critically Endangered (VU + EN + CR)**.
37. **WWF (World Wide Fund for Nature)** is a global conservation **NGO**, not a UN agency.
38. **Headquarters** is also in **Gland, Switzerland** — same city as IUCN but **different organisation and role**.
39. **FAO (Food and Agriculture Organization)** is a UN specialised agency for **food and agriculture**.
40. **Headquarters** is in **Rome, Italy**.

📊 Quick Reference Grid

HQ / product

ITEM	LOCK
IPCC	Geneva; WMO+UNEP **1988**; Nobel Peace **2007**; ARs
UNEP	Nairobi; Emissions Gap + GEO
UNDP	New York; HDR/HDI from **1990**
IUCN	Gland; Red List (risk ≠ CITES trade)
WWF	Gland; Living Planet Report/LPI
FAO	Rome; SOFO
Threatened	VU + EN + CR
IPCC ≠ UNFCCC	Science assessment ≠ climate treaty/COP

Table: Global Environmental Reports & Flagship Publications

PUBLICATION / REPORT	PUBLISHING AGENCY	FREQUENCY / MILESTONE	CORE PRELIMS FOCUS AREA
Assessment Reports (AR1 to AR6)	**IPCC** (Intergovernmental Panel on Climate Change)	Every 6–7 years (AR6 Synthesis: 2023)	Scientific consensus on climate change, carbon budgets, 1.5°C pathways
Emissions Gap Report	**UNEP** (United Nations Environment Programme)	Annual	Gap between national NDC pledges and 1.5°C / 2°C temperature trajectories
Adaptation Gap Report	**UNEP**	Annual	Assessment of global adaptation financing, planning, and implementation deficit
Global Environment Outlook (GEO)	**UNEP**	Periodic flagship	Comprehensive state of the global environment, policy responses
Living Planet Report	**WWF** (World Wide Fund for Nature)	Biennial	Living Planet Index tracking vertebrate population declines (~69% decline)
IUCN Red List of Threatened Species	**IUCN** (Gland, Switzerland)	Continuous / Annual update	9 categories: EX, EW, CR, EN, VU, NT, LC, DD, NE
Global Biodiversity Outlook (GBO)	**CBD Secretariat**	Periodic (GBO-5 in 2020)	Progress against Aichi Biodiversity Targets / Kunming-Montreal GBF
State of the World's Forests (SOFO)	**FAO** (Food and Agriculture Organization)	Biennial	Global forest cover, deforestation rates, agroforestry and livelihoods
World Air Quality Report	**IQAir**	Annual	Global PM2.5 city rankings and WHO guidelines compliance
SDG India Index & Dashboard	**NITI Aayog**	Annual	Performance of Indian States/UTs across 16 SDGs (Kerala consistently #1)

Chapter 20: Biodiversity Conservation Methods

🎯 Consolidated Must-Score Facts

1. **In-situ** (स्थान पर) conservation protects biodiversity **in the natural habitat (वास स्थान)**. It is the preferred primary strategy because the whole ecosystem (पारिस्थितिकी तंत्र) and natural selection stay intact.
2. **Ex-situ** (स्थानांतरित) conservation protects biodiversity **outside** the natural habitat — zoos, botanic gardens, seed banks, gene banks, cryobanks. It is a backup, not a replacement for in-situ.
3. In-situ examples: National Park, Wildlife Sanctuary, Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र), Conservation/Community Reserve (सामुदायिक आरक्षित क्षेत्र), sacred grove, Ramsar (रामसर) wetland, wildlife corridor.
4. Ex-situ examples: zoo, botanical garden, seed bank, gene bank, captive breeding centre, tissue culture, cryopreservation (क्रायो-संरक्षण).
5. Four-way pattern: National Park and sacred grove are **in-situ**; zoo, seed bank, and botanic garden are **ex-situ**.
6. A **gene bank** stores genetic material broadly — seeds, DNA, tissue, pollen, sperm, embryos. A **seed bank** stores **seeds only**.
7. **Orthodox seeds** (सामान्य (रूढ़िवादी) बीज) (wheat, rice) tolerate drying and cold storage around **−18°C**. **Recalcitrant seeds** (कठिन (असहनीय) बीज) (mango, coconut) cannot take simple dry seed-bank storage and need living collections or cryo routes.
8. **Cryopreservation** stores material at about **−196°C** in liquid nitrogen.
9. India's plant gene/seed flagship is **NBPGR**, New Delhi (**1976/1978** line). India's animal gene-bank flagship is **NBAGR**, Karnal (**1984**).
10. The **Indian Botanic Garden**, Howrah (**1787**), is India's classic large botanic garden, linked with BSI administration. The **National Zoological Park**, New Delhi (**1959**), is the national zoo flagship.
11. **CZA** is the statutory zoo regulator under **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय).
12. The **Svalbard Global Seed Vault** in Norway is a duplicate backup vault — not India's NBPGR.
13. NBPGR is plants in Delhi; NBAGR is animals in Karnal. BSI in Kolkata surveys plants; it is **not** the national gene bank.
14. Dudhwa (दुधवा) and Katerniaghat-style protected areas are **in-situ**. Kukrail crocodile centre and garden collections are **ex-situ**.
15. Sacred groves (पवित्र उपवन) are community in-situ pockets — do not park them with zoos.
16. Wildlife corridors keep in-situ populations linked; they are not ex-situ cages.
17. Captive breeding is ex-situ recovery support; releasing only works if habitat (in-situ) remains.
18. Seed banks fail for recalcitrant species if the stem assumes every crop seed stores like wheat.
19. Gene bank ⊃ seed bank. Calling every gene bank a "seeds-only room" is wrong.
20. Biosphere Reserves (जैवमंडल आरक्षित क्षेत्र) and National Parks are in-situ frameworks; botanic gardens are living ex-situ plant collections.
21. UP colour: Dudhwa/Katerniaghat = in-situ; NBRI Lucknow (लखनऊ) garden collections = ex-situ; NBPGR in Delhi still serves national plant genetic resources including UP crops.
22. Community and Conservation Reserves are newer in-situ tools beside classic parks and sanctuaries.
23. Cryobanking is an ex-situ technology for germplasm that ordinary refrigeration cannot hold.
24. Priority line for prelims: **in-situ first, ex-situ supplementary** — never reverse the preference.
25. **In-situ conservation** protects species **where they naturally live**, keeping habitat, food webs, pollinators, and competitors intact.
26. **National Parks** offer strict habitat protection under **Wildlife (Protection) Act, 1972**, with human activity restricted. **Dudhwa NP (UP)** is a Terai (तराई) in-situ example.
27. **Wildlife Sanctuaries** protect habitat with slightly more flexible human use than NPs but remain **in-situ**. **Katerniaghat WLS (UP)** protects gharial and Gangetic dolphin in natural rivers.
28. **Biosphere Reserves** follow the UNESCO **MAB** model with **core**, **buffer**, and **transition** zones, combining strict protection with sustainable community use.
29. **Conservation Reserves** and **Community Reserves** (WPA amendments) protect habitat patches with community participation.
30. **Sacred groves** (deorai, sarna (सरना), kavu) are traditional **in-situ** community forests protected by local belief systems.
31. **Ramsar wetlands** and **marine protected areas** are specialised **in-situ** tools for aquatic and coastal ecosystems.
32. **Wildlife corridors** link fragmented habitats at landscape scale. The **Terai Arc Landscape** connects Dudhwa–Corbett habitats.
33. **Limitation:** In-situ alone cannot save species whose habitat is completely destroyed or whose wild population (जनसंख्या) is critically small without ex-situ backup.
34. **Ex-situ conservation** removes or copies genetic material **outside the natural ecosystem** — zoos, seed banks, gene banks, captive breeding, tissue culture.
35. **When essential:** critically small wild populations, **habitat already lost**, **reintroduction stock** needed (gharial, vulture), or **Extinct in the Wild (EW)** species surviving only in collections.
36. **Major ex-situ types:** zoological parks, botanical gardens, seed banks, gene banks, captive breeding centres, tissue culture, cryopreservation.
37. **Genetic bottleneck risk:** small captive groups lose diversity through inbreeding — zoos rotate breeding pairs and maintain studbooks.
38. **Reintroduction link:** ex-situ breeding succeeds only if **in-situ habitat** is restored. Cheetah reintroduction required **Kuno NP** habitat preparation alongside captive stock.
39. **IUCN (आईयूसीएन) Red List "Extinct in the Wild (EW)"** marks species surviving only in zoos, botanic gardens, or seed banks with no viable wild population.
40. **CBD + BDA 2002:** India regulates access to genetic resources; ex-situ collections at NBPGR/NBAGR support national biodiversity obligations.
41. **ICAR-NBPGR** at **Pusa Campus, New Delhi** (est. **1978**) is India's national **plant** gene bank. It holds one of the world's largest plant collections.
42. **NBAGR, Karnal (Haryana)** (est. **1984**) conserves **animal** genetic resources — semen, embryos, and DNA of indigenous livestock breeds (Sahiwal, Gir (गीर), Red Sindhi).

43. **Cryopreservation** uses **liquid nitrogen at -196°C**, stopping biological activity for decades-long storage of embryos, sperm, pollen, and tissue.
44. **Active collection** holds frequently accessed seeds for breeders. **Base collection** is long-term duplicate storage with minimal disturbance.
45. **Trap: NBPGR vs BSI** — NBPGR **stores** genetic resources (Delhi). **BSI** does plant **survey/taxonomy** (Kolkata).
46. **Trap: NBPGR vs NBAGR** — **NBPGR = plants (Delhi)**; **NBAGR = animals (Karnal)**.
47. **Orthodox seeds** (wheat, rice, maize, millets, pulses) tolerate desiccation and cold — suitable for **decades of storage**.
48. **Recalcitrant seeds** (mango, jackfruit, coconut, sal) **cannot tolerate drying** and need **botanical garden living collections** or **cryopreservation** instead of conventional seed banking.
49. **NBPGR** operates the **National Active Collection** (breeder access) and **National Gene Bank** (long-term base store) with **4 lakh+ accessions** including landraces and crop wild relatives.
50. **Svalbard Global Seed Vault (Norway, 2008)** stores **duplicate copies** of national collections as Arctic "doomsday backup" — not ownership transfer.

📊 Quick Reference Grid

Methods

ITEM	LOCK
In-situ	Natural habitat — preferred primary
Ex-situ	Zoo, botanic garden, seed/gene/cryobank — backup
Gene bank vs seed bank	Broad genetic material / seeds only
Orthodox / Recalcitrant	Dry+cold OK (wheat/rice) / cannot (mango/coconut)
Cryopreservation	~−196°C liquid nitrogen
NBPGR / NBAGR	Plants Delhi / animals Karnal
Indian Botanic Garden	Howrah **1787**; BSI link
Svalbard	Norway backup vault (≠ NBPGR)
CZA	Statutory zoo regulator, MoEFCC

 Biodiversity Conservation In-situ vs Ex-situ Taxonomy & IUCN Red List Hierarchy

Chapter 21: Species & Ecology (पारिस्थितिकी)

🎯 Consolidated Must-Score Facts

1. India's **Red Data Book** is a **national** catalogue — **BSI** for plants, **ZSI** for animals (both Kolkata). The **IUCN Red List** is the **global** extinction-risk system from IUCN (Gland).
2. IUCN has nine categories: EX, EW, CR, EN, VU, NT, LC, DD, NE. Risk order puts CR above EN above VU among living threatened tiers.
3. **Threatened** on IUCN means **VU + EN + CR** only — not "Endangered" alone.
4. IUCN assesses extinction risk. **CITES** (साइट्स) regulates international wildlife **trade** (पण्याध्यक्ष). Different institutions, different jobs.
5. **Ramsar (रामसर) (1971)** is the wetlands treaty under Wise Use. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).
6. Uttar Pradesh (उत्तर प्रदेश) Ramsar examples include Sarsai Nawar, Samaspur, Nawabganj, Upper Ganga (गंगा), Bakhira, and Haiderpur. **Rudrasagar is Tripura**; **Sultanpur is Haryana** — not UP.
7. **Mangroves** are salt-tolerant coastal forests with **pneumatophores** (वात-मूल) and often viviparous seeds. **Sundarbans** (सुंदरबन) is the classic largest mangrove belt; Rhizophora and Avicennia are common genera.
8. **MISHTI** is the Union Budget **2023–24** mangrove restoration push. Do not date it as 2019.
9. **Coral reefs** are built from calcium-carbonate skeletons and live with symbiotic **zooxanthellae** (ज़ूज़ैंथेली). **Bleaching** happens when heat stress expels the algae.
10. India's coral belts include Gulf of Mannar, Lakshadweep (लक्षद्वीप), Andaman (अंडमान) & Nicobar (निकोबार), and Gulf of Kutch.
11. **Grasslands** are grass-dominated systems maintained by fire and grazing. UP's Terai (तराई) is a foothill grassland–forest mosaic; **Dudhwa** (दुधवा) holds barasingha (swamp deer). Kaziranga (काजीरंगा) is the floodplain grassland rhino fact.
12. Himalayan vegetation follows altitude belts: tropical → subtropical → temperate → alpine → nival. Snow leopard is a high Himalaya (हिमालय) fact; Dudhwa is Terai foothill fringe, not alpine.
13. Marine **neritic** shelf waters are the highest fisheries-productivity zone; pelagic means open ocean. India's coast is about **7516 km**.
14. The **Gangetic dolphin** is a **freshwater river** dolphin — not a marine species trap.
15. Wetland ≠ mangrove ≠ coral. Inland marsh, saline (लवणीय) coastal forest, and reef builders are three different ecosystems.
16. Pneumatophores are mangrove breathing roots for gas exchange in waterlogged saline soil.
17. Amrit Dharohar (**2023**) is India's wetland stewardship framing beside Ramsar listing.
18. Coral bleaching (प्रवाल विरंजन) is a stress response, not a synonym for mangrove dieback.
19. Terai grasslands of UP are not Himalayan alpine meadows.
20. Red Data Book (national BSI/ZSI) is not identical to IUCN Red List categories, even when species appear on both tracks.
21. World Wetlands Day is February **2**, not Environment (पर्यावरण) Day (5 June).
22. Mangrove restoration schemes do not make landlocked UP a mangrove state — UP links are scheme awareness and river–wetland ecology.
23. Zooxanthellae live inside coral polyps; losing them bleaches the coral's colour and energy supply.
24. Threatened = three IUCN tiers combined. Memorise VU, EN, CR as the band.
25. CMS covers migratory species; Ramsar covers wetlands; do not swap those treaty jobs when ecosystems are asked.
26. Dudhwa = Terai grassland–forest and barasingha; Sundarbans = mangrove; Lakshadweep/Mannar = coral — keep the flagship map clean.
27. **ZSI (Zoological Survey of India)** publishes the **Red Data Book of Indian Animals**.
28. **Red Data Book ≠ IUCN Red List:** Red Data Book is **national documentation**; IUCN Red List is a **global standardised assessment** (EX through NE).
29. **Red Data Book ≠ CITES:** Red Data Book documents status; **CITES** regulates **international trade** in listed species.
30. **Extinct (EX):** No reasonable doubt the last individual has died.
31. **Extinct in the Wild (EW):** Survives only in **captivity, cultivation, or seed banks** — no viable wild population.
32. **Critically Endangered (CR) (अति संकटग्रस्त):** Extremely high extinction risk — e.g., **gharial**, **Great Indian Bustard**.
33. **Endangered (EN) (लुप्तप्राय):** Very high risk — e.g., **Gangetic dolphin**, **tiger** (improved from CR in some assessments).
34. **Vulnerable (VU):** High risk but lower urgency than EN/CR — e.g., **sloth bear**.
35. **Near Threatened (NT):** Close to qualifying as threatened but not yet.
36. **Least Concern (LC):** Widespread, low extinction risk.
37. **Data Deficient (DD):** Insufficient data — not the same as safe.
38. **Not Evaluated (NE):** Not yet assessed — majority of world's species.
39. **Threatened = VU + EN + CR** — the most tested formula. **Endangered alone is only one-third of threatened.**
40. **Risk hierarchy** among living categories: **CR > EN > VU > NT > LC**.
41. **IUCN assesses extinction risk.CITES** regulates **trade** — separate bodies, separate functions.
42. **Wetlands** are ecosystems where **water saturates the soil** — marshes, swamps, bogs, floodplains, lake margins — permanent or seasonal.
43. **India** has **75+ Ramsar sites** (2024 milestone). **Amrit Dharohar Scheme (2023)** supports wetland conservation and livelihoods.
44. **UP Ramsar sites:Sarsai Nawar** (Etawah (इटावा)), **Samaspur Bird Sanctuary** (Raebareli), **Nawabganj** (Unnao), **Upper Ganga River**, **Bakhira**, **Haiderpur**, **Parvati Agra Bird Sanctuary**.
45. **NOT in UP:Rudrasagar Lake** = **Tripura**. **Sultanpur National Park** = **Gurugram, Haryana**. Both are Ramsar sites outside UP.
46. **Threats:** drainage for agriculture, pollution (eutrophication (सुपोषण)), invasive species (आक्रामक विदेशी प्रजाति) (**water hyacinth** chokes UP ponds), hydrological alteration.

47. **Mangroves** are **salt-tolerant** trees and shrubs growing in **intertidal mudflats** of tropical and subtropical coasts.
48. **Pneumatophores** (vertical aerial breathing roots) and **prop roots** (Rhizophora) enable gas exchange in waterlogged anaerobic soil.
49. **Viviparous germination:** seeds germinate **while attached to the parent tree** and drop as ready seedlings (propagules).
50. **Ecological roles:** coastal protection against cyclones/tsunamis, fish nursery grounds, sediment trapping, shoreline stability, **blue carbon** (नीला कार्बन) sequestration.

📊 Quick Reference Grid

Lists / wetlands

ITEM	LOCK
India Red Data	BSI plants + ZSI animals (Kolkata)
IUCN 9 categories	EX EW CR EN VU NT LC DD NE
Threatened	VU+EN+CR only
IUCN vs CITES	Extinction risk / trade regulation
Ramsar / Wetlands Day	**1971** / **2 February**
Rudrasagar / Sultanpur	Tripura / Haryana
MISHTI	Budget **2023–24** mangroves
India corals	Mannar · Lakshadweep · A&N · Kutch (**not** Sundarbans)
Gangetic dolphin	Freshwater river (≠ marine trap)

Chapter 22: Renewable Energy (नवीकरणीय ऊर्जा)

🎯 Consolidated Must-Score Facts

1. **Renewable** energy replenishes on a human timescale — sun, wind, water, biomass, geothermal. **Nuclear** is low-carbon but usually **not** counted as renewable MCQs.
2. **Solar PV** turns sunlight directly into electricity. **CSP** (सीएसपी) uses mirrors to make heat, then steam and turbines.
3. The **National Solar Mission** began in **2010** under NAPCC (JNNSM from Jan 2010). India chased about **100 GW** solar by 2022 and **500 GW** non-fossil capacity by **2030**.
4. Solar has the **greatest RE potential** in India among common options and is the most eco-friendly among solar cell vs fossils/nuclear fission options.
5. **Bhadla** in Rajasthan is India's mega solar-park fact (not Gujarat as "largest"). **PM-KUSUM** supports farmer solar pumps. **OSOWOG** is One Sun One World One Grid.
6. **Domestic Content Requirement (DCR)** in solar news means mandating domestically manufactured PV cells/modules.
7. The **International Solar Alliance** began in **2015** (India–France, COP21). Headquarters is **Gurugram**. Not all UN members — originally sun-belt countries between Tropics. First Summit **New Delhi, March 2018**. Members target mobilising about **USD 1 trillion** for solar by **2030**.
8. **Cochin International Airport** = world's first fully solar-powered airport. **Diu** = first Indian UT / smart city running **100% solar** (daytime / UT stems). **Surat** = first district with **100% solar-powered PHCs**.
9. Wind power converts kinetic energy of air; power scales roughly with the **cube** of wind speed. Leading states include **Tamil Nadu (नाडु), Gujarat, and Karnataka**.
10. **Biomass** is energy from organic matter. **Biogas** (बायोगैस) from anaerobic digestion is mainly **methane (CH_4) and carbon dioxide (CO_2) (कार्बन डाइऑक्साइड)** — not a butane/carbon-monoxide pair.
11. Bagasse cogeneration, **GOBAR-Dhan**, and **SATAT** (bio-CNG) are biomass-policy neighbours.
12. **Geothermal** uses Earth's internal heat. Indian provinces include **Himalaya (हिमालय), Cambay, West Coast, SONATA (Son–Narmada (नर्मदा)–Tapi (तापी)), Godavari (गोदावरी), Mahanadi (महानदी)** — **not Ganga (गंगा) Delta (डेल्टा)**. Sites: **Puga, Manikaran, Tattapani**.
13. **Tidal** energy comes from Moon-driven sea-level change. India's best potential is often locked to the **Gulf of Khambhat**, then Gulf of Kutch.
14. **Small hydro** under MNRE is **≤25 MW**. Above that is treated as large hydro. Himalayan streams hold large SHP potential.
15. The **Ramganga (रामगंगा) Project** in Garhwal (गढ़वाल), Uttarakhand (उत्तराखंड), is about **198 MW** — a hydro fact that also matters for irrigation links toward UP.
16. **Green hydrogen** is H_2 from water electrolysis powered by **renewable** electricity. Grey hydrogen comes from natural gas; blue adds CCS (सुरक्षा कैबिनेट समिति) to gas routes.
17. The **National Green Hydrogen Mission** (January **2023**) targets about **5 MMT** per year by **2030**. National Hydrogen Energy Board dates to **3 Oct 2003**.
18. Hydrogen can power **IC engines**, be **blended with natural gas**, and run **fuel-cell vehicles** — all three are correct in recent stems.
19. Fuel cells with pure H_2 emit **heat and water**; they produce **DC**; they can power buildings and portable devices.
20. Solar ≠ wind ≠ tidal ≠ geothermal. Source and physics differ even when all are "clean."
21. ISA HQ is Gurugram, not Paris and not New Delhi as a casual guess.
22. Biogas major pair is CH_4 + CO_2. Wrong gas pairs are frequent traps.
23. Geothermal and nuclear are not "derived from the Sun" in the renewable-sun family sense used in many stems — though fossil-fuel energy ultimately traces to the Sun.
24. India still lacks a large commercial tidal plant even where potential maps look strong.
25. Green / grey / blue hydrogen colours mark production route and carbon intensity — not the colour of the gas.
26. UP colour: Mirzapur (मिर्ज़ापुर) solar collaboration history (इतिहास), western UP bagasse cogeneration, limited wind, Ramganga water–power links.
27. CSP is heat-first; PV is photon-to-electron. Do not describe CSP as a simple rooftop panel.
28. Wind leadership is southern and western coastal/plateau states — not a default UP fact.
29. Small hydro ceiling **25 MW** is the MNRE number; do not invent a different cutoff.
30. OSOWOG is a grid-connectivity vision around solar; it is not a biomass scheme.
31. Non-fossil **500 GW by 2030** is a capacity ambition fact beside older Solar Mission milestones.
32. Tariffs for solar are set by **CERC / SERCs**, not by SECI alone; India is **not** a top silicon-wafer manufacturer (panels often assembled from imported cells).
33. Microbial fuel cells and bioasphalt appear as advanced sustainable-energy neighbours (wastewater electricity; non-petroleum road binder).
34. Algae biofuel limits for developing countries: high expertise + large facilities with ecological/social concerns — **not** "seas only."
35. **Solar energy** is radiant light and heat from the Sun — harnessed via **photovoltaic (PV)** panels or **concentrated solar power (CSP)** (संकेंद्रित सौर ताप).
36. **PV cells** (silicon semiconductors) convert photons to DC electricity; an inverter converts to AC for the grid.
37. **CSP** uses mirrors to heat fluid, produce steam, and drive a turbine — can store heat for after-sunset generation.
38. **National Solar Mission (2010)** is a pillar of **NAPCC (2008)** — also called **JNNSM** from January 2010 — target scaled to **100 GW solar**, achieved around **2022–2024**.
39. **500 GW non-fossil fuel capacity by 2030** (COP26, 2021) includes solar, wind, hydro, nuclear, and biomass — solar remains the largest growth component.
40. **International Solar Alliance (ISA)** (अंतरराष्ट्रीय सौर गठबंधन) launched **30 November 2015** at **COP21 Paris** by **India and France** — treaty-based IGO, HQ **Gurugram, Haryana**.
41. **OSOWOG (One Sun One World One Grid)** and **World Solar Bank** concepts are linked to ISA's cross-border solar grid vision.

42. **PM-KUSUM (2019):** solar pumps for farmers and grid-connected solar on barren/fallow land — MNRE nodal.
43. **Cochin International Airport** was the **first airport in the world** fully powered by solar energy.
44. **Diu** became the first Indian UT / smart-city example to run on **100% solar** (daytime / UT stems). **Surat** became the first district with **100% solar-powered Primary Health Centres**.
45. **Bhadla (Rajasthan)** is India's mega solar park fact — not "Gujarat has the largest solar park" in newer stems. India's large floating solar example: **Ramagundam (Telangana)**.
46. **Wind energy** converts the **kinetic energy of moving air** into electricity using **wind turbines**.
47. **Tamil Nadu** is historically India's leading wind state (**Muppandal wind farm**). **Gujarat, Karnataka, Maharashtra, Rajasthan** follow.
48. **Onshore** turbines dominate India. **Offshore wind** potential on **Gujarat and Tamil Nadu coasts** — stronger winds but higher cost.
49. **Intermittency:** wind is variable — needs **grid integration**, **hybrid wind+solar**, or **storage** (batteries, pumped hydro).
50. **Capacity factor:** wind farms typically produce **~25–35%** of nameplate capacity annually — location-dependent.
51. In Indian usage, **conventional** energy means coal, oil, gas, **large hydro**, and nuclear. **Non-conventional** means solar, wind, biomass, biogas, tidal, geothermal, ocean, and **small hydro ≤25 MW** — treated as roughly equal to **renewable** in many stems.
52. **Energy crisis** in bank language means the danger of exhaustion of **fossil fuels** (coal, oil, gas) — not merely "short hydro" or malnutrition.
53. UNGA declared **2014–2024** the **Decade of Sustainable Energy for All** (supports SDG 7). Initiative is of the **United Nations**, not World Bank alone.
54. Ultimate source of energy stored in fossil fuels is the **Sun** (ancient photosynthesis).
55. Among coal / oil-gas / nuclear / hydro options, **hydroelectricity** is often the best answer for electricity generation under **sustainable development** (water not depleted by the generation cycle).
56. **Uranium is not a fossil fuel** — nuclear fuel, non-renewable. Coal / petroleum / natural gas are fossils.
57. **Bioethanol** (maize, sugarcane, etc.) blends with petrol. **Biodiesel** (Jatropha, Pongamia, UCO) blends with diesel. Biofuels are eco-friendly and can ease crisis but are often **not cost-effective** vs fossils.
58. Biomass gasification of shells/husk yields **producer gas** (CO, H_2, traces of CH_4 — not CO_2-only); can run IC engines.
59. Solar facts include National Solar Mission **2010**, ISA at **Gurugram**, **500 GW** non-fossil by **2030**, and **PM-KUSUM**.
60. Wind leadership often names Tamil Nadu (नाडु) and Gujarat. **SHP ≤25 MW**. **Tidal** is Moon-driven. **Ocean energy** is a wider set that includes wave, OTEC, and tidal.
61. **OTEC** uses the temperature difference between warm surface water and cold deep water — not the same as tidal barrage power.
62. Energy security: India imports a very large share of crude oil (often taught near **~85%**), while coal is mostly domestic. Strategic Petroleum Reserves sit at **Vizag, Mangalore, and Padur**.
63. **Energy Conservation Act 2001** created **BEE**. **PAT (2012)** trades **ESCerts**. Star labels and UJALA LEDs are efficiency/conservation tools. PAT is **not** CCTS carbon credits.
64. Coal is fossil non-renewable; biomass is renewable organic matter — do not call coal "biomass." Nuclear is **not** a biomass energy (बायोमास ऊर्जा) source.
65. Conventional includes large hydro; calling all hydro "non-conventional" is wrong under the Indian split.
66. E20 is the higher ethanol blend ambition beyond E10 — do not freeze policy forever at E10.
67. BEE sits under the EC Act **2001** efficiency architecture.
68. SPR locations are Vizag, Mangalore, Padur — three coastal storage facts.
69. Western UP sugarcane supports bioethanol; bagasse supports cogeneration; UP wind remains limited; rooftop solar is the distributed play.
70. **Hydrogen** is the least-polluting common fuel option (burns to water); called a **future fuel**. National Hydrogen Energy Board set up **3 October 2003**.
71. Microbial fuel cells use living microbes as catalysts; useful in wastewater plants — substrates are mainly **organic**, not "only inorganic."
72. **Bioasphalt** is non-petroleum / renewable-based paving binder — eco-friendlier road surfacing.
73. ISA is solar diplomacy/IGO; BEE is domestic efficiency regulation — different lanes.
74. PAT certificates measure energy savings trading; carbon markets measure CO_2 units — related climate politics, different instruments.
75. Nuclear's classification is usually non-renewable/conventional, not "renewable because low carbon."
76. Energy conservation (using less) and energy efficiency (more service per unit energy) are neighbours under BEE/PAT, not synonyms for renewable capacity addition alone.
77. **SATAT** promotes **compressed biogas (CBG)** for transport; oil-marketing companies procure CBG. **GOBAR-Dhan** is the community/dung biogas neighbour scheme — do not invent butane as a major biogas gas.
78. Biogas forms by **anaerobic** digestion; slurry is useful bio-fertiliser after gas capture.
79. **Nuclear** is usually classified as **non-renewable** (finite uranium + waste).
80. **Non-renewable** sources form over geological time — use depletes the stock.
81. **Coal:** India's primary electricity fuel; large domestic reserves in Jharkhand, Odisha, Chhattisgarh, West Bengal — eastern belt.
82. **Natural gas:** cleaner than coal; used in power, fertiliser, CNG; partly imported (LNG).
83. **Bagasse** cogeneration is major in western UP sugar mills.

📊 Quick Reference Grid

Solar / wind

ITEM	LOCK
NSM / JNNSM	**2010** under NAPCC
Targets class	~100 GW solar by 2022; **500 GW** non-fossil by 2030
Bhadla	Rajasthan mega solar park
ISA	**2015** India–France COP21; HQ **Gurugram**
PM-KUSUM / OSOWOG	Farmer solar pumps / One Sun One World One Grid
Wind leaders	TN, Gujarat (power ∝ speed^3)
Biogas majors	**CH_4 + CO_2** (≠ butane/CO)
Tidal best India	Gulf of Khambhat → Gulf of Kutch
SHP (MNRE)	≤**25 MW**

Chapter 23: Disaster & Environment (पर्यावरण)

🎯 Consolidated Must-Score Facts

1. Environmental degradation (पर्यावरणीय क्षरण) raises disaster risk (deforestation worsens floods and landslides). Disasters also damage ecosystems through erosion, pollution, and forest loss.
2. Flood types include riverine floods on the Ganga (गंगा) plains, flash floods/cloudbursts in Himalaya–Terai (तराई), and urban (नगरीय) waterlogging. UP river facts include Ganga, Yamuna (यमुना), Ghaghara (घाघरा), Rapti (राप्ती), and Gandak (गंडक).
3. **NDMA** (आपदा प्राधिकरण) is under the Disaster Management Act **2005**. The **Prime Minister** chairs NDMA. **NDRF** is the response force. Sendai Framework for disaster risk reduction is **2015**.
4. Drought progresses from **meteorological** (low rain) to **agricultural** (soil moisture) to **hydrological** (rivers and wells). Bundelkhand's seven UP districts are a chronic drought hotspot.
5. **DPAP** (Drought Prone Areas Programme) is **1973**. Do not date DPAP as **1952**. A rough programme spine is CDP (सीडीपी) 1952 (सामुदायिक विकास) → DPAP 1973 → FWP 1977 → TRYSEM 1979.
6. Landslides need gravity, water, weak slopes, and often vegetation loss. Himalaya and north-eastern hills are core belts. Kedarnath (केदारनाथ) **2013** is the cloudburst–landslide–flood chain fact. Roads, mining, and deforestation are human triggers.
7. Tropical cyclones need warm seas around **26–27°C**. The **Bay of Bengal** produces more than the Arabian Sea. The **eye** is calm; the **eyewall** has the strongest winds; **storm surge** (तूफानी महोर्मि) is the coastal flood killer.
8. Mangroves reduce storm-surge damage. Named cyclone memory often uses Fani (**2019**) and Amphan (**2020**).
9. IMD plains heat-wave idea: about **≥40°C** with departure **+4.5°C to +6.4°C** as heat wave; severe when departure is **≥+6.4°C** or temperature **≥47°C**. Peak is May–June in North India. Urban heat island amplifies cities but is not identical to a heat-wave definition.
10. Forest fires need fuel, oxygen, and ignition. Peak season is roughly March–June. Chir pine needles are a Himalayan fuel fact. FSI issues satellite fire alerts.
11. **Piral Lao–Paise Pao** in Uttarakhand (उत्तराखंड) pays for pine-needle collection to cut fire risk and raise income. It is an Uttarakhand flagship, not a default UP scheme name.
12. NDMA Chair = Prime Minister — a recurring institution fact.
13. Eye ≠ strongest winds. Eyewall holds the violence; eye is the calm core.
14. Landslides are not "purely natural" when options ignore slope cutting and deforestation.
15. Heat wave (IMD criteria) ≠ urban heat island (city microclimate effect), though cities worsen heat stress.
16. DPAP year is **1973**, not the older CDP **1952** tag.
17. Cloudburst is extreme local rain in a short time — often the Himalayan flood trigger neighbour to landslides.
18. Storm surge is sea pushed ashore by cyclone winds — distinct from ordinary astronomical tide.
19. Meteorological drought (मौसमी (मौसम विज्ञान) सूखा) is rainfall failure; agricultural drought (कृषि सूखा) is crop-moisture failure; hydrological drought (जल-वैज्ञानिक सूखा) is stream/groundwater failure. Keep the sequence.
20. Sendai **2015** is the global DRR framework fact beside India's NDMA architecture.
21. Forest-fire season and chir-pine fuel matter more in Uttarakhand hills than in deep Gangetic floodplain districts.
22. Cyclone naming and Bay of Bengal dominance are coastal facts; UP's disaster map is flood–drought–heat first.
23. NDRF is the specialised response force; NDMA is the apex policy body chaired by the PM.
24. Bundelkhand drought and Terai/Ganga floods are the two UP geography disaster anchors on this sheet.
25. **Riverine floods** swell slowly over days–weeks on plains. **Flash floods** follow cloudbursts in hills within hours. **Urban flooding** is waterlogging from impervious surfaces and clogged drains.
26. **UP flood rivers:** Ganga, Yamuna, Ghaghara (Saryu), Rapti, Gandak, Gomti** — monsoon inundation of the Gangetic floodplain.
27. **Human amplification:** floodplain encroachment, embankments that raise silted beds, and **deforestation** in catchments that speeds runoff.
28. **Terai flash risk:** Himalayan/Nepal-side catchments can inundate **Pilibhit, Lakhimpur Kheri, Bahraich** rapidly after extreme rain.
29. **Natural flood benefit:** controlled flooding deposits **alluvial (जलोढ़) silt** that renews soil fertility — the problem is uncontrolled flooding of encroached zones.
30. **Institutions:NDMA (Disaster Management Act 2005)** — **Prime Minister** is Ex-officio Chair; **NDRF** for response; **IMD/CWC (केंद्रीय जल आयोग)** for forecasting; **Sendai Framework 2015** for disaster risk reduction.
31. **Mitigation:** floodplain zoning, wetland restoration (natural sponges), catchment afforestation, early warning.
32. **Drought** develops slowly — weeks to months — unlike sudden floods.
33. **Meteorological drought:** rainfall below long-term normal.
34. **Agricultural drought:** soil moisture insufficient for crops (even if some rain occurs).
35. **Hydrological drought:** rivers, lakes, and groundwater below normal after prolonged dry spell.
36. **Bundelkhand (UP):** seven districts — **Jhansi (झांसी), Lalitpur, Jalaun, Banda, Hamirpur, Mahoba, Chitrakoot (चित्रकूट)** — chronic drought–desertification (मरुस्थलीकरण) hotspot.
37. **Drought Prone Areas Programme (DPAP)** launched in **1973** — soil conservation, water harvesting, afforestation in dryland districts.
38. **Feedback loop:** drought kills vegetation → soil exposed → erosion → worse water retention → next drought worse.
39. **Human triggers:** deforestation (roots bind soil), road cutting/blasting, unplanned hillside construction, mining — landslides are **not purely natural**.
40. **Earthquake link:** Himalayan seismic shaking can trigger widespread mass failures.

41. **Downstream risk:** landslide debris can dam (दाम -) rivers; dam-break floods then hit plains including UP Terai.
42. **Mitigation:** bio-engineering, retaining walls, afforestation, land-use zoning on steep slopes, GSI hazard zonation maps.
43. **Eye:** calm, clear, low-pressure centre. **Eyewall:** ring of **strongest winds and rain** around the eye. Trap: strongest winds are **not** in the eye.
44. **Storm surge:** cyclone winds push seawater onshore — often deadlier than wind on flat coasts (Odisha, WB, Bangladesh).
45. **Bay of Bengal** produces **more frequent and intense** cyclones than the **Arabian Sea** (warmer, shallower, moister waters).
46. **Mangroves** (e.g. Sundarbans (सुंदरबन)) absorb surge energy — coastal buffer. Deforestation of mangroves increases damage.
47. **IMD plains criteria:** maximum temperature **≥40°C** with departure **+4.5°C to +6.4°C** = heat wave; departure **≥ +6.4°C** or absolute (निरपेक्ष) temp **≥47°C** = **severe heat wave**.
48. **Urban heat island:** cities stay hotter than surrounding rural areas due to concrete, asphalt, waste heat, and lost greenery — amplifies heat-wave mortality.
49. **Climate link:** warming makes heat waves more frequent, longer, and hotter (IPCC (आईपीसीसी) AR6).
50. **Compound risk:** heat + drought raises crop failure and forest-fire fuel dryness; heat + power surge risks blackouts.

📊 Quick Reference Grid

Institutions / hazards

ITEM	LOCK
NDMA	DM Act **2005**; Chair = **PM**; NDRF response
Sendai	Disaster risk reduction framework
Drought ladder	Meteorological → agricultural → hydrological
DPAP	**1973** (≠ 1952)
Cyclone needs	Warm sea ~26–27°C; BoB > Arabian Sea
Eye / Eyewall	Calm core / strongest winds
Heat wave (plains)	≥40°C + departure rules; severe ≥47°C or +6.4°C
Forest fire season	Roughly Mar–Jun; chir pine fuel
Piral Lao–Paise Pao	UK pine-needle scheme (≠ Delhi)

Chapter 24: Current Environmental Issues

🎯 Consolidated Must-Score Facts

1. **Microplastics** are plastic fragments **smaller than 5 mm**. Primary sources include microbeads; secondary sources come from larger plastic breaking down. They are not "fast biodegradable."
2. Microplastics enter food webs from plankton to fish to humans. India's identified single-use plastic bans tightened around **2022**. UN plastics treaty talks (INC) also run from that period.
3. One **carbon credit** equals **1 tonne of CO_2-equivalent** reduced, removed, or avoided. Kyoto (क्योटो)'s CDM produced CERs; voluntary markets use standards such as Verra and Gold Standard.
4. **Carbon trading** is the market that buys and sells allowances or credits (cap-and-trade). The **EU ETS** is the largest compliance market. India's **CCTS (2023)** under the Energy Conservation Act line is steered with **BEE**.
5. **PAT** creates energy-efficiency certificates (**ESCerts**). PAT is **not** identical to carbon credits under CCTS.
6. **Green Credit Programme (2023)** rewards eco-actions. It is **not** the same instrument as a carbon credit.
7. **Net zero** means anthropogenic emissions balanced by removals. India's target is **2070** (COP26 Glasgow **2021**). China is often locked at **2060**; EU/UK at **2050**. Net zero is not gross zero.
8. India's NDC track includes cutting emission intensity and raising non-fossil capacity share toward the **2030** horizon (including the ~50% non-fossil capacity language in later updates).
9. **Green finance** is capital for climate and environment (पर्यावरण) projects — green bonds, climate funds, GCF. India issued **Sovereign Green Bonds** in FY **2022–23**; SEBI's Green Bond Framework is **2023**.
10. The **$100 billion** a year climate-finance pledge is the developed-to-developing finance politics fact. Loss and Damage Fund operationalisation is linked to **COP28 (2023)**.
11. **Blue economy** is sustainable ocean and coastal use — fisheries, ports, tourism, offshore renewables — aligned with **SDG 14**. **Blue carbon** (नीला कार्बन) is carbon stored in mangroves and seagrass.
12. **MISHTI (2023 Budget)** restores mangroves and sits beside Sagarmala and Deep Ocean Mission in the blue–coastal policy neighbourhood.
13. **LiFE** (Lifestyle for Environment (जीवनशैली पर्यावरण)) is a MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) behaviour-change mission. The idea was pitched at **COP26 Glasgow (2021)** and launched in **June 2022**. It is **not** a COP-25 Madrid product.
14. LiFE catalogues pro-planet actions for citizens; it is lifestyle awareness, not a carbon-market statute.
15. Carbon credit (unit) ≠ carbon trading (market) ≠ Green Credit (eco-action incentive) ≠ green finance (capital).
16. India net zero is **2070**, not 2050.
17. Microplastics are **<5 mm**, not 5 cm.
18. Kyoto flexible mechanisms are climate tools; Montreal is ozone — do not swap treaty families in "current issues" stems.
19. Nature-based credits often come from forestry and mangrove (blue carbon) projects.
20. UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम)'s Emissions Gap Report (उत्सर्जन अंतराल रिपोर्ट) tracks the gap between pledges and pathways — neighbour to net-zero politics.
21. SUP item bans are domestic plastic regulation; they do not by themselves equal a global plastics treaty.
22. CCTS is India's carbon market architecture; PAT remains an efficiency-certificate neighbour under BEE.
23. Blue economy is ocean wealth with sustainability; it is not a synonym for inland wetland policy alone.
24. Glasgow **2021** facts both India's 2070 net-zero announcement track and the LiFE idea pitch.
25. Offset means using credits to compensate one's own emissions — related to, but not identical with, "trading" as a market design.
26. Keep MISHTI dated **2023**, not 2019, when mangrove restoration is asked.
27. **Microplastics** are plastic particles **smaller than 5 mm** — primary (manufactured small) or secondary (fragmented from larger plastic).
28. **Primary:** microbeads in cosmetics, plastic **nurdles**, industrial abrasives.
29. **Secondary:** bags, bottles, fishing nets, and **tyre wear** breaking down — most environmental mass is secondary.
30. **Major sources:** single-use packaging, **synthetic textile fibres** (laundry effluent), tyre abrasion, fishing gear, agricultural plastic mulch.
31. **Pathways:** land → rivers → sea. **Ganga and Yamuna (यमुना)** studies show microplastic contamination downstream of urban (नगरीय) centres (Kanpur (कानपुर), Varanasi (वाराणसी), Lucknow (लखनऊ)).
32. **Health concern:** detected in blood, lungs, placenta — potential carrier for toxins; research ongoing.
33. **Persistence trap:** polyethylene/polypropylene persist **decades to centuries** — photodegradation makes **smaller** particles, not harmless minerals.
34. **Global policy:** UNEA **2022** resolution launched **Global Plastics Treaty** negotiations (INC process).
35. **India:** Plastic Waste Management Rules (EPR); **ban on identified single-use plastic items (July 2022)**; National Dashboard for EPR.
36. **Carbon credit** is a tradable certificate representing **one tonne of CO_2 equivalent (tCO_2e)** reduced, avoided, or removed.
37. **Project-based generation:** verified emission-reduction project (solar displacing coal, biogas (बायोगैस), afforestation) → accredited body issues credits → sold to buyer.
38. **Kyoto CDM (Clean Development Mechanism):** developed countries fund projects in developing countries → **Certified Emission Reductions (CERs)**. India hosted many wind/solar CDM projects.
39. **Voluntary Carbon Market:** companies buy credits for net-zero pledges — **Verra (VCS), Gold Standard** standards.
40. **Nature-based credits:** afforestation, REDD+, **mangrove restoration** — permanence risk if forest burns or is cleared.

41. **Avoidance vs removal:** avoidance prevents emissions that would occur; removal pulls CO_2 from atmosphere (afforestation, DAC).
42. **Additionality:** project must prove reduction would not happen without credit revenue.
43. **Carbon trading** is the **market mechanism** to buy and sell emission **allowances or credits** under a regulatory cap.
44. **Cap-and-trade:** regulator sets total emissions cap → firms get allowances → efficient firms sell surplus to high emitters → cap met at lowest cost.
45. **Carbon credit vs carbon trading:** credit = **unit** (1 tCO_2e); trading = **market** exchanging those units.
46. **EU Emissions Trading System (EU ETS)** is the world's **largest compliance carbon market**.
47. **Kyoto flexible mechanisms:** (1) Emissions Trading among Annex I countries, (2) **CDM**, (3) Joint Implementation.
48. **India Carbon Credit Trading Scheme (CCTS), 2023** under **Energy Conservation Act 2001** — **Bureau of Energy Efficiency (BEE)** (ऊर्जा दक्षता ब्यूरो) as designated authority.
49. **PAT (Perform, Achieve and Trade (पण्याध्यक्ष)), 2012:** trades **energy efficiency certificates (ESCerts)** — parallel market-based compliance, **not identical to carbon credits**.
50. **Compliance vs voluntary:** compliance markets are legally mandated (EU ETS, future CCTS); voluntary markets are optional corporate purchases.
51. A **carbon footprint** totals greenhouse gases in **tCO_2e**. Scope **1** is direct, Scope **2** is purchased energy, Scope **3** is the wider value chain.
52. India's **per capita** carbon footprint is about **2 tCO_2e** — low versus global and USA averages — even though absolute (निरपेक्ष) emissions are high.
53. An **ecological footprint** (पारिस्थितिक पदचिह्न) measures resource demand in **global hectares (gha)**. It is **not** the same unit as tCO_2e. The **Social Cost of Carbon** is the monetary long-term damage of **one tonne of CO_2**.
54. NDC-style facts include about **45%** intensity cut, about **50%** non-fossil capacity, and a large forest sink goal.
55. A **green economy** (UNEP framing) is low-carbon, resource-efficient, and socially inclusive.
56. A **blue economy** is sustainable ocean use linked to **SDG 14**. **MISHTI** mangrove restoration is from the **2023** Budget, **not 2019**.
57. A **circular economy** closes material loops through 3R/9R and EPR. It sits inside the broader green-economy idea and links to **SDG 12**.
58. **Green hydrogen** uses electrolysis powered by renewables. India's mission (January **2023**) targets about **5 MMT** by **2030**.
59. **Grey hydrogen** reforms natural gas. **Blue hydrogen** adds CCS (सुरक्षा कैबिनेट समिति) to grey pathways. Do not call grey "green."
60. The **SUP ban** push of **2022** targets single-use plastics that feed microplastic pathways.
61. **Climate refugees** are mostly **internally** displaced. They are **not** a separate legal category under the **1951** Refugee Convention.
62. **Nature-based Solutions** protect, manage, or restore ecosystems (IUCN (आईयूसीएन) framing). They are not geoengineering (भू-इंजीनियरिंग) (cirrus thinning / sulphate aerosols are geoengineering ideas for cooling, not NbS).
63. Net zero allows balancing removals. Gross zero means nearly eliminating emissions without relying on offsets.
64. MISHTI is a mangrove initiative. Pairing it with 2019 is wrong.
65. Microplastic size fact is **<5 mm**, not 5 cm.
66. NbS examples include mangroves, wetlands, and urban (नगरीय) forests — ecosystem (पारिस्थितिकी तंत्र) actions, not aerosol injection.
67. LiFE is lifestyle and demand-side behaviour change, not a substitute for the entire NDC package.
68. Blue economy is oceans and coasts. Green hydrogen is energy. Do not merge the two labels.
69. **Scope 1** covers **direct** emissions from owned sources (boilers, fleet fuel).
70. **Scope 2** covers **purchased electricity, heat, or steam**.
71. **Scope 3** covers other indirect emissions (supply chain, product use, travel, waste) — often the largest share and hardest to measure.
72. **Biocapacity** is nature's regenerative capacity; if footprint exceeds biocapacity, there is an **ecological deficit**.
73. **Earth Overshoot Day** marks when annual demand exceeds that year's regeneration.
74. **Net zero** means anthropogenic emissions are **balanced by anthropogenic removals** over a period — residual emissions may remain if matched by sinks or removals.
75. **Green finance** (हरित वित्त) (sovereign green bonds, SEBI green-bond rules) funds this pathway — detailed market tools sit with Topic 24.
76. **Blue economy** is sustainable use of ocean and coastal resources for growth and jobs while protecting marine ecosystems.
77. **SDG 14** = **Life Below Water**.
78. **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) was announced in **Union Budget 2023–24** to restore mangroves for shoreline protection and income.
79. **Sagarmala** is port-led coastal development; **Deep Ocean Mission** covers deep-sea exploration (Samudrayaan).
80. **Plastic Waste Management Rules 2016** (amended 2022) and related EPR rules make producers responsible for end-of-life collection.
81. **Green hydrogen** is H_2 from **water electrolysis** powered by **renewable electricity**.
82. **Grey hydrogen** from natural-gas reforming dominates today's supply and is **not** climate-friendly.
83. **Blue hydrogen** is grey production plus **CCS** — a debated transition label.
84. **Primary** types are manufactured small (microbeads, nurdles).
85. **Climate refugees / climate migrants** are people forced to move by climate impacts — cyclones, floods, drought (सूखा), sea-level rise, desertification (मरुस्थलीकरण).
86. **Nature-based Solutions** protect, sustainably manage, or restore ecosystems to address societal challenges (climate, disasters, food, water) while aiding biodiversity and well-being.
87. **LiFE (Lifestyle for Environment)** nudges individual and community behaviour toward mindful consumption.

📊 Quick Reference Grid

Markets / targets

ITEM	LOCK
Microplastics	<**5 mm**; primary beads / secondary fragments
1 carbon credit	1 tCO_2e reduced/removed/avoided
PAT	ESCerts — ≠ carbon credit under CCTS
Green Credit	**2023** eco-action rewards (≠ carbon credit)
India net zero	**2070** (COP26 Glasgow)
LiFE	Idea COP26 **2021**; launch Jun **2022**
MISHTI	Budget **2023** mangroves
Blue economy	Sustainable ocean use; SDG **14**

Chapter 25: Global Environmental Geography

🎯 Consolidated Must-Score Facts

1. Global environmental geography asks **where and why** Earth's climate–biome patterns occur. Climate (temperature and rainfall) shapes biomes, biodiversity, and degradation belts.
2. Major biomes to ratta: tropical rainforest (Amazon **Selva** (सेल्वा)), tropical deciduous (Indian monsoon forests), desert (**<25 cm** rain), savanna (सवाना) (scattered trees with wet–dry seasons), temperate grassland (**Prairie (प्रेयरी), Steppe (स्टेपी), Pampas, Puszta (पुस्ज़टा)**), taiga/boreal, tundra (टुंड्रा), Mediterranean chaparral, and coastal mangrove as a transitional belt.
3. **Köppen** letters: **A** tropical, **B** arid, **C** temperate, **D** cold, **E** polar. India is largely monsoon (**Am** type language); Thar (थार) is hot desert (**BWh**); Gobi is cold desert (**BWk**).
4. **Sahara** is the largest hot desert. **Atacama** is among the driest. **Gobi** spans **Mongolia and China**.
5. Australia's interior desert is explained mainly by **subtropical high pressure and continentality** — not by calling northern Australia "temperate."
6. Brazil facts: **Selva** = Amazon equatorial rainforest name; **Terra Roxa** = fertile dark volcanic soil. Both can be true together in match stems.
7. Floral matching classics: Planting Hearth → **C.O. Sauer**; World Floral Regions → **Adolf Engler**; Cryophyte Steppe → Arctic tundra setting; **Puszta** → **Hungary**.
8. Tribe–region facts: Bushman–Kalahari; Eskimo–northern Canada; Pygmy–Congo. **Masai are East Africa**, not West Africa.
9. **Permafrost** (स्थायी हिम मृदा) appears in **taiga and tundra**. It is **not** a savanna feature.
10. Global biome counts (often **14** in textbook lists) are **not** the same as India's **ten** biogeographic zones (Rodgers and Panwar).
11. Biome is a large climate-defined vegetation–fauna (प्राणीजात) region. Ecosystem is a local biotic–abiotic functional unit. Do not treat them as identical scales.
12. Temperate grassland names: Pampas (Argentina), Prairie (North America), Steppe (Eurasia), Puszta (Hungary).
13. Taiga is boreal conifer forest; tundra is moss–lichen cold treeless belt. Savanna is tropical grassland with trees.
14. Mangrove is coastal transitional vegetation — not a purely inland terrestrial biome.
15. Hotspot (Myers) means high endemism plus severe habitat (वास स्थान) loss (often taught as **≥70%** loss). It is not merely "any rich forest."
16. Gobi is cold desert Asia — not Sahara Africa and not Atacama Chile.
17. Selva ≠ Terra Roxa. Forest name versus soil name.
18. Masai wrong-option trap is West Africa; correct region is East Africa.
19. India's monsoon deciduous belt is not Amazon Selva.
20. Permafrost continuous cover is strongest in tundra; taiga may show patches — still not savanna.
21. Mediterranean biome is winter-rain shrubland (chaparral type), distinct from monsoon India.
22. Desert rainfall fact **<25 cm** is the quick arid filter in many MCQs.
23. Engler floral regions and Sauer planting hearth are cultural–biogeography matchs, not climate-classification letters.
24. Keep "14 biomes vs 10 Indian biogeographic zones" as a numbers trap pair.
25. **Global environmental geography** studies *where* environmental systems occur and *why* — linking physical geography (climate, relief, oceans) to biomes, biodiversity, and degradation.
26. **Climate** (temperature + precipitation) is the primary control on biome distribution.
27. **Latitude:** equatorial regions receive direct sunlight year-round → high productivity and biodiversity; polar regions receive oblique rays → low productivity.
28. **Continentality:** interior continents (central Australia, central Asia) have extreme temperatures and aridity.
29. **Altitude:** Himalaya (हिमालय), Andes, Alps show vertical zonation — tropical forest at base, alpine/nival at summit.
30. **Ocean currents:** warm currents (Gulf Stream) mild climate; cold currents (Humboldt) support fisheries and coastal deserts (Atacama).
31. **Subtropical highs (horse latitudes (हॉर्स अक्षांश) ~20–35° N/S)** suppress rainfall — major desert belt (Sahara, Arabian, Thar, Australian interior).
32. **Latitudinal biodiversity gradient:** species richness highest near equator, decreases toward poles.
33. **Tropical rainforest:** hot, wet year-round; layered canopy; highest biodiversity — Amazon (**Selva**), Congo, SE Asia, Western Ghats (पश्चिमी घाट).
34. **Tropical deciduous:** wet–dry monsoon — sal/teak forests of peninsular India.
35. **Desert:** typically **<25 cm** rain; sparse xerophytes — Sahara, Arabian, Gobi, Thar, Atacama.
36. **Savanna:** tropical grassland with **scattered trees** and distinct wet/dry seasons — East Africa ("Land of Big Games"); Gir (गीर) (Gujarat) is India patch.
37. **Temperate grassland:** few trees; fertile soils — Prairies (N America), Steppes (Eurasia), **Pampas (Argentina)**, **Puszta (Hungary)**.
38. **Taiga (boreal):** cold winters; spruce/fir/pine; **permafrost patches** — Siberia, Canada, Scandinavia.
39. **Tundra:** short summer; moss/lichen; **continuous permafrost** — Arctic coasts; no trees.
40. **Mediterranean (chaparral):** wet winters, dry summers; fire-adapted shrubs — California, Mediterranean Basin, Cape Town.
41. **Mangrove / wetland:transitional** terrestrial–aquatic — not purely terrestrial.
42. **Köppen Group A (Tropical):** all months ≥18°C — Af rainforest, **Am monsoon**, Aw savanna. India mainly **Am**.
43. **Group B (Arid):** evaporation > precipitation — BW desert, BS steppe. Thar **BWh** (hot); Gobi **BWk** (cold).
44. **Group C (Temperate):** mild winters — Mediterranean Csa, humid subtropical Cwa.
45. **Group D (Cold/Continental):** cold winters — taiga belt Siberia/Canada.
46. **Group E (Polar):** always cold — tundra ET, ice cap EF (Antarctica).
47. **ITCZ** seasonal shift drives monsoon and equatorial rain; failure causes Sahel drought (सूखा).

48. **Sahara (Africa):** largest **hot desert**.
49. **Atacama (Chile):** among the **driest** deserts on Earth — cold Humboldt Current + rain shadow (छाया).
50. **Gobi:** cold desert extending mainly into **Mongolia and China**.

📊 Quick Reference Grid

Biomes / climate

ITEM	LOCK
Köppen letters	A tropical · B arid · C temperate · D cold · E polar
India / Thar / Gobi	Monsoon Am-type / BWh hot desert / Mongolia–China
Largest hot / driest	Sahara / Atacama
Selva / Terra Roxa	Amazon rainforest name / fertile volcanic soil (Brazil)
Temperate grasslands	Pampas AR · Prairie NA · Steppe Eurasia · Puszta HU
Taiga / Tundra / Savanna	Boreal conifer / moss–lichen treeless / tropical grass+trees
Permafrost	Taiga–tundra (≠ savanna)
Biome ≠ ecosystem	Large climate region / local functional unit
India biogeographic zones	**10** (Rodgers & Panwar) ≠ global biome count

 Global Biomes Distribution, Climatic Pressure Belts & World Deserts

Chapter 26: Water Resources & Water Conservation

🎯 Consolidated Must-Score Facts

1. World water rough shares: oceans about **97.2%**, ice/glaciers about **2%**, groundwater about **0.6%**, rivers about **0.01%**. Descending order is **Oceans > Polar ice > Groundwater > Rivers**.
2. India holds roughly **4%** of world freshwater for about **18%** of world population (जनसंख्या). Per-capita availability around **1400 m^3** marks water stress in many readings.
3. The **water cycle** is driven primarily by **solar radiation** through evaporation from oceans, lakes, and rivers. Humidity is measured with a **hygrometer** or psychrometer.
4. Amount of rainfall depends strongly on **humidity** in the atmosphere — more vapour means greater chance of cloud and rain when temperature falls.
5. Many transplanted seedlings fail because most **root hairs** are lost during transplantation; root hairs absorb water and mineral nutrients.
6. **Freshwater is not a biotic resource.** Biotic resources come from living/organic material (forests, animals, fish); **coal and petroleum** are classed biotic because they form from decayed organic matter.
7. **World Water Day (विश्व जल दिवस) = 22 March** (UN observance from **1993**; UN-Water since **2003**). Theme **2024** = *Leveraging Water for Peace* (also taught as prosperity and peace).
8. **Rajendra Singh** ("Jal Purush" / Waterman of India; **Tarun Bharat Sangh**, Alwar) won **Ramon Magsaysay 2001** and **Stockholm Water Prize 2015**.
9. **IWRM** follows Dublin Principles (**1992**) and treats the **river basin** as the planning unit. National Water Policy **2012** priority language puts drinking first, then ecology (पारिस्थितिकी), agriculture, and industry.
10. **National Water Mission** (NAPCC, from **30 June 2008**) aims at conservation, less wastage, equitable distribution via IWRM. Five goals include public water database + climate impact assessment, citizen/state conservation action, focus on over-exploited areas, **+20% water-use efficiency**, and basin-level IWRM. Urban (नगरीय) wastewater recycling and ocean-water tech for coastal cities fit NWM; **Himalayan–peninsular river linking and free borewell reimbursement do not**.
11. The **Ministry of Jal Shakti (शक्ति)** (**2019**) merged water resources and drinking-water departments.
12. **Rainwater harvesting** captures rain for use or recharge — rooftops, pits, check dams, johads. Tamil Nadu (नाडु)'s mandatory RWH (**2003**) is a state flagship fact.
13. A **watershed** is land draining to a common outlet. **Hariyali** is the GOI watershed project for rural water conservation (≠ Pani Sansad / Jal Kranti as the 2022 key).
14. Desert plants conserve water via hard/waxy leaves, tiny/absent leaves, and thorns instead of leaves.
15. River cleaning spine: **Central Ganga (गंगा) Authority / GAP 1985 → NRCA 1995 → NGRBA 20 Feb 2009 (EPA; PM chair) → Namami Gange 2014** (NMCG). **STP** = domestic; **CETP** = industrial.
16. **Yamuna (यमुना) Action Plan** launched **1993** (YAP-I to ~2003; YAP-II Dec 2004; **YAP-III 7 May 2016** under Namami Gange).
17. **Ganges River Dolphin** (*Platanista gangetica*) = **National Aquatic Animal (2009)**; dams, nets, agrochemicals threaten it — crocodile increase is **not** a standard decline cause in UPSC (संघ लोक सेवा आयोग) framing.
18. Kanpur (कानपुर) is the Ganga tannery pollution fact. **Gomti** is Lucknow (लखनऊ)'s river — do not put Gomti's main story on Kanpur.
19. **NLCP (June 2001)** restored urban/semi-urban lakes; merged with NWCP into **NPCA (Feb 2013)**. NLCP lakes include Bhoj, Sukhna, Pichola, Bhimtal — **Chilika is not NLCP**.
20. Wetlands follow **Ramsar (रामसर) 1971**, Wise Use, World Wetlands Day (विश्व आर्द्रभूमि दिवस) on **2 February** (रामसर), and Wetland Rules **2017**. UP Ramsar sites exist; **Rudrasagar** and **Sultanpur** are **not** UP.
21. **CGWB** prepares the Annual Groundwater Quality Report. **CGWA** (groundwater regulation) is under **EPA 1986** (पर्यावरण संरक्षण 1986). India has the **largest groundwater-irrigated area** in the world. Arsenic drinking-water teaching often points to **Kolkata**.
22. Disinfection: **chlorine**, chloramine, chlorine dioxide, ozone; **UV** kills microbes without changing taste/odour — UV does **not** remove odour or settle turbidity.
23. Water Act **1974** (amended **1988**); Water Cess Act **1977** (enforced **1 Apr 1978**).
24. **Atal Bhujal Yojana (2019)** is a groundwater management scheme tag.
25. **Jal Jeevan Mission** launched in **August 2019** for rural household taps (Har Ghar Jal). Do not date JJM as **2014**.
26. Drinking-water standard fact **IS 10500:2012**: fluoride about **≤1.5 mg/L**, arsenic about **≤0.01 mg/L**, pH about **6.5–8.5**.
27. Heavy sand mining lowers water table and can pollute groundwater — it does **not** decrease river salinity as a standard consequence.
28. **Lisbon Declaration** (UN Ocean Conference) is about **oceans**, not freshwater glaciers.
29. Rivers are **not** second after oceans in global freshwater ranking — ice/glaciers are.
30. CGWB ≠ CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) on the groundwater quality report; CGWA ≠ CGWB.
31. Namami Gange is **2014**; GAP is **1985**. NGRBA chair is the **Prime Minister**, not rotating CMs.
32. IWRM is basin coordination; watershed is the local drainage unit used in ridge-to-valley works.
33. Aquifer is the underground storage layer; it is not a surface canal.
34. BOD measures organic pollution strength in water (mg/L) — a quality metric, not a volume share.
35. Bundelkhand scarcity, Kanpur tanneries, Gomti–Lucknow, and arsenic plains are the UP colour set.
36. JJM is rural tap connectivity; it is not Namami Gange's river-cleaning mission.
37. STP vs CETP: domestic vs industrial effluent trains — swap them and consent logic breaks.
38. National Water Policy priority order starts with **drinking water**, not industry.
39. **Surface water:** rivers, lakes, reservoirs — highly seasonal under monsoon.

40. **Groundwater:** India's largest **extractable** freshwater source (~45% irrigation, ~80% rural domestic).
41. **Ganga–Brahmaputra (ब्रह्मपुत्र)** system carries about **40%** of India's river water.
42. **Solar radiation** plays the main role in driving the water cycle by evaporating water from oceans, lakes, and rivers.
43. **Humidity** measures water-vapour content of air and is measured with a **hygrometer** or psychrometer.
44. **Biotic resources** come from living or organic material (forests, animals, fish) and from materials obtained from them. **Coal and petroleum** are counted as biotic because they form from decayed organic matter.
45. **Freshwater / pure water is not a biotic resource** — it is an abiotic natural resource, and usable drinking freshwater is scarce, so conservation techniques matter.
46. **World Water Day (22 March)** highlights freshwater and sustainable management (UN designation; first observed **1993**; **UN-Water** coordination from **2003**). Theme **2024**: *Leveraging Water for Peace*.
47. **Rajendra Singh** of Alwar (Rajasthan), called **Jal Purush / Waterman of India**, runs **Tarun Bharat Sangh** for community water harvesting; awards include **Ramon Magsaysay (2001)** and **Stockholm Water Prize (2015)**.
48. **IWRM** coordinates water, land, and related resources within a **river basin** to balance economy, society, and ecosystems.
49. **Dublin Principles (1992):** freshwater is finite; participatory management; women central; water has **economic value**.
50. **National Water Policy 2012:** river basin as planning unit; allocation priority **drinking water > ecology > agriculture > industry**.

📊 Quick Reference Grid

Shares / policy

ITEM	LOCK
World water order	Oceans ~97.2% > ice ~2% > GW ~0.6% > rivers ~0.01%
India freshwater share	~**4%** water for ~**18%** population
Water cycle driver	Solar radiation / evaporation
World Water Day	**22 March**
Rajendra Singh	Jal Purush; Magsaysay 2001; Stockholm Water Prize 2015
IWRM / Dublin	**1992**; basin as planning unit
NWM	NAPCC; from **30 Jun 2008**
Jal Shakti Ministry	**2019** merger
Freshwater resource type	**Abiotic** (≠ biotic)


Global Water Partitioning, Aquifer Hydrogeology & Water Stress

Chapter 27: Environmental Research & Institutions

🎯 Consolidated Must-Score Facts

1. Headquarters map: **Kolkata** — BSI and ZSI; **Dehradun** (देहरादून) — FSI, FRI, and WII; **New Delhi** — NBPGR; **Chennai** — NBA.
2. **BSI (1890, Kolkata)** surveys **plants** (*Flora* (वनस्पति), plant Red Data Book (रेड डेटा बुक)). **ZSI (1916, Kolkata)** surveys **animals** (*Fauna* (प्राणीजात), animal Red Data Book).
3. **FSI (1981, Dehradun)** maps forest cover (वन आवरण) and publishes **ISFR every two years**. **FRI (1906, Dehradun)** does forestry **research** under the ICFRE world — it does **not** publish ISFR.
4. **WII (1982, Dehradun)** does wildlife research, IFS/officer training, and tiger-census protocols.
5. **NBA** under the Biological Diversity Act sits in **Chennai**, not Delhi. It handles foreign access and ABS at the national level.
6. Biodiversity pyramid (पिरामिड): **NBA (national) → SBB (state) → BMC (local)**. BMCs prepare the **People's Biodiversity Register (PBR)**.
7. **NBPGR (1978, New Delhi, ICAR)** is the plant gene bank. It is **not** Kolkata BSI and **not** Karnal **NBAGR** (animals).
8. ISFR **2023** class figures often cited: India about **21.76%** forest cover; Uttar Pradesh (उत्तर प्रदेश) about **6.24%**. UP districts with **>20%** forest cover in that reading (रीडिंग) were **Chandauli and Shravasti only** — **Bahraich** was not in that pair.
9. BSI ≠ animals; ZSI ≠ plants; FRI ≠ ISFR; NBA ≠ Delhi; PBR ≠ NBA headquarters function.
10. SBB is the state biodiversity board — not the State Pollution Control Board.
11. NBPGR ≠ Kolkata. Gene bank is Delhi; plant survey is Kolkata.
12. ISFR is biennial, not annual.
13. UPSBB sits in the Lucknow (लखनऊ) state-biodiversity lane; WII supports Terai (तराई) wildlife science links.
14. BMC is local and owns PBR documentation of local biodiversity knowledge.
15. SBB supervises state-level commercial access and BMC work under BDA — domestic track beside NBA's foreign-access focus.
16. FRI research ≠ FSI satellite forest-cover accounting, even though both are Dehradun forestry institutions.
17. WII is wildlife science/training — not the ISFR publisher and not the plant gene bank.
18. Red Data Books from BSI/ZSI are national catalogues; IUCN (आईयूसीएन) Red List remains the global risk ladder (different chapter neighbour).
19. NBA year fact rides on BDA **2002** institutionalisation — headquarters city remains Chennai.
20. Gene-bank NBPGR serves crop genetic resources for all states, including UP, from New Delhi.
21. Forest-cover percentage traps for UP hinge on Chandauli + Shravasti, not a long distractor list of Terai districts.
22. PBR is prepared locally by BMCs; it is not an NBA HQ publication series.
23. Match institution → city before matching function. Wrong city is enough to kill a match list.
24. Dehradun trio: FSI (cover/ISFR), FRI (research), WII (wildlife). Keep the three jobs separate.
25. **BSI** is the national organisation for survey, taxonomy, and documentation of **Indian plant diversity**.
26. **Does not** survey animals (ZSI), map forest cover (FSI), or store crop gene banks (NBPGR).
27. **ZSI** documents **Indian animal (faunal) diversity** — taxonomy and inventories.
28. **ZSI ≠ WII:** ZSI = species inventory/taxonomy; WII = applied wildlife ecology (पारिस्थितिकी) and training.
29. **FSI** monitors India's **forest and tree cover (वृक्ष आवरण)** using satellite imagery and ground verification.
30. **ISFR 2023:** India forest cover **21.76%**; UP **6.24%**.
31. **FRI** is India's premier **forestry research** institute — silviculture, pathology, entomology, wood science, NTFPs.
32. **FRI ≠ FSI:** both in Dehradun — FRI = **how to grow/manage forests**; FSI = **how much forest exists on the map**.
33. **WII** is MoEFCC's autonomous institute for **wildlife research, training, and conservation science**.
34. **Does not** do national plant/animal taxonomy (BSI/ZSI) or publish ISFR (FSI).
35. **NBA** is the national statutory body under the **Biological Diversity Act, 2002**.
36. **Does not** prepare People's Biodiversity Registers — that is **BMC**.
37. **Uttar Pradesh State Biodiversity Board (UPSBB)** sits at **Lucknow** (लखनऊ).
38. **SBB ≠ SPCB:** SBB = biodiversity (BDA); SPCB = pollution (Water/Air Acts).
39. **BMC** is the **local** committee under BDA 2002 — at panchayat, municipality, or cantonment level.
40. **PBR is not prepared by NBA or BSI**.
41. **NBPGR** conserves **plant genetic resources** — seeds, tissue, DNA — for food and agriculture.
42. **Ex-situ** (स्थानांतरित) conservation — complementary to in-situ (स्थान पर) protected areas.
43. **NBPGR ≠ BSI:** BSI surveys/taxonomy (Kolkata); NBPGR stores germplasm (Delhi).
44. **NBPGR ≠ NBAGR:** animals/livestock gene bank is **NBAGR, Karnal**.

📊 Quick Reference Grid

HQ map

ITEM	LOCK
Kolkata	BSI **1890** plants; ZSI **1916** animals
Dehradun	FSI **1981** ISFR; FRI **1906**; WII **1982**
New Delhi	NBPGR plant gene bank
Chennai	NBA (≠ Delhi)
Karnal	NBAGR animals
ISFR	Biennial (≠ annual); FSI product
BDA ladder	NBA → SBB → BMC → PBR
SBB ≠ SPCB	Biodiversity board ≠ pollution board

Chapter 28: Indian Environmental Movements

🎯 Consolidated Must-Score Facts

1. **Bishnoi** (बिश्नोई) resistance at **Khejarli, Rajasthan (1730)** is led in memory by **Amrita Devi** (अमृता देवी) and the **363** martyrs who died protecting khejri trees. It is centuries older than Chipko.
2. **Chipko (1973)** began at **Reni, Uttarakhand** (then Uttar Pradesh (उत्तर प्रदेश)). Leaders include **Sunderlal Bahuguna, Chandi Prasad Bhatt (चंडी प्रसाद भट्ट), and Gaura Devi**. Method: embracing trees against logging.
3. **Silent Valley (1970s)** (शांत घाटी) in **Kerala (Palakkad)** opposed a hydro project in rainforest. **KSSP** and scientists were central. It is **not** a Tamil Nadu (नाडु) movement.
4. **Jungle Bachao Andolan (1982)** (जंगल बचाओ) in **Singhbhum, Jharkhand** was tribal (आदिवासी) resistance for forest rights.
5. **Appiko (1983)** in **Karnataka (Sirsi)** was led by **Panduranga Hegde**. It is the southern tree-embrace movement. Appiko is **Karnataka only** — not Uttarakhand, UP, or Kerala.
6. **Narmada Bachao Andolan (from 1985)** spans **Madhya Pradesh, Gujarat, and Maharashtra**, with **Medha Patkar** (मेधा पाटकर) and **Baba Amte** as leader facts. It is anti-large-dam displacement politics.
7. **Save Western Ghats (from about 2011)** is multi-state ecology (पारिस्थितिकी) politics around the **Gadgil** and **Kasturirangan** reports, mining, and ESZ questions.
8. Method families: Chipko, Appiko, and Bishnoi = forest/tree protection. Silent Valley and Narmada Bachao = anti-dam. Save Western Ghats = report/policy ecology protests.
9. Chipko ≠ Appiko. Chipko is Uttarakhand; Appiko is Karnataka.
10. Bishnoi **1730** ≠ Chipko **1973**. Century trap matters.
11. "NBA" as Narmada Bachao Andolan is **not** the National Biodiversity Authority. Same letters, different bodies.
12. Silent Valley is Kerala rainforest–dam (दाम -) conflict, not a Western Ghats mining-only slogan.
13. Jungle Bachao is Jharkhand Singhbhum tribal forest rights — not Chipko's Garhwal (गढ़वाल) story.
14. Narmada Bachao is multi-state Narmada valley politics — not a single-state Kerala story.
15. Save Western Ghats leans on expert committee reports; Chipko leans on village tree hugging.
16. UP link: Chipko's Reni lay in erstwhile UP until Uttarakhand statehood (**2000**). Appiko and Silent Valley are **not** UP movements.
17. Gaura Devi is a Chipko village leadership fact; Panduranga Hegde is Appiko; Medha Patkar is Narmada.
18. Khejarli martyrdom is Rajasthan Bishnoi memory — not Uttarakhand Chipko.
19. Anti-dam movements still differ: Silent Valley saved a Kerala valley rainforest project path; Narmada contested a large multi-state dam-displacement model.
20. Gadgil report is often read as stronger ecological protection; Kasturirangan as a later recalibration — both sit inside Save Western Ghats politics.
21. Tree-hugging method travelled from Chipko north to Appiko south — same method family, different states and years.
22. Match movement → year → state → leader before answering "which state" stems; state alone often decides the key.
23. **Maiti Movement** (मैती आंदोलन) (Uttarakhand; pioneered by **Kalyan Singh Rawat**, mid-1990s) asks newly married couples to plant a sapling at the bride's parental home — separate from Chipko.
24. Assertion (A): Chipko began at Reni in then Uttar Pradesh (now Uttarakhand).
25. Reason (R): Appiko and Silent Valley are UP movements.
26. Assertion (A): "NBA" as Narmada Bachao Andolan is not the National Biodiversity Authority.
27. Reason (R): Same letters can mean different bodies.
28. Assertion (A): Tree-hugging method travelled from Chipko north to Appiko south.
29. Reason (R): Same method family, different states and years.
30. Assertion (A): Gadgil is often read as stronger ecological protection; Kasturirangan as later recalibration.
31. Reason (R): Both sit inside Save Western Ghats politics.
32. Assertion (A): Match movement → year → state → leader before answering "which state" stems.
33. Reason (R): State alone often decides the key.
34. Assertion (A): Khejarli martyrdom is Rajasthan Bishnoi memory.
35. Reason (R): It is Uttarakhand Chipko's founding event.

Chronological Sequence of Indian Environmental Movements

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1730 ⟶ Bishnoi Movement (Khejarli, Rajasthan; Amrita Devi sacrifice protecting sacred Khejri trees)
1973 ⟶ Chipko Movement (Reni, Chamoli, Uttarakhand; Sunderlal Bahuguna, Chandi Prasad Bhatt, Gaura Devi)
1978 ⟶ Silent Valley Movement (Palakkad, Kerala; saved tropical evergreen rainforest from hydroelectric dam)
1982 ⟶ Jungle Bachao Andolan (Singhbhum, Bihar/Jharkhand; tribal protest against replacing Sal with Teak)
1983 ⟶ Appiko Movement (Uttara Kannada, Karnataka; Pandurang Hegde; hugging trees in Western Ghats)
1985 ⟶ Narmada Bachao Andolan (Gujarat/MP/Maharashtra; Medha Patkar, Baba Amte; protesting Sardar Sarovar)
1990 ⟶ Tehri Dam Anti-Dam Movement (Uttarakhand; Sunderlal Bahuguna hunger strikes against high seismic risk)
```

📊 Quick Reference Grid

Movement ↔ place / year

ITEM	LOCK
Bishnoi / Khejarli	**1730** Rajasthan; Amrita Devi; 363 martyrs
Chipko	**1973** Reni, Uttarakhand; Bahuguna / Bhatt / Gaura Devi
Silent Valley	1970s **Kerala** rainforest–dam; KSSP
Jungle Bachao	**1982** Singhbhum, Jharkhand
Appiko	**1983** Sirsi, **Karnataka**; Panduranga Hegde
Narmada Bachao	From **1985**; MP–GJ–MH; Medha Patkar / Baba Amte
Chipko ≠ Appiko	UK / Karnataka
NBA letters trap	Narmada Bachao ≠ National Biodiversity Authority


Cartographic Spatial Map of Historic Indian Environmental Movements

Chapter 29: Environmental Literature & Awareness

🎯 Consolidated Must-Score Facts

1. Four (चातुर्याम) pillars on this sheet: **literature** (landmark books/reports), **education** (formal EE), **awareness** (public campaigns), and **communication** (science to citizens).
2. ***Silent Spring* (1962)** by **Rachel Carson** exposed pesticide/DDT harms. It is not Hardin's commons essay.
3. **"Tragedy of the Commons" (1968)** by **Garrett Hardin** is about overuse of shared resources.
4. ***Limits to Growth* (1972)** is the **Club of Rome** systems study. ***Our Common Future* (1987)** is the **Brundtland Report** that popularised sustainable development. Brundtland ≠ Limits to Growth.
5. ***Small Is Beautiful* (1973)** by **E.F. Schumacher** critiques mass-production economics — "economics as if people mattered." It is not Carson's pesticide book.
6. Assertion–reason traps often make Brundtland and Limits to Growth look interchangeable — they are different books/years/authors.
7. Environmental education landmarks include **Tbilisi (1977)**, MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) **NGC/ECO Clubs**, **GSDP (2017)**, and **NEP 2020** ESD language.
8. Environmental awareness hooks include **Article 51A(g)**, **NEAC**, **Van Mahotsav** (July tree planting), and **LiFE** (launched **June 2022** from a COP26 idea).
9. LiFE is **not** a COP-25 product. Idea pitch is Glasgow COP26; launch is June 2022.
10. Environmental communication flagships include CSE's ***Down To Earth***, Gobar Times, and IEC components inside schemes. **KSSP** also sits in science–public communication memory from Kerala.
11. Carson ≠ Hardin. Pesticides book versus commons essay.
12. Brundtland ≠ Limits to Growth. SD definition report versus Club of Rome growth study.
13. Literature shapes thought; education is curriculum and clubs; awareness is campaigns and duties; communication is media/IEC — keep the four jobs distinct.
14. NGC/ECO Clubs are MoEFCC school eco-action programmes — not a Bharat Nirman infrastructure scheme.
15. Agenda 21 is a sustainable-development action plan from Rio; it is not "only environmental education."
16. Article **51A(g)** is the constitutional citizen duty to protect the environment (पर्यावरण) — an awareness/duty fact.
17. Van Mahotsav is the July greening festival fact for schools and public tree planting.
18. NEP 2020 supports education for sustainable development language; it does not replace Tbilisi as the classic EE conference fact.
19. Down To Earth is CSE journalism for environment — a communication pillar example.
20. **Eco-mark (1991)** is the BIS earthen-pot label for certified environment-friendly consumer goods. It is **not** AGMARK and not a FAO mark.
21. **Indira Gandhi Paryavaran Puraskar (1987, MoEF)** honours measurable contribution to environment protection. **Rajiv Gandhi Environment Award** recognises clean technology and industrial pollution reduction.
22. **Tyler Prize** (University of Southern California) is often called the "Nobel of Environment" for environmental science, health, and energy.
23. UNEP's **Global 500 Award** (from **1987**) was later succeeded in spirit by **Champions of the Earth** (from **2005**). Narendra Modi received Champions of the Earth in **2018**.
24. The **Borlaug Award** is for outstanding agricultural research. Norman Borlaug's Nobel was the **Peace** Prize (Green Revolution (हरित क्रांति)), not medicine.
25. UP colour: NEAC district campaigns, Van Mahotsav in schools, NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) city awareness (Kanpur (कानपुर)), and ECO Clubs in UP schools.
26. Silent Spring's year is **1962**; Limits to Growth is **1972**; Brundtland is **1987** — three different decades.
27. Hardin's commons logic is shared-resource overuse; it is not Carson's pesticide narrative.
28. Match pillar → flagship before matching author names: Carson/Hardin/Brundtland/Schumacher for literature; Tbilisi/ECO Club for education; 51A(g)/LiFE for awareness; Down To Earth for communication; awards for recognition traps.
29. **Environmental literature** includes books and landmark reports that shaped ecological thought and policy.
30. **Rachel Carson, *Silent Spring* (1962):** exposed DDT/pesticide harm to birds, food chains, and health — triggered the modern environmental movement.
31. **Garrett Hardin, "Tragedy of the Commons" (1968):** argued individuals overuse shared resources (pasture, fisheries, air) for private gain.
32. **Club of Rome, *The Limits to Growth* (1972):** modelled exponential growth hitting planetary resource limits — **not** the Brundtland Report.
33. **Brundtland Commission, *Our Common Future* (1987):** defined and **popularised Sustainable Development** — meeting present needs without compromising future generations.
34. **W.M. Adams, *Green Development* (first edition 1990):** environment and sustainability in the developing world.
35. **Masanobu Fukuoka** was a Japanese farmer-philosopher who pioneered **natural farming** and re-vegetation of desertified lands.
36. **Aldo Leopold, *A Sand County Almanac* (1949):** "Land Ethic" — moral duty to the biotic community.
37. **Paul Ehrlich, *The Population (जनसंख्या) Bomb* (1968):** population growth vs resource limits debate.
38. **Indian voices:** Gandhi (गांधी) *Hind Swaraj (स्वराज)* (हिंद स्वराज) (1909) — restraint vs reckless industrialism; Tagore/Shantiniketan — nature-linked education; Bahuguna — "ecology (पारिस्थितिकी) is permanent economy"; CSE (Anil Agarwal, Sunita Narain) — *Down To Earth*.
39. **Environmental Education (EE)** builds knowledge, attitudes, skills, and participation for environmental protection.
40. **Stockholm 1972** (स्टॉकहोम सम्मेलन) put EE on the global agenda; **Tbilisi Declaration (1977)** became the first intergovernmental EE framework.
41. **Rio 1992 Agenda 21 Chapter 36** called for reorienting education toward sustainable development.
42. **NEP 2020** integrates environmental awareness and ESD across stages through experiential, multidisciplinary learning.
43. **National Green Corps (NGC) (नेशनल ग्रीन कॉर्प्स) / ECO Clubs (MoEFCC):** school/college clubs for plantation, waste management, biodiversity surveys, energy audits.

44. **Paryavaran Mitra (मित्र -):** CEE Ahmedabad (अहमदाबाद) + MoEFCC — student/teacher green ambassadors.
45. **Green Skill Development Programme (GSDP), 2017:** MoEFCC + Skill Ministry — vocational green skills.
46. **Environmental awareness** is public understanding and concern that drives behaviour change and civic action.
47. **Article 51A(g)** (42nd (42वां) Amendment, 1976): Fundamental Duty to protect and improve forests, lakes, rivers, and wildlife — pairs with **Art 48A** (state duty).
48. **National Environment Awareness Campaign (NEAC):** MoEFCC-funded NGO/school campaigns since the 1980s.
49. **Van Mahotsav:** tree-plantation week in **July**, popularised by **K.M. Munshi (1950)**.
50. **LiFE Mission:** idea at **COP26 (2021)**; launched **June 2022** — 75 pro-planet lifestyle actions (awareness/nudge, not school curriculum).

📊 Quick Reference Grid

Books / campaigns

ITEM	LOCK
Silent Spring	Rachel Carson **1962**; pesticides/DDT
Tragedy of the Commons	Garrett Hardin **1968**
Limits to Growth	Club of Rome **1972**
Our Common Future	Brundtland **1987**
Tbilisi	EE landmark **1977**
LiFE	COP26 idea **2021**; launch Jun **2022**
NGC / ECO Clubs	MoEFCC school programmes
CSE flagship	*Down To Earth* / Gobar Times

Chapter 30: Environmental Days

🎯 Consolidated Must-Score Facts

1. **World Wetlands Day** falls on **2 February** (रामसर) and marks the **Ramsar (रामसर) Convention (1971)** (रामसर कन्वेंशन). It is not 2 March or 3 March.
2. **World Wildlife Day** falls on **3 March** and marks the adoption of **CITES (साइट्स) (1973)** (साइट्स). It is **not** 22 March.
3. **International Day of Forests** falls on **21 March**. It was proclaimed by the UN General Assembly in **2012**.
4. **World Water Day** falls on **22 March**. The classic trap is pairing Wildlife Day with 22 March.
5. **Earth Day** falls on **22 April**. It began in the United States in **1970** as a civic teach-in movement.
6. **International Day for Biological Diversity** falls on **22 May** and commemorates the CBD text adopted (अंगीकृत) at Nairobi in **1992**.
7. **World Environment Day** falls on **5 June**. It marks the opening of the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) and is run by **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम).
8. **World Ozone Day** falls on **16 September** and marks the **Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) (1987)** (मॉन्ट्रियल प्रोटोकॉल).
9. In calendar order the eight syllabus days run **2 Feb → 3 Mar → 21 Mar → 22 Mar → 22 Apr → 22 May → 5 Jun → 16 Sep**.
10. The March ladder is **Wildlife (3) → Forests (21) → Water (22)**. Never put Wildlife on 22 March.
11. World Environment Day is **not** Earth Day and **not** Wetlands Day. The dates are **5 June**, **22 April**, and **2 February**.
12. India hosted World Environment Day in **2011** (Forests theme) and **2018** (Beat Plastic Pollution).
13. Biological Diversity Day shifted from an older **29 December** link to **22 May** from **2000**; papers use **22 May** only.
14. World Water Day is coordinated under **UN-Water**. Wetlands Day is a **Ramsar** (रामसर) day, not a Water Day synonym.
15. World Wildlife Day was proclaimed by the UN in **2013**, but the date itself is the **1973 CITES** anniversary.
16. Ozone Day belongs to the **Montreal Protocol** (मॉन्ट्रियल प्रोटोकॉल) family. Do not date it as 2 October, 21 April, or 30 January.
17. Earth Day is a civic global day. It is **not** the same UN treaty day as World Environment Day.
18. Biodiversity Day is broader than Wildlife Day. Wildlife Day is fauna (प्राणीजात)–flora (वनस्पति) trade (पण्याध्यक्ष) and crime awareness under the CITES date.
19. Van Mahotsav is a separate **July** plantation campaign in India. It is **not** International Day of Forests.
20. Earth Hour (अर्थ आवर) (last Saturday of March, WWF (डब्ल्यूडब्ल्यूएफ)) is **not** on this syllabus day list and is not Earth Day.
21. Uttar Pradesh (उत्तर प्रदेश) has **10** Ramsar sites — useful Wetlands Day geography, but the day itself stays **2 February**.
22. A correct Ozone–WED pair is **16 September** and **5 June**. A Wildlife–**22 March** pair is always wrong.
23. Stockholm **1972** anchors WED. Ramsar **1971** anchors Wetlands. Montreal **1987** anchors Ozone. CITES **1973** anchors Wildlife.
24. Same-number traps: **22** March is Water, **22** April is Earth, **22** May is Biodiversity — three different months.
25. **World Toilet Day = 19 November** (UN, from 2013). **World No Tobacco Day = 31 May** (not 5 May). **World Habitat (वास स्थान) Day** = first Monday of October (not a fixed 5 October).
26. **Swachh Bharat Mission** launched **2 October 2014** (Gandhi (गांधी) Jayanti). Sierra Leone observes national cleanliness on the last Saturday of every month.
27. Earth Day theme teaching: **2024 = Planet vs. Plastics**. WED **2023 = Beat Plastic Pollution**.
28. **International Tiger Day** falls on **29 July** and marks the **St Petersburg Tiger Summit (2010)**. It is **not** World Wildlife Day (3 March).
29. **World Environment Day (WED)** is the UN's main annual day for environmental awareness (पर्यावरण जागरूकता) and action.
30. **2022** also marked **50 years** since Stockholm 1972, which designated **5 June** as World Environment Day.
31. **World Wetlands Day** marks adoption of the **Ramsar Convention on Wetlands** at Ramsar, Iran (**2 February 1971**).
32. **Earth Day** promotes environmental protection, climate action, and sustainability every **22 April**.
33. **International Day of Forests** raises awareness of forests and sustainable forest management every **21 March**.
34. **World Water Day** focuses on freshwater resources, access, and sustainable water management every **22 March**.
35. **World Ozone Day** (International Day for the Preservation of the Ozone Layer) is observed every **16 September**.
36. **World Wildlife Day** celebrates wild fauna and flora and fights wildlife crime every **3 March**.
37. **Highest-yield trap:** listing Wildlife Day as **22 March** (that is Water Day) — wrong pair in **
38. **International Tiger Day** is observed every **29 July**.
39. **World Toilet Day = 19 November** (UNGA 2013; sanitation for all / end open defecation messaging).
40. **World No Tobacco Day = 31 May** (WHO) — not 5 May. Themes rotate (e.g. tobacco–environment / grow food not tobacco).
41. **World Habitat Day** = **first Monday of October** (UN; not a fixed calendar date like 5 October).
42. **World Cities Day = 31 October**. **World Population Day = 11 July**. **World Health Day = 7 April**.
43. **Earth Overshoot Day** marks when humanity's annual demand exceeds Earth's regenerative capacity — an environmental-degradation awareness marker (Global Footprint Network partnership framing).

📊 Quick Reference Grid

Day ↔ date

ITEM	LOCK
Wetlands	**2 February** (Ramsar)
Wildlife	**3 March** (CITES) — ≠ 22 Mar
Forests	**21 March**
Water	**22 March**
Earth	**22 April** (US 1970)
Biological Diversity	**22 May** (CBD)
Environment	**5 June** (Stockholm/UNEP)
Ozone	**16 September** (Montreal)
March ladder	Wildlife 3 → Forests 21 → Water 22
India WED hosts	**2011** Forests; **2018** Beat Plastic

Chapter 31: National Parks & Protected Areas (Advanced)

🎯 Consolidated Must-Score Facts

1. Uttar Pradesh has **one National Park — Dudhwa (दुधवा)** — in **Lakhimpur Kheri** on the Indo-Nepal Terai (तराई).
2. Dudhwa moved sanctuary (**1958**) → National Park (**1977**) → core of **Dudhwa Tiger Reserve (1987)** with Kishanpur and Katarniaghat. Rhino rehabilitation began **1984** (later phases including ~**2018**).
3. **Valmiki** (वाल्मीकि) National Park is in **Bihar**. **Sultanpur** National Park is in **Haryana**. **Corbett** is in **Uttarakhand** (उत्तराखंड). None is a UP national park.
4. Uttar Pradesh is **landlocked**, so it has **zero** marine or coastal national parks.
5. UP has **25 Wildlife Sanctuaries**. **Hastinapur** (हस्तिनापुर) is the largest. **Chandraprabha** is in **Chandauli**. **Katarniaghat** was established in **1975**.
6. **Rudrasagar** Lake is in **Tripura**. **Sultanpur** NP is in **Haryana**. **Samaspur** and **Sarsai Nawar** are in Uttar Pradesh.
7. India's first National Park is **Hailey / Jim Corbett (जिम कॉर्बेट) (1936)** in Uttarakhand. The world's first is **Yellowstone (USA, 1872)** — Old Faithful geyser; Wyoming–Montana–Idaho.
8. India's **largest** National Park is **Hemis** in **Ladakh** (लद्दाख). **Madhya Pradesh** has the most national parks (~**11**). **Andaman (अंडमान) & Nicobar (निकोबार)** has the most wildlife sanctuaries.
9. **Gir (गीर) (Gujarat)** holds the only wild **Asiatic lions**. **Kaziranga (काजीरंगा) (Assam)** is the classic one-horned rhino stronghold.
10. A National Park is the strictest **Wildlife Protection (वन्यजीव सुरक्षा) Act** category (Section **35**). Boundary change needs the **State Legislature**.
11. A Wildlife Sanctuary (Section **18**) is more flexible than a National Park. National Parks are **in-situ** (स्थान पर) conservation.
12. India's first Marine National Park is **Gulf of Kutch (Gujarat)**. Others include **Gulf of Mannar (Tamil Nadu (नाडु))** and the Andaman marine parks.
13. Marine NP facts: **Mahatma Gandhi (महात्मा गांधी) Marine** and **Rani Jhansi (झांसी) Marine (1996)** are in the **Andaman** (अंडमान) group.
14. Natural World Heritage (विश्व धरोहर) overlaps include Kaziranga, Keoladeo (केवलादेव), Manas, Sundarbans (सुंदरबन), Nanda Devi (नंदा देवी)–Valley of Flowers (फूलों की घाटी), Western Ghats (पश्चिमी घाट), Great Himalayan (**2014**), and Khangchendzonga (mixed **2016**).
15. Uttar Pradesh has **no** natural UNESCO World Heritage site of its own.
16. Classic match set: **Dachigam–J&K–Hangul**, **Keoladeo–Rajasthan–birds**, **Kanha (कान्हा)–MP–hard-ground barasingha / Bhoorsingh mascot**, **Periyar (पेरियार)–Kerala–elephant/tiger**.
17. **Pilibhit (2014)** and **Ranipur (2023)** are tiger reserves. They are **not** additional full state national parks of UP.
18. National Chambal (चंबल) Sanctuary spans UP–MP–Rajasthan and is famous for **gharial** and dolphin — **not Haryana**.
19. River–park traps: Corbett = **Ramganga (रामगंगा) + Kosi (कोसी)**; Kaziranga = **Brahmaputra (ब्रह्मपुत्र) / Diphlu**; Silent Valley (साइलेंट वैली) = **Kunthipuzha**.
20. Species–park traps: **Keoladeo Ghana (घन) ≠ lion (सिंह)**; **Eravikulam = Nilgiri (नीलगिरि) Tahr** (not Hoolock); **Hoolock = NE Assam**; **Bhitarkanika = saltwater crocodile**; **Desert NP = Great Indian Bustard** (Jaisalmer–Barmer; human habitation exists).
21. Dudhwa flagship fauna (प्राणीजात) include tiger, reintroduced one-horned rhino, and **barasingha** (swamp deer).
22. Terai Arc pieces with Dudhwa are **Kishanpur** (Lakhimpur Kheri) and **Katarniaghat** (Bahraich).
23. **Rajaji (राजाजी) NP (UK)** = Dehradun (देहरादून)–Haridwar (हरिद्वार)–Pauri (पौड़ी) (Rajaji + Motichur + Chilla merged **1983**); Tiger Reserve **2015**. Named after C. Rajagopalachari.
24. **Pin Valley = Himachal (हिमाचल) (Lahaul–Spiti)**, not J&K. **Van Vihar = Bhopal**. **Salim Ali NP = Srinagar (ex–City Forest)**. **Nagarhole = Rajiv Gandhi NP (Karnataka)**.
25. **Indravati = Bastar (Chhattisgarh)** NP 1981 / TR 1983. **Bandhavgarh (Umaria, MP)** famous for white tiger association. **Betla/Palamu = Jharkhand**.
26. **Keibul Lamjao (Manipur)** on **Loktak** is the only floating NP (**phumdis**; Sangai (सांगाइ) deer). **Valley of Flowers** (फूलों की घाटी) lies wholly in temperate alpine zone.
27. **Govind NP (Uttarkashi (उत्तरकाशी))** includes the source region of the **Tons** (टौंस) (Yamuna (यमुना) tributary). **Great Himalayan NP = Kullu (HP)**.
28. **Wild Ass Sanctuary (Rann of Kachchh, Gujarat)** is India's largest wildlife sanctuary. **Kuno–Palpur (MP)** is the Asiatic lion second-home site.
29. UK west→east sanctuary line: **Kedarnath (केदारनाथ) → Nanda Devi → Binsar → Askot** (Askot/Kedarnath = musk deer country).
30. **Sathyamangalam TR** links Eastern and Western Ghats. **Nagarjunsagar–Srisailam** holds the largest critical tiger habitat (वास स्थान) among common options.
31. Bird/butterfly facts: first butterfly park = **Bannerghatta (Karnataka)**; first State butterfly = **Blue Mormon (Maharashtra)**; Great Indian Hornbill (हॉर्नबिल) = **Western Ghats** (पश्चिमी घाट).
32. World-first versus India-first is a fixed pair: **Yellowstone 1872** versus **Corbett 1936**.
33. Early world peers: **Royal National Park, Australia (1879)**; **Banff NP, Canada (1885)**.
34. Continent icons: **Serengeti (Tanzania)**, **Kruger (South Africa)**, **Kakadu (Australia)**, **Galápagos (Ecuador)**, plus USA icons Yosemite / Everglades / Grand Canyon.
35. **Pilibhit TR (2014)** and **Ranipur TR (2023)** are tiger reserves — not additional full state NPs.
36. **Terai trio with Dudhwa:Katarniaghat** (Bahraich) and **Kishanpur** (Lakhimpur Kheri) — tiger, gharial, dolphin corridor.
37. **Hastinapur WLS** (~2,073 sq km) is UP's **largest**, spanning Meerut (मेरठ)–Muzaffarnagar–Ghaziabad–Bijnor–Amroha.
38. **National Chambal WLS** (Agra–Etawah (इटावा) + MP–Rajasthan) — gharial, dolphin, Indian skimmer.
39. **Chandraprabha WLS** is in **Chandauli** (1957; among earliest in UP).
40. **Katarniaghat** was established in **1975**.
41. **Largest NP:Hemis**, Ladakh (~3,350 sq km) — snow leopard country.
42. **Most NPs:Madhya Pradesh** (~11).

43. **Gulf of Mannar Marine NP, Tamil Nadu** — coral islands, dugong; sits inside Gulf of Mannar Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र).
44. **Mahatma Gandhi Marine NP (Wandoor)** and **Rani Jhansi Marine NP** are in the **Andaman** Islands.
45. **World's first National Park** is **Yellowstone NP, USA (1872)** — signed under President **Ulysses S. Grant**.
46. **Keoladeo** (केवलादेव) often carries **NP + Ramsar + UNESCO WH** together.
47. **Manas** was removed from the WH danger list in **2011** after recovery.
48. **Uttar Pradesh has no natural UNESCO World Heritage Site** — nearest are Valley of Flowers / Nanda Devi in Uttarakhand.
49. **Coastal NPs** protect mangrove, estuary, delta (डेल्टा), intertidal, and reef ecosystems; all marine NPs are coastal, but not every coastal NP is a "Marine NP" label.
50. **Gahirmatha** = marine **sanctuary** (turtle rookery), not NP.

📊 Quick Reference Grid

UP / India firsts

ITEM	LOCK
UP National Park	**Only Dudhwa** (Lakhimpur Kheri)
Dudhwa path	Sanctuary 1958 → NP **1977** → TR core **1987**
Valmiki / Sultanpur / Corbett	Bihar / Haryana / Uttarakhand (**not** UP NP)
UP marine NP	**Zero** (landlocked)
UP sanctuaries	**25**; Hastinapur largest
India first NP	Hailey/Corbett **1936** (UK)
World first NP	Yellowstone **1872**
Largest India NP	Hemis (Ladakh)
Most NPs / most WLS	MP / A&N
First Marine NP (प्रथम समुद्री उद्यान)	Gulf of Kutch **1980**
Gir / Kaziranga	Only Asiatic lions / one-horned rhino classic

Chapter 32: Biosphere Reserves (जैवमंडल आरक्षित क्षेत्र)

🎯 Consolidated Must-Score Facts

1. A Biosphere Reserve is a **UNESCO MAB (1971)** landscape with **core → buffer → transition**. It is **not** a Wildlife Protection (वन्यजीव सुरक्षा) Act category.
2. India has **18** notified national Biosphere Reserves. About **13** sit on the UNESCO World Network.
3. India's **first** Biosphere Reserve is **Nilgiri (नीलगिरि) (1986)**. The **largest** is **Great Rann of Kutch (Gujarat)**. The **smallest** is **Nokrek (Meghalaya (मेघालय))**.
4. **Nokrek** is in **Meghalaya** (मेघालय). Pairing Nokrek with **Manipur** is always wrong.
5. **Manas (Assam)** remains a national Biosphere Reserve but was **removed from the UNESCO World Network in 2011**.
6. **Uttar Pradesh has zero Biosphere Reserves.** Nanda Devi (नंदा देवी) BR is in **Uttarakhand** (उत्तराखंड), not UP.
7. Dudhwa (दुधवा) is a National Park. Valmiki (वाल्मीकि) is a Bihar National Park. Neither is a UP Biosphere Reserve.
8. Match fact: **Simlipal–Odisha**, **Nokrek–Meghalaya**, **Agasthyamalai–Kerala**, **Manas–Assam**.
9. Match fact: **Simlipal–Odisha**, **Pachmarhi–Madhya Pradesh**, **Nanda Devi–Uttarakhand**, **Nokrek–Meghalaya**.
10. **Panna (Madhya Pradesh, 2011)** is India's newest notified Biosphere Reserve.
11. Madhya Pradesh has **three** Biosphere Reserves: Pachmarhi, Achanakmar–Amarkantak (with Chhattisgarh), and Panna.
12. Multi-state classics: **Nilgiri** (नीलगिरि) (Tamil Nadu (नाडु)–Karnataka–Kerala), **Agasthyamalai** (Kerala–Tamil Nadu), **Achanakmar–Amarkantak** (MP–Chhattisgarh).
13. Core allows **no exploitation**. Buffer allows limited research and education use. Transition supports livelihoods.
14. A National Park is the strictest Wildlife Protection Act site. A Biosphere Reserve adds sustainable use and research under MAB.
15. World Network membership is a UNESCO subset of national Biosphere Reserves. National listing alone is not enough for WNBR.
16. Bihar and Rajasthan also have **zero** Biosphere Reserves. Valmiki remains an NP, not a BR.
17. Gulf of Mannar is a **marine** Biosphere Reserve in Tamil Nadu. Sundarbans (सुंदरबन) BR is in West Bengal.
18. Cold Desert BR is in **Himachal (हिमाचल) Pradesh**. Khangchendzonga BR is in **Sikkim**.
19. Seshachalam Hills BR is in **Andhra Pradesh**. Great Nicobar (निकोबार) BR is in the Andaman (अंडमान) and Nicobar Islands.
20. Periodic UNESCO review can delist a site from the World Network — India's example is **Manas 2011**.
21. Yellowstone, Serengeti–Ngorongoro, Galápagos, Lake Baikal, Doñana, and Mount Kenya are classic world Biosphere icons.
22. Do not place Nanda Devi in Uttar Pradesh on a match list. After bifurcation it is an **Uttarakhand** site.
23. Agasthyamalai paired only with Kerala is accepted as correct in recent papers even though the landscape also spans Tamil Nadu.
24. Biosphere Reserves can overlap National Parks and World Heritage (विश्व धरोहर) tags, but the MAB label is still a separate tool.
25. Among common options, **biosphere reserves** are the best strategy for conserving biodiversity **together with traditional human life / cultural diversity**. **Gulf of Mannar** is often cited as India's richest marine biodiversity BR mosaic.
26. **Three zones: Core** (no exploitative use / genetic diversity (आनुवंशिक विविधता)) → **Buffer** (research, education, regulated eco-tourism and grazing) → **Transition** (settlements, crop cultivation, livelihoods).
27. **Triple mandate:** conservation + sustainable development + logistic support (research/monitoring/education).
28. **Agasthyamalai** includes Neyyar, Peppara, Shendurney WLS and **Kalakad–Mundanthurai Tiger Reserve**.
29. **Gulf of Mannar** is often the largest **UNESCO-certified** Indian BR by area; **Great Rann of Kutch** is the largest **national** BR.
30. **Bundala** Biosphere Reserve is in **Sri Lanka**, not India.
31. **Largest BR:Great Rann of Kutch**, Gujarat. **Smallest:Nokrek**, Meghalaya.
32. **Newest notified:Panna**, Madhya Pradesh (2011).
33. **Manas (Assam)** remains a **national BR** but was **removed from UNESCO WNBR in 2011**.
34. **Madhya Pradesh** has three BRs: Pachmarhi, Achanakmar–Amarkantak (with Chhattisgarh), Panna.
35. **Uttar Pradesh has no notified Biosphere Reserve.**

📊 Quick Reference Grid

Frame

ITEM	LOCK
BR model	UNESCO MAB **1971**; core → buffer → transition
India count	**18** national; ~**13** UNESCO WNBR
First / largest / smallest	Nilgiri **1986** / Great Rann of Kutch / Nokrek
Nokrek state	**Meghalaya** (≠ Manipur)
Manas UNESCO	Removed from WNBR **2011** (still national BR)
UP BRs	**Zero**; Nanda Devi = Uttarakhand
Newest notified	Panna **2011** (MP)
MP trio	Pachmarhi · Achanakmar–Amarkantak · Panna
Multi-state	Nilgiri TN–KA–KL; Agasthyamalai KL–TN

Table: Complete Master Matrix of 18 Biosphere Reserves in India

S.NO.	BIOSPHERE RESERVE	STATE / UT	YEAR NOTIFIED	UNESCO WNBR STATUS	KEY FAUNA / UNIQUE ECOLOGICAL FEATURE
1	**Nilgiri**	Tamil Nadu, Kerala, Karnataka	1986	**Yes (2000 — 1st in India)**	Lion-tailed Macaque, Nilgiri Tahr; tri-state junction
2	**Nanda Devi**	Uttarakhand	1988	**Yes (2004)**	Snow Leopard, Himalayan Musk Deer, Valley of Flowers
3	**Nokrek**	Meghalaya	1988	**Yes (2009)**	Red Panda, Asian Elephant, Citrus indica (wild mother orange); Garo Hills
4	**Great Nicobar**	Andaman & Nicobar Islands	1989	**Yes (2013)**	Saltwater Crocodile, Megapode, Leatherback Turtle
5	**Gulf of Mannar**	Tamil Nadu	1989	**Yes (2001)**	Dugong (Sea Cow), Coral Reefs, Seagrass beds
6	**Manas**	Assam	1989	**No** (Removed 2011)	Golden Langur, Pygmy Hog, One-horned Rhino; Bodo territory
7	**Sundarbans**	West Bengal	1989	**Yes (2001)**	Royal Bengal Tiger, Mangrove swamps, Estuarine crocodile
8	**Simlipal**	Odisha	1994	**Yes (2009)**	Tiger, Elephant, Gaur; Mayurbhanj elephant reserve
9	**Dihang-Dibang**	Arunachal Pradesh	1998	**No**	Mishmi Takin, Musk Deer; Eastern Himalayas
10	**Pachmarhi**	Madhya Pradesh	1999	**Yes (2009)**	Giant Squirrel, Flying Squirrel; Satpura Range
11	**Achanakmar-Amarkantak**	Madhya Pradesh, Chhattisgarh	2005	**Yes (2012)**	Four-horned Antelope, Wild Dog; Maikal Hills origin of Narmada/Son
12	**Great Rann of Kutch**	Gujarat	2008	**No** (Largest BR in India)	Indian Wild Ass (Khur), Greater Flamingo; seasonal salt marsh
13	**Cold Desert**	Himachal Pradesh	2009	**No**	Snow Leopard, Tibetan Gazelle; includes Pin Valley NP & Kibber
14	**Khangchendzonga**	Sikkim	2010	**Yes (2018)**	Snow Leopard, Red Panda, Musk Deer; Mixed World Heritage Site
15	**Agasthyamalai**	Kerala, Tamil Nadu	2001	**Yes (2016)**	Nilgiri Tahr, Asian Elephant, Kanikaran indigenous tribe
16	**Panna**	Madhya Pradesh	2011	**Yes (2020 — Newest in WNBR)**	Bengal Tiger, Chital, Chinkara, Sambhar; Ken river valley
17	**Seshachalam Hills**	Andhra Pradesh	2010	**No**	Red Sanders (Pterocarpus santalinus), Slender Loris
18	**Dibru-Saikhowa**	Assam	1997	**No** (Smallest BR in India)	Feral Horses, White-winged Wood Duck; Brahmaputra-Lohit floodplains

Chapter 33: Atmosphere

🎯 Consolidated Must-Score Facts

1. From bottom to top the layers run **Troposphere → Stratosphere → Mesosphere → Thermosphere → Exosphere**.
2. The **troposphere** (क्षोभमंडल) holds weather. Temperature **falls** with height in this layer.
3. The **stratosphere** (समतापमंडल) holds the **ozone layer**. Temperature **rises** with height here.
4. The **mesosphere** is the **coldest** layer and is where most meteors burn.
5. The **thermosphere** is very hot but thin. Auroras, the ISS path, and much of the ionosphere sit here.
6. The **exosphere** is the outermost fringe and merges into space.
7. Dry air composition is about **N_2 78%**, **O_2 21%**, **argon ~0.93%**, and **CO_2 ~0.04%**.
8. **Nitrogen gas (N_2) is not a greenhouse gas.Nitrous oxide (N_2O)** is a greenhouse gas. Do not confuse the two.
9. **Argon is not a greenhouse gas.** Oxygen and helium are also not greenhouse gases.
10. Earth's air is heated mainly by **long-wave terrestrial radiation (भौमिक विकिरण)** from the warmed surface, not by direct shortwave alone.
11. The Sun sends **shortwave** insolation (सूर्यताप). The surface emits **longwave** radiation that warms the air.
12. Maximum ozone concentration is in the **stratosphere**, not the troposphere.
13. The **homosphere** (सममंडल) is well mixed to about **80 km**. Above that the **heterosphere** (विषममंडल) separates gases by mass.
14. Tropospheric ozone (क्षोभमंडलीय ओजोन) is a pollutant and greenhouse gas. Stratospheric ozone (समतापमंडलीय ओजोन) is the UV shield.
15. Temperature inversion and weather systems live in the troposphere. Commercial jets often cruise near the tropopause.
16. Do not place the ozone layer in the troposphere on a match list.
17. Mesosphere cold and thermosphere hot can both be true because density differs — thin hot air is not the same as surface heat.
18. Exosphere atoms can escape to space. It is not a weather layer.
19. Composition percentages are for **dry air**. Water vapour varies strongly by place and season.
20. Atmospheric studies for climate start from this layer ladder plus greenhouse-gas composition facts.
21. Ionosphere radio reflection is mainly a thermosphere–upper-atmosphere feature, not a troposphere fact.
22. Stratospheric heating with height comes largely from ozone absorbing ultraviolet radiation.
23. Never reverse the order of troposphere and stratosphere in a bottom-to-top ladder.
24. Long-wave terrestrial radiation is the standard answer for how the lower atmosphere is heated after surface warming.
25. **Dry air near the surface: Nitrogen ~78.08%**, **Oxygen ~20.95%**, **Argon ~0.93%**, **CO_2 ~0.04%**.
26. **Permanent gases** (N_2, O_2, Ar) stay roughly constant in the homosphere.
27. **Variable gases** include water vapour (0–4%), CO_2, CH_4, O_3, and aerosols.
28. **N_2 is not a greenhouse gas**; **N_2O** is. Do not confuse N_2 with nitrogen oxides or nitrous oxide.
29. **Argon is not a GHG** (also covered in Topic 34 greenhouse-gas questions).
30. **Water vapour** is the strongest natural tropospheric greenhouse gas by volume, but mainly a climate **feedback**.
31. Assertion (A): Dry air is about N_2 78%, O_2 21%, argon ~0.93%, CO_2 ~0.04%.
32. Reason (R): Argon is a major greenhouse gas.
33. Assertion (A): Stratospheric temperature rises with height largely because ozone absorbs UV.
34. Reason (R): The ozone layer's maximum concentration is in the stratosphere.
35. Assertion (A): Commercial jets often cruise near the tropopause.
36. Reason (R): The tropopause sits atop the troposphere neighbourhood.
37. Assertion (A): Do not place the ozone layer in the troposphere on a match list.
38. Reason (R): Maximum ozone concentration is in the stratosphere.
39. Assertion (A): Temperature falls with height in the troposphere.
40. Reason (R): Temperature rises with height in the stratosphere.
41. Assertion (A): Oxygen and helium are not greenhouse gases.
42. Reason (R): Only CO_2 among all atmospheric gases can ever affect radiation.

📊 Quick Reference Grid

Structure

ITEM	LOCK
Order (bottom→top)	Tropo → Strato → Meso → Thermo → Exo
Weather / ozone layer	Troposphere / Stratosphere
Coldest / meteors	Mesosphere
Aurora / ISS	Thermosphere
Dry air	N_2 ~78% · O_2 ~21% · Ar ~0.93% · CO_2 ~0.04%
N_2 vs N_2O	Not GHG / is GHG
Not GHGs	Ar, N_2, O_2, He
Air heating	Mainly longwave from warmed surface
Homosphere	Well mixed to ~**80 km**

Chapter 34: Environmental Monitoring (पर्यावरणीय निगरानी)

🎯 Consolidated Must-Score Facts

1. The **PSR** model runs **Pressure → State → Response** for environmental indicators.
2. Air indicators include **PM2.5, PM10, SO_2, NOx, and AQI**. Water indicators include **BOD, COD, DO, pH, and coliform**.
3. Noise is measured in **decibels**. Smoke opacity uses the **Ringelmann** scale **0–5**.
4. **Lichens** are classic bio-indicators for sulphur dioxide stress.
5. **Monitoring** means systematic repeated measurement against standards. It is not the same as an audit.
6. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) runs **NAMP** (manual air), **CAAQMS** (real-time air), and **NWMP** (surface water).
7. **CGWB** monitors groundwater quality and levels and publishes the annual groundwater quality report.
8. **SAFAR** (सफर) is the IITM air-quality and weather forecast system associated with Harsh Vardhan's launch messaging in **July 2018**.
9. **NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) (2019)** expands city monitoring and action. **SAMEER** is an AQI public app.
10. An **environmental audit** verifies compliance with law or clearance conditions — it is more than raw (रॉ) data logging.
11. **Form V** Environmental Statement under EPA Rule **14** goes to the SPCB yearly.
12. **ISO 14001** is the Environmental Management System standard.
13. Consent to Operate links industry permission to both monitoring data and audit compliance.
14. **BOD** is microbial oxygen demand. **COD** is chemical oxygen demand. COD is usually ≥ BOD.
15. NAMP shows long trend with manual stations. CAAQMS gives continuous real-time feeds.
16. CPCB is the air–surface-water regulator home. CGWB is the groundwater home. Do not swap them.
17. Green audits and EC compliance reports sit in the audit family, not in casual one-off sampling.
18. Indicators turn raw numbers into decision signals for policy and public warning.
19. Bio-indicators supplement instruments when living organisms reveal chronic pollution stress.
20. Ringelmann is about visible smoke shade, not a water-quality number.
21. Do not call an audit "just another AQI reading (रीडिंग)." Audit checks legal conformity.
22. Surface-water networks (NWMP) do not replace groundwater networks (CGWB).
23. **Environmental indicators** turn raw data into decision signals for policy and public warning.
24. **PSR model:Pressure** (emissions, resource use) → **State** (ambient quality) → **Response** (treatment capacity, action plans).
25. **Ringelmann scale (रिंगेलमैन पैमाना) (0–5)** measures **smoke plume density/opacity**, not water, fog, or noise.
26. **Environmental monitoring** is systematic, repeated measurement of air, water, soil, noise, and related parameters against standards.
27. **CPCB** runs national **air** and **surface-water** programmes and sets ambient/effluent standards with SPCBs.
28. **NAMP** = manual air stations for long-term trends; **CAAQMS** = continuous real-time ambient air (feeds AQI boards).
29. **NWMP** = National Water Quality Monitoring Programme for rivers/lakes.
30. **CGWB** monitors **groundwater** levels and quality; publishes the **Annual Groundwater Quality Report** — not CPCB.
31. **SAFAR** (IITM) combines air-quality monitoring with weather forecasting; inaugurated by **Dr. Harsh Vardhan** (July 2018).
32. **NCAP (2019)** expands monitoring and action in non-attainment cities (including multiple UP cities).
33. **SAMEER** app disseminates CPCB AQI; **PARIVESH** tracks EC compliance reporting.
34. **UPPCB** issues Consent to Establish/Operate and inspects industries in Uttar Pradesh (उत्तर प्रदेश).
35. **Environmental audit** is a systematic, documented check of whether an organisation meets environmental laws, consent conditions, and internal EMS goals.
36. **ISO 14001** is the voluntary international standard for an **Environmental Management System (EMS)**.

📊 Quick Reference Grid

Tools

ITEM	LOCK
PSR	Pressure → State → Response
Noise / smoke	dB / Ringelmann **0–5**
Lichens	SO_2 bio-indicators
CPCB networks	NAMP · CAAQMS · NWMP
CGWB	Groundwater quality + levels report
SAFAR	IITM forecast (2018 launch messaging)
SAMEER	AQI public app
Form V	EPA Rule 14 Environmental Statement → SPCB yearly
ISO 14001	EMS standard
Monitoring ≠ audit	Repeated measurement / compliance verification

Chapter 35: International Environmental Organizations

🎯 Consolidated Must-Score Facts

1. **UNEP** was created after **Stockholm 1972** (स्टॉकहोम सम्मेलन). Its HQ is **Nairobi**. Flagship products include the **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) and **GEO**.
2. **UNDP** sits in **New York** and publishes the **Human Development Report / HDI** (from **1990**).
3. **FAO** sits in **Rome** and publishes **SOFO**. It co-hosts Rotterdam work with UNEP.
4. **UNESCO** sits in **Paris** and runs World Heritage (विश्व धरोहर) and the **MAB** Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र) programme.
5. **IUCN** (आईयूसीएन) sits in **Gland** and publishes the **Red List** of extinction risk.
6. **WWF** also sits in **Gland** and publishes the **Living Planet Report / LPI**.
7. **IPCC** sits in **Geneva**. It was created by **WMO + UNEP** in **1988** and shared the **2007** Nobel Peace Prize.
8. Emissions Gap Report = **UNEP**, not IPCC or UNDP. HDR = **UNDP**, not UNEP.
9. IPCC does science assessment. **UNFCCC** (यूएनएफसीसीसी) is the climate treaty. Do not merge the two.
10. IUCN assesses extinction risk. **CITES** (साइट्स) regulates wildlife **trade** (पण्याध्यक्ष). Different jobs.
11. IUCN and WWF share a city but differ: assessment union versus conservation NGO campaigning and Living Planet metrics.
12. UNEP coordinates environment (पर्यावरण). UNDP coordinates development. HQ and report facts separate them.
13. UNESCO heritage–education–MAB is not the same as UNEP environment coordination.
14. The **GEF (1991)** is a multilateral funder. Donors are not "developed countries only" as a rigid fact.
15. IPCC Assessment Reports synthesize climate science. They do not set national NDCs.
16. FAO's food–forest–agriculture mandate explains why SOFO and pesticide PIC work sit there.
17. Red List categories are IUCN risk ranks, not CITES Appendix numbers.
18. Living Planet Index tracks vertebrate abundance trends for WWF messaging.
19. Nairobi versus Geneva versus New York versus Rome versus Paris versus Gland is a high-yield HQ ladder.
20. Do not credit IPCC with the Emissions Gap Report.
21. Do not credit UNEP with HDI.
22. Treaty secretariats hosted or supported by UNEP (Montreal, Basel, Stockholm POPs, Minamata) are still treaty bodies, not UNEP synonyms.
23. WWF is not the Red List author. IUCN is.
24. FAO HQ is Rome, not Paris or Geneva.
25. **WMO** headquarters is **Geneva**. It grew from the International Meteorological Organization (**1873**) and became WMO on **17 March 1951**. It co-created the **IPCC** with UNEP in **1988**.
26. **Greenpeace International** is an environmental NGO headquartered in **Amsterdam**, Netherlands.
27. The **US EPA** was established on **2 December 1970**. Do not confuse it with India's **Environment (Protection) Act, 1986**.
28. **OPCW** implements the **Chemical Weapons Convention** (force **29 April 1997**); HQ is **The Hague**. It won the Nobel Peace Prize in **2013**. It is **not** an EU, NATO, or WHO body.
29. **UN-Habitat** (UN Human Settlements Programme; HQ **Nairobi**, from **1978**) promotes sustainable towns and shelter. Partners are **not** governments-only.
30. **UNEP (United Nations Environment Programme)** is the UN's principal environment coordination body, created after the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन).
31. **HQ: Nairobi, Kenya** — not Geneva or New York.
32. **GEF (Global Environment Facility, 1991)** finances projects under major environment conventions; UNEP is an implementing agency. Donors are **not** "developed countries only."
33. **UNDP (United Nations Development Programme)** is the UN's development network agency.
34. **FAO (Food and Agriculture Organization)** is the UN specialised agency for food and agriculture.
35. **UNESCO** leads education, science, and culture — environment enters via heritage and biosphere programmes.
36. **Limits to Growth** is **Club of Rome**, not UNESCO.
37. **IUCN (International Union for Conservation of Nature)** is a global conservation union of governments and NGOs, founded **1948**.
38. **HQ: Gland, Switzerland**.
39. **Threatened** categories commonly = **VU + EN + CR**.
40. **WWF (World Wide Fund for Nature)** is a global conservation **NGO**, not a UN agency.
41. **HQ: Gland, Switzerland** — same city as IUCN, different organisation.
42. **IPCC (Intergovernmental Panel on Climate Change (जलवायु परिवर्तन))** was established in **1988** by **WMO** and **UNEP**.
43. **UNFCCC/COP** is the policy–treaty track; IPCC informs it.
44. **WMO headquarters is Geneva, Switzerland.** It co-created the **IPCC** with UNEP in **1988**.
45. **OPCW (Organisation for the Prohibition (प्रतिषेध) of Chemical Weapons)** implements the **Chemical Weapons Convention** (force **29 April 1997**); HQ **The Hague**, Netherlands. It verifies destruction/non-re-emergence of chemical weapons and assists states against chemical threats — **not** an EU/NATO/WHO body. Nobel Peace Prize **2013**.

📊 Quick Reference Grid

Org ↔ HQ / product

ITEM	LOCK
UNEP	Nairobi; Emissions Gap + GEO; post-Stockholm 1972
UNDP	New York; HDR/HDI
FAO	Rome; SOFO
UNESCO	Paris; WH + MAB
IUCN	Gland; Red List
WWF	Gland; Living Planet
IPCC	Geneva; WMO+UNEP 1988; Nobel 2007
IUCN ≠ CITES	Risk assess / trade regulate
GEF	**1991** multilateral funder